Complex Number

DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Complex Number

INTRODUCTION

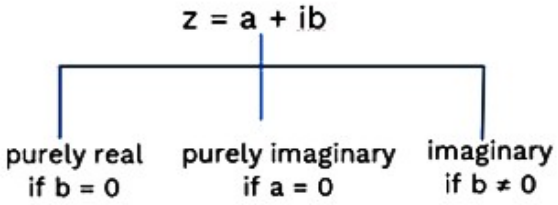
Indian mathematician Mahavira(850A.D.) was first to mention in his work 'Ganitasara Sangraha' As in nature of things a negative (quantity) is not a square (quantity), it has, therefore, no square root'. Hence there is no real number x which satisfies the polynomial equation $x^2 + 1 = 0$

A symbol $\sqrt{-1}$, denoted by letter i was introduced by Swiss Mathematician, Leonhard Euler (1707-1783) in 1748 to provide solutions of equation $x^2 + 1 = 0$. i was regarded as a fictitious or imaginary number which could be manipulated algebraically like an ordinary real number, except that its square was -1. The letter i was used to denote $\sqrt{-1}$, possibly because i is the first letter of the Latin word 'imaginarius'.

To permit solutions of such polynomial equations, the set of complex numbers is introduced. We can consider a complex number as having the form a + ib where a and b are real number. It is denoted by z i.e. z = a + ib. 'a' is called as real part of z which is denoted by Re(z) and 'b' is called as imaginary part of z which is denoted by Im(z)

Classification of complex number:

In fact, every complex can be classified as

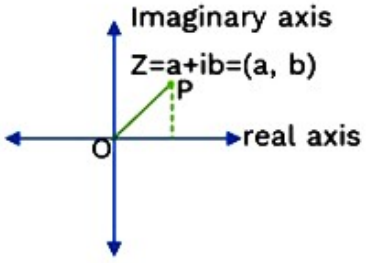


Hence, 0+0 i is both a purely real as well as a purely imaginary but not imaginary.

Geometrical representation of a complex number:

Master Argand had done a systematic study on complex numbers and represented every complex number as a set of ordered pair (a, b) on a plane called complex plane/argand plane.

All complex numbers lying on the real axis were called as purely real and those lying on imaginary axis as purely imaginary.



Integral Powers of i:

We have $i = \sqrt{-1}$ so $i^2 = -1, i^3 = -i, \quad i^4 = 1$

Or $\quad i^{4n+1} = i, i^{4n+2} = -1 \quad$ for any $n \in I$

$i^{4n+3} = -i, i^{4n} = 1$

Thus any integral power of i can be expressed as ± 1 or $\pm i$.

Q. **Find the value of** $\dfrac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} + 2$

Sol. $\dfrac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} + 2 = \dfrac{i^{584}}{i^{574}} + 2$

$= i^{10} + 2 = -1 + 2 = 1$

Q. **Show that the polynomial** $x^{4p} + x^{4q+1} + x^{4r+2} + x^{4s+3}$ **is divisible by** $x^3 + x^2 + x + 1$ **where** $p, q, r, s \in N$

Sol. Let $f(x) = x^{4p} + x^{4q+1} + x^{4r+2} + x^{4s+3}$

$x^3 + x^2 + x + 1 = \left(x^2 + 1\right)(x + 1) = (x + i)(x - i)(x + 1)$

$f(i) = i^{4p} + i^{4q+1} + i^{4r+2} + i^{4s+3} = 1 + i^1 + i^2 + i^3 = 1 + i - 1 - i = 0$

$f(-i) = (-i)^{4p} + (-i)^{4q+1} + (-i)^{4r+2} + (-i)^{4s+3}$

$= 1 + (-i)^1 + (-i)^2 + (-i)^3 = 1 - i - 1 + i = 0$

$f(-1) = (-1)^{4p} + (-1)^{4q+1} + (-1)^{4r+2} + (-1)^{4s+3} = 0$

Thus by division theorem f(x) is divisible by $x^3 + x^2 + x + 1$.

Q. **Find** $\sum_{r=1}^{2006} i^r$

Sol. $S = \sum_{r=1}^{2006} i^r$

$= i + i^2 + i^3 + ... + i^{2006}$

$= \left(i + i^2 + i^3 + i^4\right) + \left(i^5 + i^6 + i^7 + i^8\right) + ... + \left(i^{2001} + i^{2002} + i^{2003} + i^{2004}\right) + i^{2005} + i^{2006}$

$= 0 + 0 + ... + 0 + i^{2005} + i^{2006} \left(\because i^{4r} + i^{4r+1} + i^{4r+2} + i^{4r+3} = 0\right)$

$= i + i^2 \qquad = -1 + i$

Q. **Find** $\sum_{r=1}^{2020} r.i^r$

Sol. $s = \sum_{r=1}^{2020} r.i^r$

$$\Rightarrow s = i + 2i^2 + 3i^3 + 4i^4 + 5i^5 + \ldots + 2020i^{2020}$$

$$= i(1 + 5 + 9 + \ldots + 2017) + i^2(2 + 6 + 10 + \ldots + 2018)$$

$$+ i^3(3 + 7 + 11 + \ldots + 2019) + i^4(4 + 8 + \ldots + 2020)$$

$$= i \times \frac{505}{2}(1 + 2017) + i^2 \times \frac{505}{2}(2 + 2018) + i^3 \times \frac{505}{2}(3 + 2019) + i^4 \times \frac{505}{2}(4 + 2020)$$

$$\Rightarrow s = \frac{505}{2}(2018i - 2020 - 2022i + 2024)$$

$$= \frac{505}{2}(4 - 4i)$$

$$\Rightarrow s = 1010(1 - i)$$

Note:

(a) The set R of real number is a proper subset of the Complex Numbers. Hence the complete number system is $N \subset W \subset I \subset Q \subset R \subset C$

(b) Zero is purely real as well as purely imaginary but not imaginary.

(c) $\sqrt{a}\sqrt{b} = \sqrt{ab}$ only if atleast one of a or b is non-negative.

(d) $z_1^2 + z_2^2 = 0 \not\Rightarrow z_1 = 0 = z_2$ i.e., $z_1 = 1 + i$ and $z_2 = 1 - i$

ALGEBRA OF COMPLEX NUMBER

Equality of complex number:

Let there be two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

If $z_1 = z_2$ then $Re(z_1) = Re(z_2)$ and $Im(z_1) = Im(z_2)$

i.e. if $x_1 + iy_1 = x_2 + iy_2$

$\Rightarrow$ $x_1 = x_2$ and $y_1 = y_2$ simultaneously.

Q. **Find the real values of x and y for which the equation** $(x^4 + 2xi) - (3x^2 + yi) = (3 - 5i) + (1 + 2yi)$ **is satisfied.**

Sol. Given equation $(x^4 + 2xi) - (3x^2 + yi) = (3 - 5i) + (1 + 2yi)$

$\Rightarrow \left(x^4 - 3x^2\right) + i(2x - 3y) = 4 - 5i$

Equating real and imaginary parts, we get

$x^4 - 3x^2 = 4$ ...(i)

And $2x - 3y = -5$ ...(ii)

From (i) and (ii), we get $x = \pm 2$ and $y = 3, \frac{1}{3}$

Q. **If $z = x + iy, z^{\frac{1}{3}} = a - ib$ and $\frac{x}{a} - \frac{y}{b} = k\left(a^2 - b^2\right)$, then find the value of k**

Sol. $(x + iy)^6 = a - ib$

$\Rightarrow x + iy = (a - ib)^3 = \left(a^3 - 3ab^2\right) + i\left(b^3 - 3a^2b\right)$

$\Rightarrow x = a^3 - 3ab^2, y = b^3 - 3a^2b \Rightarrow \frac{x}{a} = a^2 - 3b^2$ and $\frac{y}{b} = b^2 - 3a^2$

$\Rightarrow \frac{x}{a} - \frac{y}{b} = a^2 - 3b^2 - b^2 + 3a^2 = 4\left(a^2 - b^2\right) \therefore k = 4.$

Q. **If the expression $(1 + ir)^3$ is of the form of $s(1 + i)$ for some real 's' where 'r' is also real and $i = \sqrt{-1}$ then the value of 'r' can be**

(A) $\cot\frac{\pi}{8}$ **(B)** $\sec\pi$ **(C)** $\tan\frac{\pi}{12}$ **(D)** $\tan\frac{5\pi}{12}$

Sol. **(BCD)**

We have $(1 + ri)^3 = s(1 + i)$

$1 + 3ri + 3r^2i^2 + r^3i^3 = s(1 + i)$

$1 - 3r^2 + i\left(3r - r^3\right) = s + si \Rightarrow 1 - 3r^2 = s = 3r - r^3$

Hence $1 - 3r^2 = 3r - r^3$

$\Rightarrow r^3 - 3r^2 - 3r + 1 = 0 \Rightarrow \left(r^3 + 1\right) - 3r(r + 1) = 0 \Rightarrow (r + 1)\left(r^2 + 1 - r - 3r\right) = 0$

$\therefore r = -1$ or $r^2 - 4r + 1 = 0$

$\Rightarrow r = \frac{4 \pm \sqrt{16 - 4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} \Rightarrow r = 2 + \sqrt{3}$ or $2 - \sqrt{3} \Rightarrow B, C, D$

Q. **Let z be a complex number satisfying the equation, $z^2 - (3 + i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root, then find the value of m.**

Sol. $z^2-(3+i)z+m+2i=0$

Since z is real,

$(z^2-3z+m)+(2-z)i=0$

$\Rightarrow z^2-3z+m=0$ and $z=2$

$\Rightarrow 2^2-3\times 2+m=0$

$\Rightarrow m=2$

Q. **Find the set of points on the complex plane for which z^2+z+1 is real and positive.**

Sol. Let $z=x+iy, x,y\in R$

$\Rightarrow z^2+z+1=(x+iy)^2+(x+iy)+1$

$=(x^2-y^2+x+1)+i(2xy+y)$

Since z^2+z+1 is a position real number,

$2xy+y=0$ and $x^2-y^2+x+1>0$

$(2x+1)y=0$

$\Rightarrow y=0$ or $x=-\frac{1}{2}$

Case-I: $y=0$

$\Rightarrow x^2-x+1>0$

$\Rightarrow x\in R$

Case-II: $x=\frac{-1}{2}$

$\Rightarrow \frac{1}{4}-y^2-\frac{1}{2}+1>0$

$\Rightarrow y^2<\frac{3}{4}$

$\Rightarrow y\in\left(-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$

$\therefore$ Point lie on x-axis (Real axis) or line segment joining $\left(-\frac{1}{2},-\frac{\sqrt{3}}{2}\right)$ and $\left(\frac{-1}{2},\frac{\sqrt{3}}{2}\right)$.

Note:

Inequality in complex numbers are never talked. If $x_1 + iy_1 > x_2 + iy_2$ has to be meaningful $\Rightarrow y_1 = y_2 = 0$. Equalities however in complex numbers are meaningful. Two complex numbers z_1 and z_2 are said to be equal if $\mathrm{Re}\, z_1 = \mathrm{Re}\, z_2$ and $\mathrm{Im}(z_1) = \mathrm{Im}(z_2)$ (i.e., they occupy the same position on complex plane)

Addition:

$$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2) \in C$$

It is easy to observe that the sum of two complex numbers is a complex number whose real (imaginary) part is the sum of the real (imaginary) parts of the given numbers:

$$\mathrm{Re}(z_1 + z_2) = \mathrm{Re}(z_1) + \mathrm{Re}(z_2)$$

$$\mathrm{Im}(z_1 + z_2) = \mathrm{Im}(z_1) + \mathrm{Im}(z_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2) \in C$$

That is

$$\mathrm{Re}(z_1 - z_2) = \mathrm{Re}(z_1) - \mathrm{Re}(z_2)$$

$$\mathrm{Im}(z_1 - z_2) = \mathrm{Im}(z_1) - \mathrm{Im}(z_2)$$

Multiplication:

$$z_1 \cdot z_2 = (x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1) \in C.$$

In other words

$$\mathrm{Re}(z_1z_2) = \mathrm{Re}(z_1) \cdot \mathrm{Re}(z_2) - \mathrm{Im}(z_1) \cdot \mathrm{Im}(z_2)$$

And $\mathrm{Im}(z_1z_2) = \mathrm{Im}(z_1) \cdot \mathrm{Re}(z_2) + \mathrm{Im}(z_2) \cdot \mathrm{Re}(z_1)$

For a real number λ and a complex number $z = x + iy$
$\lambda \cdot z = \lambda(x + iy) = \lambda x + i\lambda y \in C$ is the product of a real number with a complex number. The following properties are obvious:

(a) $\lambda(z_1 + z_2) = \lambda z_1 + \lambda z_2$

(b) $\lambda_1(\lambda_2 z) = (\lambda_1\lambda_2)z$;

(c) $(\lambda_1 + \lambda_2)z = \lambda_1 z + \lambda_2 z$ for all $z, z_1, z_2 \in C$ and $\lambda, \lambda_1, \lambda_2 \in R$

Actually, relations (a) and (c) are special cases of the distributive law and relation (b) comes from the associative law of multiplication for complex numbers.

Division of Complex Number:

Let $z_1 = x_1 + iy_1$ & $z_2 = x_2 + iy_2$

Then $\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} \Rightarrow \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)}$

$\Rightarrow \frac{(x_1x_2 + y_1y_2) + i(y_1x_2 - x_1y_2)}{(x_2^2 + y_2^2)} \Rightarrow \left\{\frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2}\right\} + i\left\{\frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2}\right\}$

$\frac{z_1}{z_2} = \text{Re}\left(\frac{z_1}{z_2}\right) + i\,\text{Im}\left(\frac{z_1}{z_2}\right)$

Q. **Find the value θ if $\frac{(3 + 2i\sin\theta)}{(1 - 2i\sin\theta)}$ is purely real or purely imaginary.**

Sol. $z = \frac{3 + 2i\sin\theta}{1 - 2i\sin\theta}$

Multiplying numerator and denominator by conjugate,

$z = \frac{(3 + 2i\sin\theta)(1 + 2i\sin\theta)}{1 + 4\sin^2\theta} = \frac{3 - 4\sin^2\theta + 8i\sin\theta}{1 + 4\sin^2\theta}$

Now z is purely real if $\sin\theta = 0$ or $\theta = n\pi, n \in I$.

z is purely imaginary if $3 - 4\sin^2\theta = 0$

$\Rightarrow \sin\theta = \pm\frac{\sqrt{3}}{2} = \pm\sin\frac{\pi}{3} \Rightarrow \theta = n\pi \pm \frac{\pi}{3}, n \in I$

Q. **Find $\text{Re}\left(\frac{2+i}{2-i}\right)$.**

Sol. $\frac{2+i}{2-i} = \frac{(2+i)^2}{(2-i)(2+i)}$

$= \frac{4 + i^2 + 4i}{4 - i^2}$

$= \frac{3}{5} + \frac{4}{5}i$

$\Rightarrow \text{Re}\left(\frac{2+i}{2-i}\right) = \frac{3}{5}$

Q. **If $\frac{(x-2) + (y-3)I}{1+i} = 1 - 3i$, find (x, y)**

Sol. $(x-2)+(y-3)i=(1-3i)(1+i)$

$=1+i-3i-3i^2$

$=4-2i$

$\Rightarrow x-2=4 \,\&\, y-3=-2$

$\Rightarrow x=6, y=1$

Q. **Find the least positive $n \in N$ if $\left(\frac{1+i}{1-i}\right)^n=1$**

Sol. $\frac{1+i}{1-i}=\frac{(1+i)^2}{(1-i)(1+i)}$

$=\frac{1+i^2+2i}{1-i^2}$

$=\frac{2i}{2}=i$

$\Rightarrow i^n=1$

$\Rightarrow n=4r, r \in I$

$\Rightarrow$ Least positive $n=4$.

Q. **a, b, c are real numbers in the polynomial $P(Z)=2Z^4+aZ^3+bZ^2+cZ+3$. If two roots of the equation $P(Z)=0$ are 2 and i, the find the value of 'a'.**

Sol. Imaginary roots occur in conjugate pairs.

$\Rightarrow$ One of the roots is '$-i$'

$\Rightarrow$ Product of the roots $=\frac{3}{2}$

If 4th root is α, then

$2 \times i \times (-i) \times \alpha=\frac{3}{2}$

$\Rightarrow -2\alpha i^2=\frac{3}{2}$

$\Rightarrow \alpha = \frac{3}{4}$

$\Rightarrow$ Sum of the roots $= 2 + \frac{3}{4} + i - i = \frac{-a}{2}$

$\Rightarrow a = \frac{-11}{2}$.

Q. **If $f(x) = x^4 - 4x^3 + 4x^2 + 8x + 44$, find $f(3+2i)$.**

Sol. $x = 3 + 2i$

$\Rightarrow x - 3 = 2i$

$\Rightarrow x^2 - 6x + 9 = -4$

$\Rightarrow x^2 - 6x + 13 = 0$

$x^4 - 4x^3 + 4x^2 + 8x + 44 = (x^2 - 6x + 13)(x^2 + 2x + 3) + 5$

If $x = 3 + 2i$, then $x^2 - 6x + 13 = 0$

So, for $x = 3 + 2i, x^4 - 4x^3 + 4x^2 + 8x + 44 = 0 + 5 = 5$

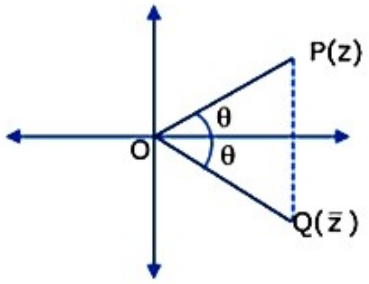
THREE IMPORTANT TERMS WITH RESPECT TO COMPLEX NUMBER

Conjugate of Complex Number:

Conjugate of a complex number $z = a + ib$ is denoted and defined by $\bar{z} = a - ib$, a and b are real numbers.

In a complex number if we replace i by $-i$ we get conjugate of the complex number. $\bar{z}$ is the mirror image of z about real axis on Argand's Plane.

Geometrical representation of conjugate of complex number



Note:

(a) $z + \bar{z} = 2\text{Re}z$

(b) $z - \bar{z} = 2i\text{Im}z$

(c) $z\bar{z} = a^2 + b^2$, where $z = a + ib$

(d) If z lies in 1st quadrant then $\bar{z}$ lies in 4th quadrant and $-\bar{z}$ in the 2nd quadrant.

(e) If $x + iy = f(a + ib)$ then $x - iy = f(a - ib)$

Further, $g(x+iy) = f(a+ib) \Rightarrow g(x-iy) = f(a-ib)$

e.g. $\sin(\alpha+i\beta) = x+iy \Rightarrow \sin(\alpha-i\beta) = x-iy$

Q. **If the biquadratic $x^4+ax^3+bx^2+cx+d=0$ $(a,b,c,d \in R)$ has 4 non real roots, two with sum $3+4i$ and the other two with product $13+i$, find the value of 'b'.**

Sol. Let the roots be $p \pm qi$ and $r \pm si$.

If $(p+qi)+(r+si) = 3+4i$,

then $(p-qi)(r-si) = 13+i$

$\Rightarrow p+r = 3$ and $q+s = 4$

and $pr-qs = 13$ and $ps+qr = -1$

$b = (p+qi)(p-qi)+(r+si)(r-si)+(p+qi)(r+si)$

$+(p+qi)(r-si)+(p-qi)(r+si)+(p-qi)(r-si)$

$= p^2+q^2+r^2+s^2+4rp$

$= (p+r)^2+2pr+(q+s)^2-2qs$

$= 3^2+4^2+2(pr-qs)$

$= 25+2(13)$

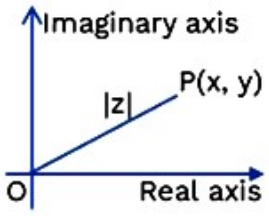
$\Rightarrow b = 51$.

Modulus of Complex Number:

Modulus of complex number is a distance of the point on the argand plane representing the complex number z from the origin.

If P denotes a complex number $z = x+iy$

Then $OP = |z| = \sqrt{x^2+y^2}$



Note:

(i) $|z| \geq 0$

(ii) All complex numbers having the same modulus lie on a circle with centre as origin and radius $r = |z|$.

Q. **Find the modulus of the following complex numbers:**

(i) $\frac{1}{2}+i\frac{\sqrt{3}}{2}$ **(ii)** $\frac{\sqrt{3}+1}{2\sqrt{2}}-i\frac{\sqrt{3}-1}{2\sqrt{2}}$ **(iii)** $1+\cos\alpha+i\sin\alpha, \alpha\in\left(\pi,\frac{3\pi}{2}\right)$

Sol. **(i)** $z=\frac{1}{2}+i\frac{\sqrt{3}}{2}$

$$\therefore |z|=\sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}=1$$

(ii) $|z|=\sqrt{\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)^2+\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2}\Rightarrow\sqrt{\frac{3+1+2\sqrt{3}+3+1-2\sqrt{3}}{8}}=\sqrt{1}=1$

(iii) $z=1+\cos\alpha+i\sin\alpha=2\cos^2\frac{\alpha}{2}+2i\sin\frac{\alpha}{2}\cdot\cos\frac{\alpha}{2}=2\cos\frac{\alpha}{2}\left[\cos\frac{\alpha}{2}+i\sin\frac{\alpha}{2}\right]$

$$|z|=-2\cos\frac{\alpha}{2}\quad\Rightarrow\quad\left\{\frac{\alpha}{2}\in\left(\frac{\pi}{2},\frac{3\pi}{4}\right)\Rightarrow\cos\frac{\alpha}{2}<0\right\}$$

Q. **If the complex number z satisfies $z+|z|=1+7i$ then find the value of $|z|^2$**

Sol. $z = x + iy$

$\therefore\ x+iy+\sqrt{x^2+y^2}=1+7i$

$x+\sqrt{x^2+y^2}=1$ ... (1)

and $y = 7$ ...(2)

$\therefore\ x+\sqrt{x^2+49}=1$

$x^2+49=1+x^2-2x$

$2x=-48$

$x=-24$

$\therefore |z|^2=x^2+y^2=625$

Previous Year's Question

Let z_1 **and** z_2 be two numbers satisfying $|z_1|=9$ **and** $|z_2-3-4i|=4$. Then the minimum value of $|z_1-z_2|$ is: **[JEE Main 2019]**

(A) 0 (B) $\sqrt{2}$ (C) 1 (D) 2

Previous Year's Question

If the equation, $x^2 + bx + 45 = 0 (b \in k)$ has complex conjugate roots and they satisfy $|z+1| = 2\sqrt{10}$, then: **[JEE Main 2020]**

(A) $b^2 + b = 12$ (B) $b^2 - b = 42$ (C) $b^2 - b = 30$ (D) $b^2 + b = 72$

Q. **Describe $|z - z_1|$.**

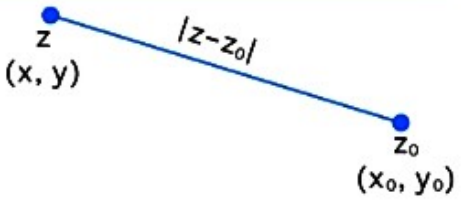
Sol. Let $z = x + yi$ & $z_1 = x_0 + y_0 i$

$\Rightarrow |z - z_1| = |(x - x_0) + i(y - y_0)|$

$= \sqrt{(x - x_0)^2 + (y - y_0)^2}$

= Distance between (x, y) and (x_0, y_0)

$\Rightarrow |z - z_1|$ represents distance between (x, y) and (x_0, y_0).

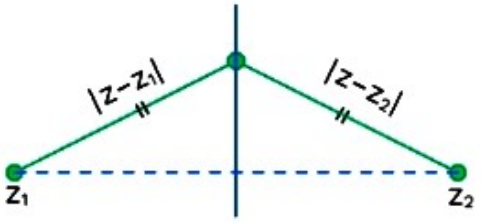


Q. **Describe $|z - z_1| = |z - z_2|$**

Sol. As explained in previous question, above expression mean distance of z from z_1 is equal to the distance of z from z_2.

$\Rightarrow$ z is equidistant from z_1 and z_2.

$\Rightarrow$ z lies on perpendicular bisector of line joining z_1 and z_2.

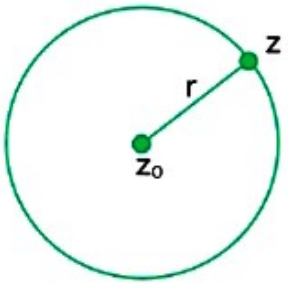


Q. **Describe $|z - z_0| = r$.**

Sol. $|z - z_0|$ = distance between z and z_0 = r.

$\Rightarrow$ z is a point whose distance from z_0 is always constant and equal to r.

$\Rightarrow$ z represents a circle with z_0 as center and r as radius.



Q. **On the complex plane, locus of a point z satisfying the inequality $2 \le |z-1| < 3$ denotes**
(A) region between the concentric circles of radii 3 and 1 centred at (1, 0)
(B) region between the concentric circles of radii 3 and 2 centred at (1, 0) excluding the inner and outer boundaries.
(C) region between the concentric circles of radii 3 and 2 centred at (1, 0) including the inner and outer boundaries.
(D) region between the concentric circles of radii 3 and 2 centred at (1, 0) including the inner boundary and excluding the outer boundary.

Sol. **(D)**

$|z-1|$ represents distance of z from (1, 0).

$\Rightarrow |z-1| < 3$ represents region inside the circle centred at (1, 0) with radius 3 excluding the boundary.

$\Rightarrow |z-1| \ge 2$ represents region outside the circle centered at (1, 0) with radius 2 and including the boundary

Q. **The complex number x satisfies $z + |z| = 2 + 8i$. The value of $|z|$ is**

(A) 10 **(B) 13** **(C) 17** **(D) 23 d**

Sol. Let $z = x + yi$

$$z + |z| = 2 + 8i$$

$$\Rightarrow x + yi + \sqrt{x^2 + y^2} = 2 + 8i$$

$$\Rightarrow x + \sqrt{x^2 + y^2} = 2$$

and $y = 8$

$$\Rightarrow x + \sqrt{x^2 + 64} = 2$$

$$\Rightarrow x^2 + 64 = 4 + x^2 - 4x$$

$$\Rightarrow x = -15$$

$$\Rightarrow z = -15 + 8i$$

$$\Rightarrow |z| = 17$$

Previous Year's Question

Let z be a complex number such that $|z| + z = 3 + i$ (where $i = \sqrt{-1}$). Then $|z|$ is equal to

(JEE Main 2019)

(A) $\frac{\sqrt{34}}{3}$ (B) $\frac{5}{3}$ (C) $\frac{\sqrt{41}}{4}$ (D) $\frac{5}{4}$

Previous Year's Question

The region represented by $\{z = x + iy \in C : |z| - Re(z) \le 1\}$ is also given by the inequality

(JEE Main 2020)

(A) $y^2 \le 2\left(x + \frac{1}{2}\right)$ (B) $y^2 \le x + \frac{1}{2}$ (C) $y^2 \ge 2(x+1)$ (D) $y^2 \ge x + 1$

Previous Year's Question

Let $z = x + iy$ be a non-zero complex number such that $z^2 = i|z|^2$, where $i = \sqrt{-1}$, then z lies on the

(JEE Main 2020)

(A) line y = x (B) real axis (C) imaginary axis (D) line, y = –x

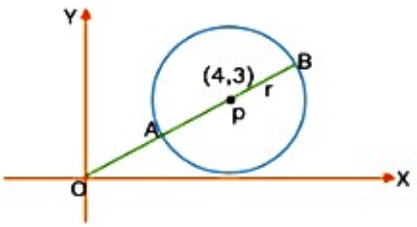
Q. **A point 'z' moves on the curve $|z - 4 - 3i| = 2$ in an argand plane. The maximum and minimum value of $|z|$ are :**

(A) 2, 1 **(B) 6, 5** **(C) 4, 3** **(D) 7, 3**

Sol. **(D)**

$|z - 4 - 3i| = 2 \Rightarrow |z - (4 + 3i)| = 2$

$\Rightarrow$ It represents circle with radius 2 centred



at (4, 3).

$|z|$=distance from origin.

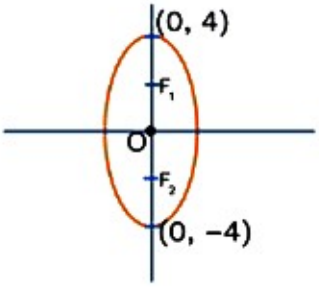
$\Rightarrow |z|_{min} = OA = OP - r = 5 - 2 = 3$

$|z|_{max} = OB = OP + r = 5 + 2 = 7$

Q. **If z is a complex number satisfying the equation $|z+i| + |z-i| = 8$, on the complex plane then maximum value of $|z|$ is**

(A) 2 **(B) 4** **(C) 6** **(D) 8**

Sol. $|z+i| + |z-i| = 8$



Let z be represented by P.

$\Rightarrow PF_1 + PF_2 = 8$

Thus above equation represents an ellipse with foci (0, –1) and (0, 1) and center (0, 0)

$\Rightarrow$ From above figure it is clear that maximum value of $|z| = 4$ when

$z = 4i$ or $-4i$.

Q. **Number of complex numbers z satisfying $z^3 = \bar{z}$ is/are**

(A) 1 **(B) 2** **(C) 4** **(D) 5**

Sol. $z^3 = \bar{z}$

$\Rightarrow z \times z^3 = z \times \bar{z}$

$\Rightarrow z^4 = |z|^2$

Taking modulus both side,

$|z|^4 = |z|^2$

$\Rightarrow |z| = 0$ or 1

$\Rightarrow z^4 = 0$ or 1

If $z^4 = 0 \Rightarrow z = 0$

If $|z| = 1$,

$z^3 = \bar{z}$

$\Rightarrow \bar{z} \times z^3 = (\bar{z})^2$

$\Rightarrow |z|^2 z^2 = (\bar{z})^2$

$\Rightarrow z^2 = (\bar{z})^2$

If $z = x + yi, x^2 + y^2 = 1$

and $(x + yi)^2 = (x - yi)^2$

$\Rightarrow x^2 + y^2i^2 + 2xyi = x^2 + y^2i^2 - 2xyi$

$\Rightarrow 4xyi = 0$

$\Rightarrow xy = 0$

$\Rightarrow x = 0 \Rightarrow y = \pm 1 \quad (\because x^2 + y^2 = 1)$

and

$y = 0 \Rightarrow x = \pm 1$

$\Rightarrow$ possible z are 0 + 0i, i, −i, 1 & −1.

$\Rightarrow$ There are 5 possible complex numbers of z.

Q. **If $iz^3 + z^2 - z + i = 0$, then show that $|z| = 1$**

Sol. $iz^3 + z^2 - z + i = 0$

$\Rightarrow iz^3 + z^2 + i^2z + i = 0$

$\Rightarrow z^2(iz + 1) + i(iz + 1) = 0$

$\Rightarrow (z^2 + i)(iz + 1) = 0$

$\Rightarrow z^2 = -i$ or $z = i$

Taking Modulus both side,

$|z|^2 = |-i|$ or $|z| = |i|$

$\Rightarrow |z| = 1$ in both cases. **Hence Proved.**

Previous Year's Question

A Let $\left(-2 - \frac{1}{3}i\right)^3 = \frac{x + iy}{27}\left(i = \sqrt{-1}\right)$, where x and y are real numbers, then $y - x$ equals

(JEE Main 2019)

(A) 91 (B) −85 (C) 85 (D) −91

Q. **The minimum value of $|1+z|+|1-z|$ where z is a complex number is**

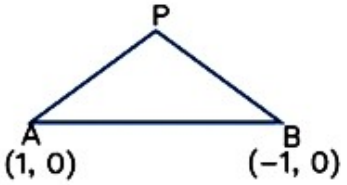
(A)2 **(B)** $\frac{3}{2}$ **(C) 1** **(D) 0**

Sol. $|1+z| \rightarrow$ distance between z and (–1, 0).

$|1-z| \rightarrow$ distance between z and (1, 0).

Let A be (1, 0), B be (–1, 0) and P be z.

We have to find $(PA+PB)_{min}$.



From figure, we know $PA+PB \geq AB$ (Equality when P lies on AB).

$\Rightarrow (PA+PB)_{min} = 2$

Q. **The system of equations** $\left.\begin{matrix}|z+1-i|=2 \\ \text{Re}\, z \geq 1\end{matrix}\right\}$ **where z is a complex number has**

(A) no solution **(B) exactly one solution**

(C) two distinct solution **(D) infinite solution**

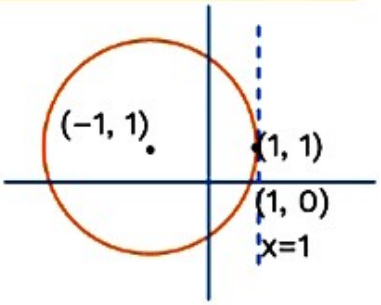
Sol. $|z+1-i|=2 \Rightarrow |z-(-1+i)|=2$

Z lies on circle centred at (–1, 1) with radii 2.

In above figure, we can see that x = 1 is a tangent to circle.

$\therefore$ only (1, 1) satisfies $\text{re}(z) \geq 1$ on the circle.

$\Rightarrow$ Given system of equations has exactly one solution.



Square Root of Complex Number:

Let z = x + iy be the given complex number and we have to obtain its square root. (x and y are real numbers and xy > 0)

Let $a+ib=(x+iy)^{1/2} \Rightarrow a^2-b^2+2iab=x+iy$

$\Rightarrow x=a^2-b^2$ and $y=2ab$

$\Rightarrow x^2=(a^2+b^2)^2-4a^2b^2 \Rightarrow x^2+y^2=(a^2+b^2)^2$

$\Rightarrow a^2+b^2=|z|$ ...(i) $\Rightarrow a^2-b^2=x$ ...(2)

$\Rightarrow a^2=\frac{|z|+x}{2} \Rightarrow a=\pm\sqrt{\frac{|z|+x}{2}}$ $\Rightarrow b^2=\frac{|z|-x}{2} \Rightarrow b=\pm\sqrt{\frac{|z|-x}{2}}$

$\therefore \sqrt{x+iy}=a+ib=\pm\left(\sqrt{\frac{|z|+\text{Re}(z)}{2}}+i\sqrt{\frac{|z|-\text{Re}(z)}{2}}\right)$

Replacing i by –i, we get

$$\sqrt{x-iy}=\pm\left(\sqrt{\frac{|z|+Re(z)}{2}}-i\sqrt{\frac{|z|-Re(z)}{2}}\right)$$

Q. **Find the square root of 3 + 4i**

Sol. Let $\sqrt{3+4i}=a+ib \Rightarrow 3+4i=a^2-b^2+2abi \Rightarrow a^2-b^2=3, 2ab=4$

$\therefore \; a^2+b^2=\sqrt{\left(a^2-b^2\right)^2+4a^2b^2}=\sqrt{9+16}=5$

$\therefore \; a+ib=\pm(2+i)$

Alternative method:

Using formula $\sqrt{3+4i}=\pm\left(\sqrt{\frac{5+3}{2}}+i\sqrt{\frac{5-3}{2}}\right)=\pm(2+i)$

Q. **Find the root of z = 9 – 40i**

Sol. $\Rightarrow |z|=\sqrt{81+1600}=41$

$$\Rightarrow \sqrt{z}=\pm\left(\sqrt{\frac{41+9}{2}}-i\sqrt{\frac{41-9}{2}}\right)$$

$$=\pm(5-4i)$$

$\Rightarrow \sqrt{z}=(5-4i)$ or $(-5+4i)$

Q. **If $z^2+2(1+2i)z-(11+2i)=0$, find z in the form of a + ib.**

Sol. $z^2+2(1+2i)z-(11+2i)=0$

$$\Rightarrow z=\frac{-2(1+2i)\pm\sqrt{4(1+2i)^2+4(11+2i)}}{2}$$

$\Rightarrow z=-1-2i\pm\sqrt{8+6i}$

$$\sqrt{8+6i}=\pm\left(\sqrt{\frac{10+8}{2}}+i\sqrt{\frac{10-8}{2}}\right)=\pm(3+i)$$

$\Rightarrow z=-1-2i\pm(3+i)$

$\Rightarrow z=(2-i)$ or $(-4-3i)$

Q. If $z = (x, y) \in C$ then find z, satisfying $z^2(1,1) = (-1,7)$

Sol. $z^2(1,1) = (-1,7)$

$$\Rightarrow (x+yi)^2 = \frac{-1+7i}{1+i}$$

$$= \frac{(-1+7i)(1-i)}{2}$$

$$= \frac{6+8i}{2}$$

$$\Rightarrow z^2 = 3+4i$$

$$\Rightarrow |3+4i| = 5$$

$$\Rightarrow z = \pm\left(\sqrt{\frac{5+3}{2}} + i\sqrt{\frac{5-3}{2}}\right)$$

$$\Rightarrow z = \pm(2+i)$$

$$\Rightarrow z = (2+i) \text{ or } (-2-i).$$

Q. If $z = \dfrac{\sqrt{9+40i} + \sqrt{9-40i}}{\sqrt{9+40i} - \sqrt{9-40i}}$, find $|z|$

Sol. $$z = \frac{\sqrt{9+40i} + \sqrt{9-40i}}{\sqrt{9+40i} - \sqrt{9-40i}} \times \frac{\left(\sqrt{9+40i} + \sqrt{9-40i}\right)}{\left(\sqrt{9+40i} + \sqrt{9-40i}\right)}$$

$$\Rightarrow z = \frac{9+40i+9-40i+2\sqrt{9+40i}\sqrt{9-40i}}{9+40i-(9-40i)}$$

$$= \frac{18+2\sqrt{81+1600}}{80i}$$

$$= \frac{18+82}{80i} = \frac{5}{4i}$$

$$\Rightarrow z = \frac{5}{4i} \times \frac{i}{i}$$

$$\Rightarrow z = \frac{-5i}{4}$$

$$\Rightarrow |z| = \frac{5}{4}$$

Previous Year's Question

A If $a > 0$ and $z = \frac{(1+i)^2}{a-i}$, has magnitude $\sqrt{\frac{2}{5}}$, then $\bar{z}$ is equal to: **(JEE Main 2019)**

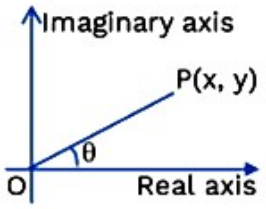
(A) $-\frac{3}{5}-\frac{1}{5}i$ (B) $-\frac{1}{5}+\frac{3}{5}i$ (C) $-\frac{1}{5}-\frac{3}{5}i$ (D) $\frac{1}{5}-\frac{3}{5}i$

Previous Year's Question

A **If** $\left(\frac{1+i}{1-i}\right)^{m/2} = \left(\frac{1+i}{i-1}\right)^{n/3} = 1 (m, n \in N)$ then the greatest common divisor of the least values of m and n is ... **[JEE (Main) 2020]**

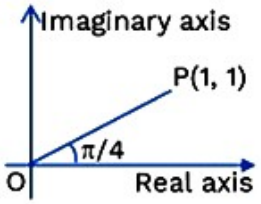
Argument of Complex Number:

Angle (θ) made by the line segment, joining the point on the complex plane representing the complex number z to the origin, from the positive real axis is called argument of complex number z which is denoted as arg (z)= θ



(i) If OP makes an angle θ z with real axis then θ is called one of the argument of z. General values of argument of z are given by $2n\pi + \theta, n \in I$ Note that any two arguments of the same complex number z differ by 2π.

e.g. If z = 1 + i then arg(z) = $\frac{\pi}{4}$



General value of argument of $z = 2n\pi + \frac{\pi}{4}, n \in I$

Note that by specifying the modulus and argument, a complex number is completely defined. However, **for the complex number 0+0i the argument is not defined** and this is the only complex number which is completely defined by talking in terms of its modulus. i.e., |z| = 0

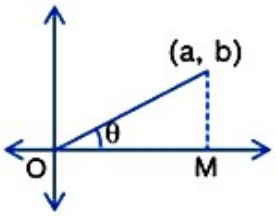
(ii) Amplitude (Principal value of argument):

The unique value of θ such that $-\pi < \theta \leq \pi$ is called principal value of argument. Unless otherwise stated, z refers to the principal value of argument.

Working Rule for finding principal argument of Complex number Z :

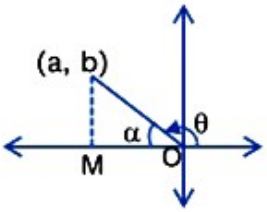
Let $z = a + ib$

First compute $\alpha = \tan^{-1}\left(\frac{|b|}{|a|}\right)$



Case I:

If z lies in I quadrant i.e. a, b>0, then amp(z)= $\theta = \alpha$.

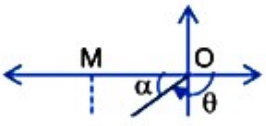


Case II:

If z lies in II quadrant i.e. a<0, b>0, then amp(z) = $\theta = (\pi - \alpha)$

Case III:

If z lies in III quadrant i.e. a < 0, b < 0, then amp(z) = $\theta = -(\pi - \alpha)$



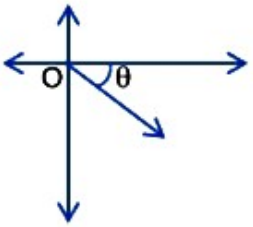
Case IV:

If z lies in IV quadrant i.e. a>0, b<0, then amp(z) = $\theta = -\alpha$

Note:

(i) If z is purely real positive complex number, then amp(z) = 0

(ii) If z is purely imaginary positive complex number, then amp(z) = $\frac{\pi}{2}$



(iii) If z is purely real negative complex number, then amp(z) = π

(iv) If z is purely imaginary negative complex number then amp(z) = $\frac{-\pi}{2}$

Q. **Find the amplitude of**

(a) $-1-i\sqrt{3}$ **(b)** $\dfrac{1+\sqrt{3}i}{\sqrt{3}+i}$ **(c)** $\sin\alpha+i(1-\cos\alpha), 0<\alpha<\pi$

Sol. **(a)** Let, $z=-1-i\sqrt{3}$. The $\alpha=\tan^{-1}\left|\dfrac{b}{a}\right|=\tan^{-1}\left|\dfrac{\sqrt{3}}{1}\right|=\dfrac{\pi}{3}$

Clearly, z is in third quadrant.

Therefore, argument is $\theta=-(\pi-\alpha)=-\left(\pi-\dfrac{\pi}{3}\right)=-\dfrac{2\pi}{3}$

(b) $\text{amp}\left(\dfrac{1+\sqrt{3}i}{\sqrt{3}+i}\right)=\text{amp}\left(\dfrac{1+\sqrt{3}i}{\sqrt{3}+i}\times\dfrac{\sqrt{3}-i}{\sqrt{3}-i}\right)=\text{amp}\left(\dfrac{2\sqrt{3}+2i}{4}\right)=\text{amp}\left(\dfrac{\sqrt{3}+i}{2}\right)$

Complex number $\dfrac{\sqrt{3}+i}{2}$ lies in 1st quadrant

$\therefore\ \text{amp}(z)=\theta=\alpha=\dfrac{\pi}{6}$

(c) $z=\sin\alpha+i(1-\cos\alpha), 0<\alpha<\pi$

$z=\sin\alpha+i(1-\cos\alpha)$

$\Rightarrow\ \text{amp}(z)=\tan^{-1}\left(\dfrac{1-\cos\alpha}{\sin\alpha}\right)=\tan^{-1}\left(\dfrac{2\sin^2\frac{\alpha}{2}}{2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}\right)$

$=\tan^{-1}\tan\left(\dfrac{\alpha}{2}\right)=\dfrac{\alpha}{2}\qquad\left\{0<\dfrac{\alpha}{2}<\dfrac{\pi}{2}\right\}$

(iii) **Least positive argument:**

The value of θ such that $0<\theta\le 2\pi$ is called the least positive argument.

Q. **Find general argument, principal argument and least positive argument of the following complex numbers**

(1) $z_1=\sqrt{3}-2i$ **(2)** $z_2=-1+I$ **(3)** $z_3=-2-3i$

(4) $z_4=(\sqrt{7}-2)i$ **(5)** $z_5=2-\sqrt{7}$ **(6)** z_6 ℵ e

Sol.

S.No	Complex No.	General Argument	Principal Argument	Least Positive Argument
1.	$\sqrt{3}-2i$	$2n\pi-\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$	$-\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$	$2\pi-\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$
2.	$-1+i$	$2n\pi+\frac{3\pi}{4}$	$\frac{3\pi}{4}$	$\frac{3\pi}{4}$
3.	$-2-3i$	$2n\pi-\pi+\tan^{-1}\left(\frac{3}{2}\right)$	$-\left(\pi-\tan^{-1}\left(\frac{3}{2}\right)\right)$	$\pi+\tan^{-1}\left(\frac{3}{2}\right)$
4.	$(\sqrt{7}-2)i$	$2n\pi+\frac{\pi}{2}$	$\frac{\pi}{2}$	$\frac{\pi}{2}$
5.	$(2-\sqrt{7})$	$2n\pi+\pi$	π	π
6.	$(\pi-e)$	$2n\pi$	0	0

Q. **If $|z-i|=1$ and $\text{Arg}(z)=\frac{\pi}{2}$, find z.**

Sol. $\text{Arg}(z)=\frac{\pi}{2}$

$\Rightarrow$ z lies on positive y-axis

$\Rightarrow z = ai$

$\Rightarrow |ai-i|=1$

$\Rightarrow |a-1|=1$

$\Rightarrow a = 0$ or 2

But $a \neq 0$ as $a = 0 \Rightarrow z = 0 \Rightarrow \text{Arg}(z) \neq \frac{\pi}{2}$

$\Rightarrow a = 2 \Rightarrow z = 2i$.

Q. **If $\text{Arg}(z)=\frac{\pi}{4}$ and $|z+3-i|=4$, find z.**

Sol. If $z = x + yi, \frac{y}{x} = \tan(\text{Arg}(z))$

$\Rightarrow \frac{y}{x}=1 \Rightarrow y = x, x > 0$

$\Rightarrow z = x(1+i)$

$|z+3-i|=4$

$\Rightarrow |(x+3)+(x-1)i|=4$

$\Rightarrow x^2+6x+9+x^2-2x+1=16$

$\Rightarrow x^2 + 2x - 3 = 0$

$\Rightarrow x = 1 \text{ or } -3$

But $x > 0$

$\Rightarrow x = 1$

$\Rightarrow z = 1 + i$

Q. **If z_1 & $\bar{z}_1$ represents adjacent vertices of a regular polygon of n sides with center at origin & if $\frac{Im(z_1)}{Re(z_1)} = \sqrt{2} - 1$ then the value of n is equal to:**

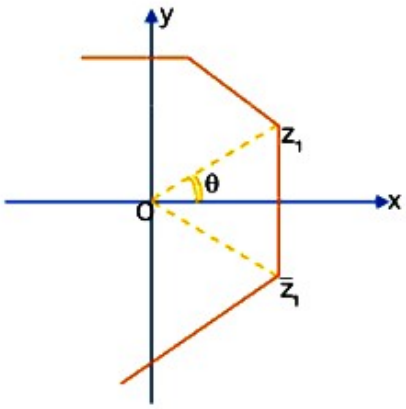
(A) 8 **(B) 12** **(C) 16** **(D) 24**

Sol. Let z_1 be $x + iy$

$\Rightarrow \tan\theta = \frac{y}{x} = \sqrt{2} - 1$

$\Rightarrow \theta = \frac{\pi}{8}$

$\Rightarrow 2\theta = \frac{\pi}{4} = \angle z_1 o \bar{z}_1$



If polygon is n – sided, then

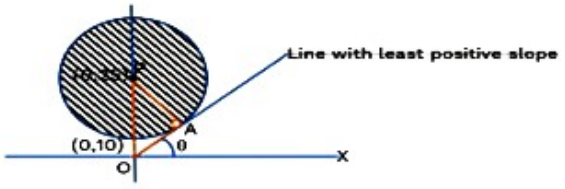
$\angle z_1 o \bar{z}_1 = \frac{2\pi}{n}$

$\Rightarrow \frac{2\pi}{n} = \frac{\pi}{4}$

$\Rightarrow n = 8$

Q. **Among the complex number z which satisfy $|z - 25i| \le 15$, find one having the least positive argument.**

Sol. $|z - 25i| \le 15$ represents a region within the circle centered at (0, 25) with radii 15 including the boundary.



From figure, it is clear that tangent OA has the least positive slope, thus A has least positive Argument.

$OA = \sqrt{OP^2 - AP^2}$

$OA = \sqrt{25^2 - 15^2}$

$\Rightarrow OA = 20$

Also, $\angle OPA = \theta$

$\Rightarrow \sin\theta = \frac{20}{25} = \frac{4}{5}$

$\Rightarrow \cos\theta = \frac{15}{25} = \frac{3}{5}$

$\Rightarrow A \equiv (20\cos\theta, 20\sin\theta)$

$\Rightarrow A \equiv (12, 16)$

$\Rightarrow A \equiv 12 + 16i$ (z with least positive argument)

Q. **Find area enclosed by $|z - 1 + i| \le 3;\ -\frac{\pi}{3} \le \arg(z + i - 1) \le \frac{\pi}{2}$**

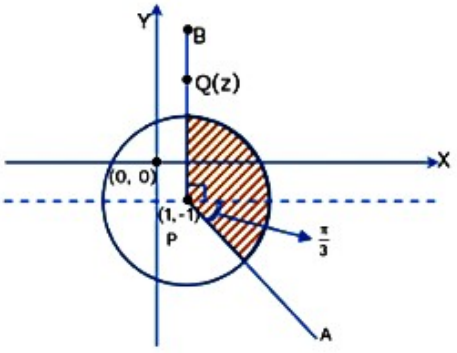
Sol. $|z - 1 + i| = |z - (1 - i)| \le 3$

It represents region inside circle with center (1, –1) and radius 3.

Let Q be z

Then AP and PQ makes angle between

$\frac{'-\pi'}{3}$ and $\frac{\pi}{2}$ with positive x-axis.



$\Rightarrow \angle APB = \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6}$.

$\Rightarrow$ Area of sector APB is required area.

$\Rightarrow$ Required area $= \pi r^2 \times \left(\frac{\frac{5\pi}{6}}{2\pi}\right) = \pi \times 3^2 \times \frac{5}{12}$

$= \frac{15\pi}{4}$

REPRESENTATION OF A COMPLEX NUMBER IN DIFFERENT FORMS

Cartesian Form/Algebraic Form:

Every complex number expressed in the form of $z = x + iy$ where $x, y \in R$ and $i = \sqrt{-1}$ is called cartesian form or algebraic form of complex number.

for $z = x + iy$, $Re(z) = x$ and $Im(z) = y$

$|z| = \sqrt{x^2 + y^2}$, $\bar{z} = x - iy$

Q. **If $Re\left(\frac{1}{z}\right) > \frac{1}{2}$ then find the locus of z.**

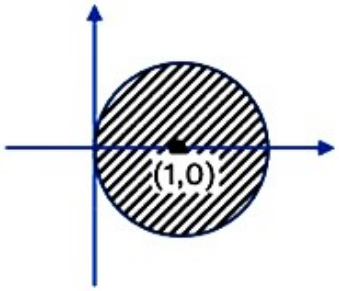
Sol. Let $z = x + iy$

$$\frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} - \frac{iy}{x^2+y^2}$$

$$\because \quad Re\left(\frac{1}{z}\right) > \frac{1}{2} \quad \Rightarrow \quad \frac{x}{x^2+y^2} > \frac{1}{2}$$

$$\Rightarrow \quad 2x > x^2 + y^2 \Rightarrow x^2 + y^2 - 2x < 0$$

$$\Rightarrow \quad (x-1)^2 + y^2 < 1.$$



Hence, locus of z represents interior region of the circle whose center is (1,0) and radius is 1 unit.

Q. **Solve the equation $z^2 + |z| = 0$ where $z \in C$.**

Sol. Let $z = x + iy$

$$z^2 + |z| = 0$$

$$x^2 - y^2 + 2xyi + \sqrt{x^2 + y^2} = 0$$

Comparing real and imaginary parts

$$\Rightarrow \quad x^2 - y^2 + \sqrt{x^2 + y^2} = 0 \text{ and } 2xy = 0 \Rightarrow x = 0 \text{ or } y = 0$$

Case-I: when $x = 0$

$$\Rightarrow \quad -y^2 + \sqrt{y^2} = 0 \quad \Rightarrow \quad y^2 = \sqrt{y^2} \quad \Rightarrow \quad y^2 = \pm y$$

$$\Rightarrow \quad y^2 \pm y = 0 \quad \Rightarrow \quad y(y \pm 1) = 0$$

$y = 0, y = \pm 1$

$\therefore$ Solutions are (0,0),(0,1) and (0,-1)

Case-II: When $y = 0$

$$x^2 + \sqrt{x^2} = 0 \Rightarrow x^2 + |x| = 0 \Rightarrow x = 0$$

Solution is (0, 0) already determined
Hence, solutions of the equation are (0, 0), (0, 1), (0, –1).

Q. **If z is a complex number satisfying the equation** $|z-(1+i)|^2=2$ and $\omega=\frac{2}{z}$, **then the locus traced by 'ω' in the complex plane is**

(A) x – y – 1 =0 **(B) x + y – 1 = 0** **(C) x – y + 1 =0** **(D) x + y + 1 = 0**

Sol. **(A)**

We have $|z-(1+i)|^2=2$

$\Rightarrow (x-1)^2+(y-1)^2=2$ (Put $z=x+iy$)

$\Rightarrow x^2+y^2=2(x+y)$ ...(1)

Let $\omega=h+ik=\frac{2}{z}=\frac{2}{x+iy}=\frac{2(x-iy)}{x^2+y^2}$, so

$h=\frac{2x}{x^2+y^2}, k=\frac{-2y}{x^2+y^2}$

$\Rightarrow h-k=\frac{2(x+y)}{x^2+y^2}=1$ (from equation (I))

$\therefore$ Locus of the point $\omega(h,k)$ will be x–y = 1

Q. **Find locus of point in complex plane satisfying** $\mathbf{Re\left(\frac{1}{z}\right)=\frac{1}{4}}$.

Sol. Let $z=x+iy, x,y\in R$

$\frac{1}{z}=\frac{1}{x+iy}=\frac{x-iy}{(x+iy)(x-iy)}$

$=\frac{x-iy}{x^2+y^2}$

$=\left(\frac{x}{x^2+y^2}\right)-i\left(\frac{y}{x^2+y^2}\right)$

$Re\left(\frac{1}{z}\right)=\frac{x}{x^2+y^2}=\frac{1}{4}$

$\Rightarrow x^2+y^2-4x=0$

$\Rightarrow (x-2)^2+y^2=4$

$\Rightarrow$ Locus of point is a circle with center (2, 0) and radius = 2.

Q. **Show that locus of a point P(w) denoting the complex number $z+\frac{1}{z}$ on the complex plane is a standard ellipse where $|z|=2$.**

Sol. Let $z = x + iy \Rightarrow x^2 + y^2 = 4 = |z|^2$.

$$z+\frac{1}{z} = z+\frac{\bar{z}}{|z|^2}$$

$$= x + iy + \frac{x-iy}{4}$$

$$\Rightarrow z+\frac{1}{z} = \frac{5x}{4}+\frac{3iy}{4} = a+bi$$

$$a = \frac{5x}{4}, b = +\frac{3y}{4}$$

$$\Rightarrow x = \frac{4a}{5}, y = \frac{4b}{3}$$

$$x^2 + y^2 = 4$$

$$\Rightarrow \frac{16a^2}{25}+\frac{16b^2}{9} = 4$$

$$\Rightarrow \frac{a^2}{\left(\frac{25}{4}\right)}+\frac{b^2}{\left(\frac{9}{4}\right)} = 1$$

Which represents the equation of an ellipse.

Q. **Find z which satisfy simultaneously $\left|\frac{z-12}{z-8i}\right| = \frac{5}{3}$ and $\left|\frac{z-4}{z-8}\right| = 1$.**

Sol. $\left|\frac{z-4}{z-8}\right| = 1$

$\Rightarrow |z-4| = |z-8|$

Z is equidistant from (4, 0) and (8, 0)

$\Rightarrow$z lies on perpendicular bisector of line joining (4, 0) and (8, 0), ie, x = 6

$\Rightarrow z = 6 + yi$

Now, $\left|\frac{z-12}{z-8i}\right| = \frac{5}{3}$

$$\Rightarrow \frac{|-6+yi|}{|6+(y-8)i|} = \frac{5}{3}$$

$$\Rightarrow 9\left(36+y^2\right) = 25\left(36+(y-8)^2\right)$$

$\Rightarrow 16y^2 - 400y + 2176 = 0$

$\Rightarrow y^2 - 25y + 136 = 0$

$\Rightarrow (y-17)(y-8) = 0$

$\Rightarrow y = 17 \text{ or } 8$

$\Rightarrow z = 6 + 17i \text{ or } 6 + 8i$

Previous Year's Question

Let z be a complex number such that $\left|\frac{z-i}{z+2i}\right| = 1$ and $|z| = \frac{5}{2}$. Then the value of $|z+3i|$ is:

(JEE Main 2020)

(A) $\sqrt{10}$ (B) $\frac{7}{2}$ (C) $\frac{15}{4}$ (D) $2\sqrt{3}$

Previous Year's Question

All points in the let $S = \left\{\frac{\alpha+i}{\alpha-i} : \alpha \in R\right\} \left(i = \sqrt{-1}\right)$ lie on a **[JEE Main 2019]**

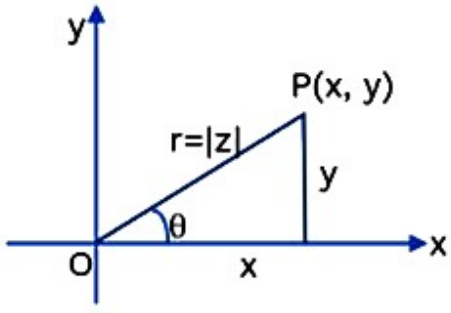
(A) Circle whose radius is 1.
(B) Straight line whose slope is 1.
(C) Straight line whose slope is –1.
(D) circle whose radius is $\sqrt{2}$

Trigonometric Form / Polar Form:

Let the given complex number be $z = x + iy$, r and θ be the modulus and amp (z) respectively. From the figure $x = r\cos\theta$, $y = r\sin\theta$

$\therefore z = x + iy = r(\cos\theta + i\sin\theta) = r(\text{cis}\theta)$

Hence, $z = r(\cos\theta + i\sin\theta)$ is called polar/ trigonometric form of the complex number.



Q. **Express the following complex number in polar form.**

(i) $z_1 = -1 - i\sqrt{3}$ **(ii)** $z_2 = -2 + 3i$

Sol. (i) $z_1 = -1 - i\sqrt{3} = r(\cos\theta + i\sin\theta)$

$$\text{amp}(z_1) = \theta = -\left(\pi - \frac{\pi}{3}\right) = \frac{-2\pi}{3}$$

$$r = \sqrt{1+3} = 2$$

$$\therefore\ z_1 = -1 - i\sqrt{3} = 2\left(\cos\left(\frac{-2\pi}{3}\right) + i\sin\left(\frac{-2\pi}{3}\right)\right) = 2\,\text{cis}\left(\frac{-2\pi}{3}\right).$$

(ii) $z_2 = -2 + 3i$

$$\text{amp}(z_2) = \theta = \pi - \tan^{-1}\frac{3}{2}$$

$$r = \sqrt{4+9} = \sqrt{13}$$

$$\therefore\ z_2 = -2 + 3i = \sqrt{13}\left(\cos\left(\pi - \tan^{-1}\frac{3}{2}\right) + i\sin\left(\pi - \tan^{-1}\frac{3}{2}\right)\right)$$

$$= \sqrt{13}\,\text{cis}\left(\pi - \tan^{-1}\frac{3}{2}\right)$$

Q. **If $z = 1 + \cos\frac{6\pi}{5} + i\sin\frac{6\pi}{5}$ then find modulus and amplitude of z**

Sol. $z = 1 + \cos\frac{6\pi}{5} + i\sin\frac{6\pi}{5}$

$$2\cos^2\frac{3\pi}{5} + i2\sin\frac{3\pi}{5}\cdot\cos\frac{3\pi}{5} \Rightarrow 2\cos\frac{3\pi}{5}\left(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}\right)$$

$$\Rightarrow 2\cos\left(\pi - \frac{2\pi}{5}\right)\left(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}\right) \Rightarrow -2\cos\frac{2\pi}{5}\left(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}\right)$$

$$\Rightarrow 2\cos\frac{2\pi}{5}\left(\cos\left(\pi - \frac{3\pi}{5}\right) - i\sin\left(\pi - \frac{3\pi}{5}\right)\right)$$

$$\Rightarrow 2\cos\frac{2\pi}{5}\left(\cos\frac{2\pi}{5} - i\sin\frac{2\pi}{5}\right) \Rightarrow 2\cos\frac{2\pi}{5}\left[\cos\left(\frac{-2\pi}{5}\right) + i\sin\left(\frac{-2\pi}{5}\right)\right]$$

$$\therefore\ |z| = 2\cos\frac{2\pi}{5} \text{ and } \text{amp}(z) = \frac{-2\pi}{5}.$$

Previous Year's Question

Let $\mathbf{A}=\left\{\theta\left(-\frac{\pi}{2},\pi\right):\frac{\mathbf{3+2i\sin\theta}}{\mathbf{1-2i\sin\theta}}\ \textbf{is purely imaginary}\right\}$. Then the sum of the elements in A is

(JEE Main 2019)

(A) $\frac{5\pi}{6}$ (B) π (C) $\frac{3\pi}{4}$ (D) $\frac{2\pi}{3}$

Q. **Describe the path of (z).**

(i) $\mathbf{Arg(z)=\frac{\pi}{4}}$. **(ii)** $\mathbf{Arg(z-1)=\frac{3\pi}{4}}$ **(iii)** $\mathbf{0\le Arg(z)\le\frac{\pi}{4}}$

Sol. **(i)** Let $z=r\cos\left(\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)=x+yi$

$$=\frac{r}{\sqrt{2}}(1+i)=x+yi$$

$\Rightarrow$ Locus, $x=y,\ x,y>0$

(ii) Let $z=x+yi$

$$z-1=(x-1)+yi$$

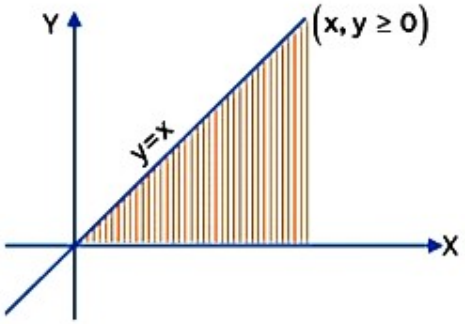
$$Arg(z-1)=\frac{3\pi}{4}$$

$\Rightarrow \tan\frac{3\pi}{4}=\frac{y}{x-1}, y>0, x-1<0$

$\Rightarrow -1=\frac{y}{x-1}$

$\Rightarrow y+x-1=0$ (Locus of z)

Where $y>0$ and $x<1$



(iii) Let $z=x+yi$

$$Arg\,z=\tan^{-1}\frac{y}{x}\in\left[0,\frac{\pi}{4}\right]$$

$\Rightarrow \frac{y}{x}\in[0,1]$ (in 1st quadrant)

$\Rightarrow y\in[0,x]$ $(x,y\ge 0)$

Z is set of all point lying between x-axis

and line $y = x$ in 1st quadrant including the boundaries.

Exponential Form:

$z = x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}$

$z = re^{i\theta}$ is called exponential form of the complex number.

where r is modulus of z and θ is amplitude of z.

Here $\cos\theta + i\sin\theta = e^{i\theta}$ ...(1)

Replacing i by –i, we get

$\cos\theta - i\sin\theta = e^{-i\theta}$ ...(2)

Adding (1) and (2)

$\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$ which is purely real

Subtracting (2) from (1)

$i\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2}$ which is purely imaginary.

Q. **If $z = -2e^{i\left(\frac{-\pi}{3}\right)}$ then find modulus and amplitude of z.**

Sol. $z = -2\left(\cos\left(\dfrac{-\pi}{3}\right) + i\sin\left(\dfrac{-\pi}{3}\right)\right)$

$= -2\left(\cos\dfrac{\pi}{3} - i\sin\dfrac{\pi}{3}\right) = -2\left(\dfrac{1}{2} - i\dfrac{\sqrt{3}}{2}\right) = -1 + i\sqrt{3}$ which lies in 2nd quadrant.

Modulus $r = \sqrt{1+3} = 2$

$\text{amp}(z) = \theta = \pi - \tan^{-1}(\sqrt{3}) = \pi - \dfrac{\pi}{3} = \dfrac{2\pi}{3}$

Q. **Find the real part of $z = e^{e^{i\theta}}, \theta \in R$**

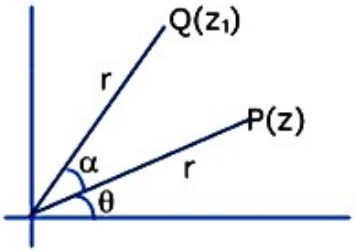
Sol. $z = e^{e^{i\theta}} = e^{(\cos\theta + i\sin\theta)} = e^{\cos\theta} \cdot e^{i\sin\theta} = e^{\cos\theta} \cdot (\cos(\sin\theta) + i\sin(\sin\theta))$

$\therefore \quad \text{Re}(z) = e^{\cos\theta} \cdot \cos(\sin\theta)$

Vectorial Representation of a Complex Number:

Every complex number can be considered as if it is the position vector of that point. If the point P represents the complex number z then, $\overrightarrow{OP} = z$ & $|\overrightarrow{OP}| = |z|$.

Geometrical meaning of $e^{i\alpha}$.

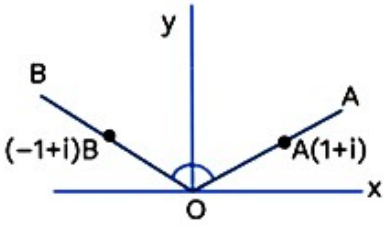


(i) If $\overrightarrow{OP} = z = re^{i\theta}$ then $\overrightarrow{OQ} = z_1 = re^{i(\theta+\alpha)} = z \cdot e^{i\alpha}$

If $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are of unequal magnitude then $\overrightarrow{OQ} = \frac{OQ}{OP} ze^{i\alpha}$

e.g., If $z = \overrightarrow{OA} = 1 + i$ and $\alpha = \frac{\pi}{2}$ then

$z_1 = \overrightarrow{OB} = i(1 + i) = -1 + i$



(ii) Using the vectorial concept and section formula, complex numbers corresponding to centroid, incentre, orthocentre and circumcentre for a triangle whose vertices are z_1, z_2, z_3 can be deduced.

PROPERTIES OF CONJUGATE, MODULUS AND ARGUMENT

Properties of conjugate of complex Numbers:

(i) $\overline{(\bar{z})} = z$

(ii) $|z| = |\bar{z}|$

(iii) $z + \bar{z} = 2\,Re(z)$

(iv) $z - \bar{z} = 2i\,Im(z)$

(v) If z is purely real, $z = \bar{z}$

(vi) If z is purely imaginary, $z = -\bar{z}$

(vii) $z\bar{z} = |z|^2 = |\bar{z}|^2$

(viii) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$. In general,

$\overline{z_1 + z_2 + \ldots\ldots + z_n} = \bar{z}_1 + \bar{z}_2 + \ldots\ldots + \bar{z}_n$

(ix) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$

(x) $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$

(xi) $\bar{z}^n = (\bar{z})^n$

(xii) $\overline{\left(\frac{z_1}{z_2}\right)} = \left(\frac{\bar{z}_1}{\bar{z}_2}\right)$

(xiii) If $\alpha = f(z)$, then $\bar{\alpha} = \overline{f(z)} = f(\bar{z})$ where

$\alpha = f(z)$ is a function in complex variable with real coefficients. In other words, if $f(x + iy) = a + ib$ then $f(x - iy) = a - ib$

Explanation:

Let $f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^2 + \ldots\ldots + a_n z,$

where $a_0, a_1, a_2, \ldots\ldots, a_n$ are real numbers and z is a complex number. Then

$$f(\bar{z}) = a_0 + a_1\bar{z} + a_2(\bar{z})^2 + a_3(\bar{z})^3 + \ldots\ldots + a_n(\bar{z})^n$$

$$= \overline{a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \ldots\ldots + a_n z^n}$$

Q. **Let z be a complex number such that $z \in C / R$ and $\frac{1+z+z^2}{1-z+z^2} \in R$ then prove that $|z| = 1$.**

Sol. $w = \frac{1+z+z^2}{1-z+z^2} \in R$

We know $z - \bar{z} = 0$ if $z \in R$

$\Rightarrow w - \bar{w} = 0 \Rightarrow w = \bar{w}$

$$\Rightarrow \frac{1+z+z^2}{1-z+z^2} = \frac{\left(\overline{1+z+z^2}\right)}{\left(\overline{1-z+z^2}\right)}$$

$$\Rightarrow \left(1+z+z^2\right)\left(1-\bar{z}+\bar{z}^2\right) = \left(1-z+z^2\right)\left(1+\bar{z}+\bar{z}^2\right)$$

$$\Rightarrow 1 - \bar{z} + \bar{z}^2 + z - |z|^2 + z\bar{z}^2 + z^2 - z^2\bar{z} + |z|^4$$

$$= 1 + \bar{z} + \bar{z}^2 - z - |z|^2 - z\bar{z}^2 + z^2 + z^2\bar{z} + |z|^4$$

$$\Rightarrow 2(z - \bar{z}) + 2z\bar{z}^2 - 2z^2\bar{z} = 0$$

$$\Rightarrow (z - \bar{z}) + |z|^2(\bar{z} - z) = 0$$

$$\Rightarrow (z - \bar{z})\left(1 - |z|^2\right) = 0$$

$\Rightarrow |z|^2 = 1$ (z is not real)

$\Rightarrow |z| = 1$

Q. **For any two complex numbers z_1, z_2 prove that:**

$$\left|\frac{z_1 - z_2}{1 - \bar{z}_1 z_2}\right| < 1 \text{ if } |z_1| < 1 \text{ and } |z_2| < 1 \text{ or } |z_1| > 1 \text{ and } |z_2| > 1$$

Sol. $$\left|\frac{z_1 - z_2}{1 - \bar{z}_1 z_2}\right| < 1$$

$$\Rightarrow \left|\frac{z_1 - z_2}{1 - \bar{z}_1 z_2}\right|^2 < 1$$

$$\Rightarrow \left(\frac{z_1 - z_2}{1 - \bar{z}_1 z_2}\right)\left(\frac{\bar{z}_1 - \bar{z}_2}{1 - z_1 \bar{z}_2}\right) < 1$$

$$\Rightarrow |z_1|^2 + |z_2|^2 - z_1\bar{z}_2 - \bar{z}_1 z_2 < 1 + |z_1|^2 |z_2|^2 - \bar{z}_1 z_2 - z_1 \bar{z}_2$$

$$\Rightarrow |z_1|^2 + |z_2|^2 < 1 + |z_1|^2 |z_2|^2$$

$$\Rightarrow 1 - |z_1|^2 - |z_2|^2 + |z_1|^2 |z_2|^2 > 0$$

$$\left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right) > 0$$

$\Rightarrow$ both $|z_1|$ and $|z_2| < 1$

Or both $|z_1|$ and $|z_2| > 1$

Q. **For any complex number z, prove that :** $\left|\frac{6z - i}{2 + 3iz}\right| \le 1$ **if** $|z| \le \frac{1}{3}$.

Sol. $$\left|\frac{6z - i}{2 + 3iz}\right| \le 1$$

$$\Rightarrow |6z - i|^2 \le |2 + 3iz|^2$$

$$\Rightarrow (6z - i)(6\bar{z} + i) \le (2 + 3iz)(2 - 3i\bar{z})$$

$$\Rightarrow 36|z|^2 + 1 + 6zi - 6\bar{z}i \le 4 + 9|z|^2 + 6iz - 6\bar{z}i$$

$$\Rightarrow 27|z|^2 \le 3$$

$$\Rightarrow |z|^2 \le \frac{1}{9}$$

$$\Rightarrow |z| \le \frac{1}{3}$$

Hence proved.

Q. **Number of complex numbers z such that** $|z| = 1$ **and** $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ **is**

(A) 4 **(B) 6** **(C) 8** **(D) more than 8**

Sol. $|z| = 1 \Rightarrow z\bar{z} = 1 \Rightarrow \bar{z} = \frac{1}{z}$ or $z = \frac{1}{\bar{z}}$

$$\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$$

$$\Rightarrow \left|z^2 + \bar{z}^2\right| = 1$$

Let $z = x + yi$

$$z^2 + \bar{z}^2 = \left(x^2 - y^2 + 2xyi\right) + \left(x^2 - y^2 - 2xyi\right)$$

$$= 2\left(x^2 - y^2\right).$$

$$\Rightarrow \left|x^2 - y^2\right| = \frac{1}{2}$$

Also $x^2 + y^2 = 1 \qquad (\because |z| = 1)$

$$\Rightarrow x^2 = \frac{3}{4} \text{ and } y^2 = \frac{1}{4} \text{ or } x^2 = \frac{1}{4} \text{ and } y^2 = \frac{3}{4}$$

$$\Rightarrow x = \pm\frac{\sqrt{3}}{2} \text{ and } y = \pm\frac{1}{2} \text{ or } x = \pm\frac{1}{2} \text{ and } y = \pm\frac{\sqrt{3}}{2}$$

$$\Rightarrow z = \pm\frac{\sqrt{3}}{2} \pm \left(\frac{1}{2}\right)i \text{ or } z = \pm\left(\frac{1}{2}\right) \pm \left(\frac{\sqrt{3}}{2}\right)i$$

$\Rightarrow$ There are 8 possible complex number z.

Properties of modulus of complex numbers:

(i) $|z| = 0 \Rightarrow z = 0 + i0$

(ii) $|z| = |-z| = |\bar{z}| = |iz|$

(iii) $-|z| \le \text{Re}(z) \le |z|$ and $-|z| \le \text{Im}(z) \le |z|$

(iv) $z\bar{z} = |z|^2$. If z is unimodular i.e. $|z| = 1$, then $\bar{z} = \frac{1}{z}$

(v) $|z_1 z_2| = |z_1||z_2|$. In general, $|z_1 \cdot z_2 \cdot z_3 \ldots\ldots z_n| = |z_1||z_2||z_3|\ldots\ldots|z_n|$

(vi) $\left|z^n\right| = |z|^n$

(vii) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} (z_2 \neq 0)$

(viii) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left(|z_1|^2 + |z_2|^2\right)$

Proof:

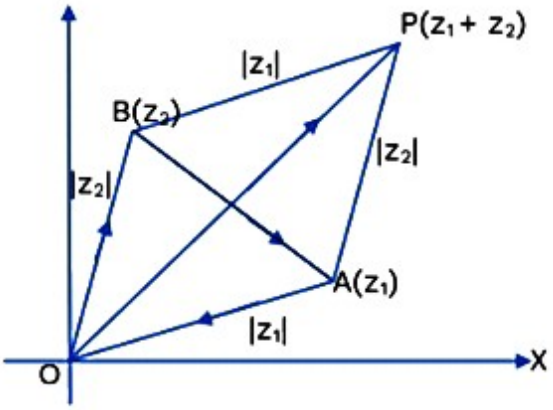
$$|z_1 + z_2|^2 + |z_1 - z_2|^2$$

$$\Rightarrow (z_1 + z_2)\overline{(z_1 + z_2)} + (z_1 - z_2)\left(\overline{z_1 - z_2}\right)$$

$\Rightarrow \quad (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) + (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$

$\Rightarrow \quad z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2 + z_1\bar{z}_1 - z_1\bar{z}_2 - z_2\bar{z}_1 + z_2\bar{z}_2$

$\Rightarrow \quad 2\left(|z_1|^2 + |z_2|^2\right)$



OAPB is a parallelogram.

By the vector law of addition

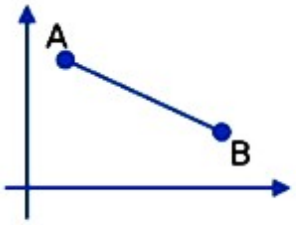
$\overline{OP} = z_1 + z_2 \Rightarrow |\overline{OP}| = |z_1 + z_2|$

$\overline{BA} = z_1 - z_2 \Rightarrow |\overline{BA}| = |z_1 - z_2|$

$\therefore \quad OP^2 + BA^2 = OA^2 + AP^2 + PB^2 + OB^2$

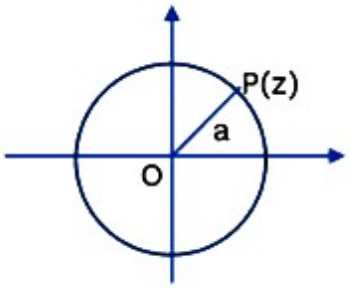
$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left(|z_1|^2 + |z_2|^2\right)$

The above identity indicates the sum of squares of diagonals of a parallelogram is equal to sum of square of its all four sides.



(ix) Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$|z_1 - z_2|$ denotes the distance between two points on the complex plane representing z_1 and z_2.

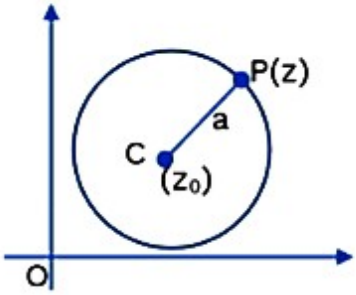


$AB = |z_1 - z_2|$

$= |x_1 + iy_1 - (x_2 + iy_2)|$

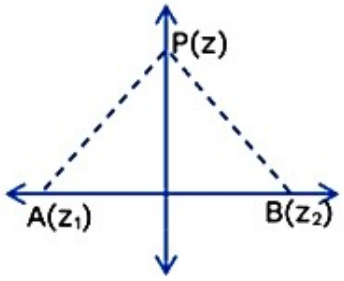
$= |x_1 - x_2 + i(y_1 - y_2)| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

(x) $|z| = a, a \in R^+ \Rightarrow$ locus of z represents a circle whose centre is origin and radius is 'a'.



(xi) $|z - z_0| = a$ where z_0 is a fixed complex number and represents a circle whose centre is z_0 and radius is a.

(xii) If $\left|\frac{z-z_1}{z-z_2}\right| = 1$ where z_1 and z_1 are fixed complex numbers then, locus of z is the perpendicular bisector of line joining the points representing z_1 and z_2.



(xiii) Triangle inequalities

$$||z_1| - |z_2|| \le |z_1 \pm z_2| \le |z_1| + |z_2|$$

Proof:

Method-I Algebraic Method:

Let

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1) \text{ and}$$

$$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

$$z_1 + z_2 = r_1\cos\theta_1 + r_2\cos\theta_2 + i(r_1\sin\theta_1 + r_2\sin\theta_2)$$

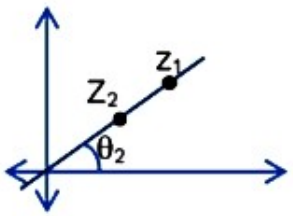
$$|z_1 + z_2| = \sqrt{(r_1\cos\theta_1 + r_2\cos\theta_2)^2 + (r_1\sin\theta_1 + r_2\sin\theta_2)^2}$$

$$= \sqrt{r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1 - \theta_2)}$$

$|z_1 + z_2|$ will be maximum when $\cos(\theta_1 - \theta_2) = 1$

$$\Rightarrow \quad \theta_1 - \theta_2 = 2n\pi$$

$$\Rightarrow \quad \theta_1 = \theta_2 + 2n\pi, n \in I$$

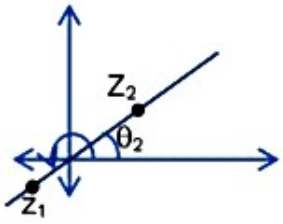


Hence, for the maximum value of $|z_1 + z_2|$ points representing complex numbers z_1, z_2 and the origin are collinear and z_1, z_2 must lie on the same side of the origin.

$|z_1 + z_2|$ will be minimum when $\cos(\theta_1 - \theta_2) = -1$

$$\Rightarrow \quad \theta_1 - \theta_2 = 2n\pi + \pi, n \in I$$

$$\Rightarrow \quad \theta_1 = \theta_2 + 2n\pi + \pi, n \in I$$



Hence, for the minimum value of $|z_1 + z_2|$ points representing the complex number z_1, z_2 and the origin are collinear and z_1, z_2 must lie on the opposite side of the origin.

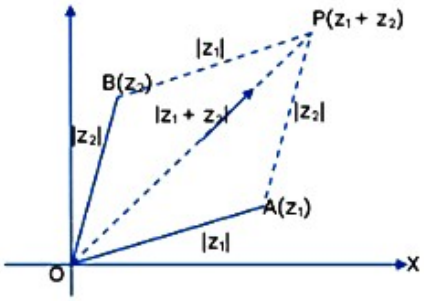
In the similar manner minimum and maximum values of $|z_1 - z_2|$ can also be determined.

Method-II Geometrical Method:

Let A and B represent complex numbers z_1 and z_2 respectively.

A parallelogram OAPB is completed $\overline{OP} = z_1 + z_2$

$\therefore \ |\overline{OP}| = |z_1 + z_2|$



In the ΔOAP, from the fact

(i) Sum of two sides is always greater than third side.

(ii) Absolute value of the difference of two sides is always less than third side.

$\therefore \ ||z_1| - |z_2|| \le |z_1 + z_2| \le |z_1| + |z_2|$

Note:

Sign of equality holds when z_1, z_2 and the origin are collinear.

Q. **If $|z_1 - 1| \le 1, |z_2 - 2| \le 2, |z_3 - 3| \le 3$ then find the greatest value of $|z_1 + z_2 + z_3|$.**

Sol. $|z_1 + z_2 + z_3| = |(z_1 - 1) + (z_2 - 2) + (z_3 - 3) + 6|$

$\le |z_1 - 1| + |z_2 - 2| + |z_3 - 3| + 6 \le 1 + 2 + 3 + 6 = 12$

Q. **If $|z_1| = |z_2| = |z_3| = 1$, prove that $|z_1 + z_2 + z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right|$.**

Sol. We know that $|z| = |\bar{z}|$

$\Rightarrow \ |z_1 + z_2 + z_3| = |\bar{z}_1 + \bar{z}_2 + \bar{z}_3|$

$$= \left|\frac{\bar{z}_1 \cdot z_1}{z_1} + \frac{\bar{z}_2 \cdot z_2}{z_2} + \frac{\bar{z}_3 \cdot z_3}{z_3}\right| = \left|\frac{|z_1|^2}{z_1} + \frac{|z_2|^2}{z_2} + \frac{|z_3|^2}{z_3}\right|$$

$$= \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| \quad \left(\because |z_1|^2 = |z_2|^2 = |z_3|^2 = 1\right)$$

Q. **If $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$**

Then find the value of $|z_1 + z_2 + z_3|$

Sol. $|z_1| = 1 = z_1\bar{z}_1 \qquad |z_2| = 2 \Rightarrow z_2\bar{z}_2 = 4, \qquad |z_3| = 3 \Rightarrow z_3\bar{z}_3 = 9$

Also, $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$

$\Rightarrow |z_1z_2z_3\bar{z}_3 + z_1z_2z_3\bar{z}_2 + z_1\bar{z}_1z_2z_3| = 12 \Rightarrow |z_1z_2z_3||\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 12$

$\Rightarrow |z_1||z_2||z_3||\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 12 \Rightarrow 6|\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 12$

$\Rightarrow \quad |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 2 \quad \Rightarrow \quad |z_1 + z_2 + z_3| = 2.$

Q. **If $|z - (5 + 7i)| = 9$ then find the greatest and least value of $|z-2-3i|$.**

Sol. $|z-2-3i| = |z-(5+7i)+(3+4i)|$

$\Rightarrow ||z - (5 + 7i)| - |3 + 4i|| \le |z - (5 + 7i) + 3 + 4i| \le |z - (5 + 7i)| + |3 + 4i|$

$\Rightarrow |9 - 5| \le |z - 2 - 3i| \le 9 + 5$

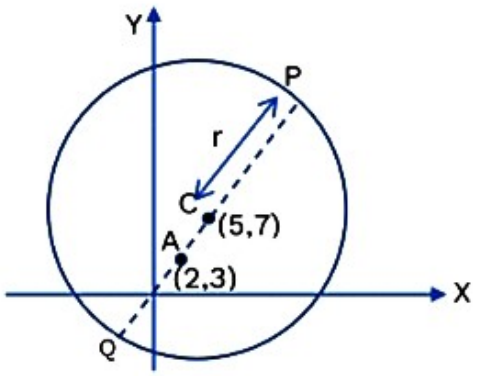
$\Rightarrow 4 \le |z - 2 - 3i| \le 14$

Alternative method:

Z lies on the circle with centre C(5,7) and radius equal to 9. Now, we have to find maximum and minimum distance of the point A(2,3) from the circle.

Maximum distance, AP = A C + C P= 5 + 9 =14

Minimum distance, AQ = CQ − AQ = 9 − 5 = 4



Q. **Find the greatest and least value of $|z|$ if z satisfies $\left|z - \frac{4}{z}\right| = 2$**

Sol. $\left||z| - \frac{4}{|z|}\right| \le \left|z - \frac{4}{z}\right| \le |z| + \frac{4}{|z|}$

$\left||z| - \frac{4}{|z|}\right| \le 2 \le |z| + \frac{4}{|z|}$

Put $|z| = r$, $r > 0$

$\left|r - \frac{4}{r}\right| \le 2 \le r + \frac{4}{r}$

$2 \le r + \frac{4}{r} \quad \Rightarrow \quad r^2 - 2r + 4 \ge 0$ always true $\Rightarrow \quad r > 0$

$\left|r - \frac{4}{r}\right| \le 2$

$$-2 \le r - \frac{4}{r} \le 2$$

$$r^2 + 2r - 4 \ge 0 \Rightarrow (r+1)^2 - 5 \ge 0$$

$$\Rightarrow (r+1-\sqrt{5})\underbrace{(r+1+\sqrt{5})}_{\text{Positive}} \ge 0$$

$$r \ge \sqrt{5} - 1$$

$$r^2 - 2r - 4 \le 0 \Rightarrow (r-1)^2 - 5 \le 0 \quad \Rightarrow \quad (r-1-\sqrt{5})(r-1+\sqrt{5}) \le 0$$

$$\Rightarrow r \in [1-\sqrt{5}, \sqrt{5}+1]$$

$$\therefore \quad r \in [\sqrt{5}-1, \sqrt{5}+1]$$

Q. **Let z_1 & z_2 be complex numbers such that $z_1 \neq z_2$ and $|z_1| = |z_2|$ then show that $\frac{z_1 + z_2}{z_1 - z_2}$ is purely imaginary.**

Sol. $$|z_1| = |z_2|, \text{Re}\left(\frac{z_1+z_2}{z_1-z_2}\right) = \frac{1}{2}\left(\frac{z_1+z_2}{z_1-z_2} + \frac{\bar{z}_1+\bar{z}_2}{\bar{z}_1-\bar{z}_2}\right) = \frac{1}{2}\left(\frac{2|z_1|^2 - 2|z_2|^2}{|z_1-z_2|^2}\right) = 0$$

So, purely imaginary.

Q. **If $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unimodular and z_2 is not unimodular, then find $|z_1|$**

Sol. $$\left|\frac{z_1-2z_2}{2-z_1\bar{z}_2}\right| = 1 \quad \Rightarrow \quad |z_1 - 2z_2|^2 = |2 - z_1\bar{z}_2|^2$$

$$\Rightarrow \quad (z_1 - 2z_2)(\bar{z}_1 - 2\bar{z}_2) = (2 - \bar{z}_1 z_2)(2 - z_1\bar{z}_2)$$

$$\Rightarrow z_1\bar{z}_1 - 2\bar{z}_1 z_2 - 2z_1\bar{z}_2 + 4z_2\bar{z}_2 = 4 - 2\bar{z}_1 z_2 - 2z_1\bar{z}_2 + z_1\bar{z}_1 z_2\bar{z}_2$$

$$\Rightarrow |z_1|^2 + 4|z_2|^2 = 4 + |z_1|^2|z_2|^2 \quad \Rightarrow \quad \left(|z_2|^2 - 1\right)\left(|z_1|^2 - 4\right) = 0$$

$$\Rightarrow \quad |z_1| = 2 \text{ as } |z_2| \neq 1$$

Q. **Let $z_1 z_2 \in C$ be complex numbers such that $|z_1 + z_2| = \sqrt{3}$ and $|z_1| = |z_2| = 1$. Compute $|z_1 - z_2|$.**

Sol. $|z_1 + z_2| = \sqrt{3}$ and $|z_1| = |z_2| = 1$

By squaring

$$(|z_1 + z_2|)^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = 3$$

$$= |z_1|^2 + |z_2|^2 + 2\text{Re}(z_1\bar{z}_2) = 3$$

or $\quad 2\text{Re}\ (z_1\bar{z}_2) = 1$

$$\therefore |z_1 - z_2|^2 = (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$$

$$= |z_1|^2 + |z_2|^2 - 2\text{Re}(z_1\bar{z}_2) = 1 + 1 - 1$$

$$\Rightarrow |z_1 - z_2|^2 = 1$$

$$\therefore |z_1 - z_2| = 1$$

Q. **Let $z_1, z_2, z_3, \ldots z_n$ are the complex numbers such that $|z_1| = |z_2| = \ldots |z_n| = 1$.**

If $z = \left(\sum_{k=1}^{n} z_k\right)\left(\sum_{k=1}^{n} \frac{1}{z_k}\right)$ then prove that :

(i) z is a real number **(ii) $0 < z \le n^2$**

Sol. $|z| = 1 \Rightarrow z = \frac{1}{\bar{z}}$ or $\frac{1}{z} = \bar{z}$

$$\therefore z = \left(\sum_{k=1}^{n} z_k\right)\sum_{k=1}^{n}\left(\frac{1}{z_k}\right)$$

$$= \left(\sum_{k=1}^{n} z_k\right)\left(\sum_{k=1}^{n} \bar{z}_k\right)$$

Let $\sum_{k=1}^{n} z_k = w$

$$\Rightarrow \sum_{k=1}^{n} \bar{z}_k = \bar{w}$$

$$\Rightarrow z = w\bar{w} = |w|^2 = \text{Real number}$$

Now $z = |w|^2$

$$= |z_1 + z_2 + z_3 + z_4 + \ldots z_n|^2$$

Since,

$0 \le |z_1 + z_2 + ... + z_n| \le |z_1| + |z_2| + ... + |z_n|$

$\Rightarrow 0 \le z \le (1+1+1+...1)^2$ n times

$\Rightarrow 0 \le z \le n^2$ **Hence proved.**

Properties of Argument of complex number:

(i) $\text{amp}(z_1 z_2) = \text{amp}(z_1) + \text{amp}(z_2) + 2k\pi, k \in I$

In general

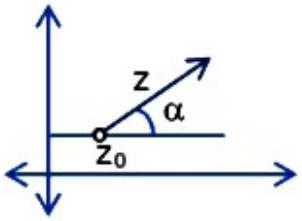
$\text{amp}(z_1 \cdot z_2 ..... z_n) = \text{amp}(z_1) + \text{amp}(z_2) + ... \text{amp}(z_n) + 2k\pi, k \in I$

(ii) $\text{amp}\left(\frac{z_1}{z_2}\right) = \text{amp}(z_1) - \text{amp}(z_2) + 2k\pi, k \in I$

(iii) $\text{amp}(z^n) = n\,\text{amp}(z) + 2k\pi, k \in I$

Note:

In the above properties, $2k\pi, k \in I$ is added in RHS and k is chosen in such a way so that value of the expression in RHS belongs to $(-\pi, \pi]$



(iv) If $\arg(z - z_0) = \alpha$ where z_0 is a fixed complex number then locus of z denotes a ray emanating from z_0 (z_0 is not included) and making an angle α from positive real axis.

Note :

For any complex number z

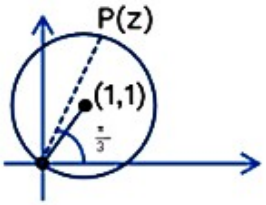
$\text{amp}(z) + \text{amp}(-\bar{z}) = \pi$ or $\text{amp}(z) + \text{amp}(-1) + \text{amp}(\bar{z}) = \pi$

Q. **Represent the locus of z satisfying given equation or inequation on the Argand's plane.**

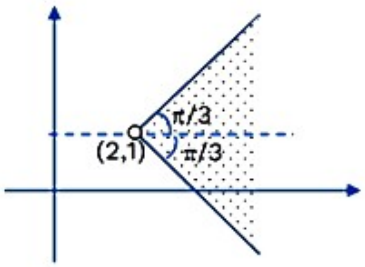
(i) $|z - 1 - i| = \sqrt{2}$ and $\text{amp}(z) = \frac{\pi}{3}$ **(ii)** $|\text{amp}(z - 2 - i)| < \frac{\pi}{3}$

(iii) $3 \le |z - 4| \le 4$ and $|\text{amp}(z - 4)| \le \frac{\pi}{4}$

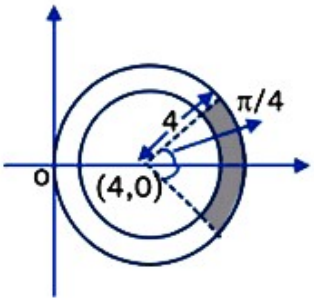
Sol. **(i)** The complex number z representing the point P is only the solution satisfying both the equations simultaneously



(ii) Complex numbers corresponding to the points lie in the shaded region satisfy the given inequality.



(iii) All the complex number z satisfying both the inequalities simultaneously lies in the shaded region



Q. **Consider two pairs of non-zero conjugate complex numbers (z_1, z_2) and (z_3, z_4)**

Find the value of $\arg\left(\frac{z_1}{z_3}\right) + \arg\left(\frac{z_2}{z_4}\right)$

Sol. $\arg\left(\frac{z_1}{z_3}\right) + \arg\left(\frac{z_2}{z_4}\right) = \arg\left(\frac{z_1 z_2}{z_3 z_4}\right)$

$= \arg\left(\frac{|z_1|^2}{|z_3|^2}\right)$

(as $z_2 = \bar{z}_1$ and $z_4 = \bar{z}_3$)

=0 (as argument of a positive real number is zero).

Q. **If z and w are complex numbers satisfying $\bar{z} + i\bar{w} = 0$ and $\text{amp}(zw) = \pi$, then $\text{amp}(w)$ is equal to**

(A) $\frac{\pi}{4}$ **(B)** $\frac{-\pi}{2}$ **(C)** $\frac{\pi}{2}$ **(D)** $\frac{3\pi}{4}$

Sol. Given $\bar{z} + i\bar{w} = 0 \Rightarrow \bar{z} = -i\bar{w}$ or $z = iw$

$\text{amp}(z) - \text{amp}(w) = \text{ampi} = \frac{\pi}{2}$ ...(i)

also $\text{amp}(zw) = \pi$

$\text{amp}(z) + \text{amp}(w) = \pi$ ...(2)

(1) + (2), gives $2\text{amp}(z) = \frac{3\pi}{2} \Rightarrow \text{amp}(z) = \frac{3\pi}{4}$; Also $\text{amp}(w) = \frac{\pi}{4}$

Q. **If $\text{Arg}(z + a) = \frac{\pi}{6}$ and $\text{Arg}(z - a) = \frac{2\pi}{3}$; $a \in R^+$, then**

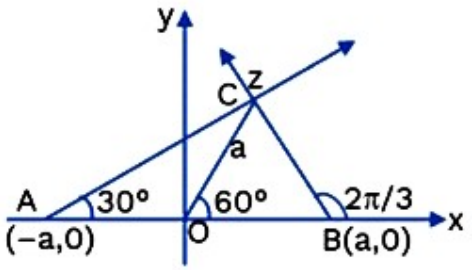
(A) z is independent of a **(B) $|a| = |z+a|$**

(C) $z = a \text{ cis } \frac{\pi}{6}$ **(D)** $z = a \text{ cis } \frac{\pi}{3}$

Sol. **(D)**

Refer the figure z lies on the point of intersection of the rays from A and B. ΔACB is a right angle and OBC is an equilateral triangle.

$\Rightarrow OC = a \Rightarrow z = a \text{ cis } \frac{\pi}{3}$



Q. **Find modulus and principal argument of the complex number:**
$z = (1 + i)(1 - i\sqrt{3})(-2 - 2i)(i)(3)$

Sol. amp of $(1 + i) = \tan^{-1} 1 = \frac{\pi}{4}$

amp of $\left(1 - i\sqrt{3}\right) = -\frac{\pi}{3}$

amp of $(-2 - 2i) = -\left(\pi - \tan^{-1} 1\right) = -\frac{3\pi}{4}$

amp of $i = \frac{\pi}{2}$

amp of 3 = 0

$$\text{amp}(z) = \frac{\pi}{4} - \frac{\pi}{3} + \left(\frac{-3\pi}{4}\right) + \frac{\pi}{2} + 0 = \frac{-\pi}{3}$$

$$|z| = |1+i||1-i\sqrt{3}||-2-2i||i||3|$$

$$= \sqrt{2}\cdot 2\cdot 2\sqrt{2}\cdot 1\cdot 3 = 24.\text{Ans}$$

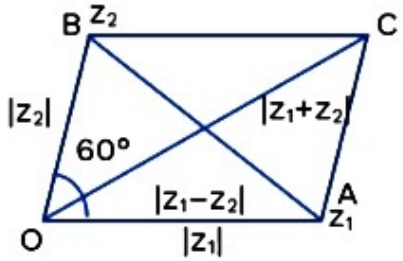
Q. **It is given that complex numbers z_1 and z_2 and satisfy $|z_1| = 2$ and $|z_2| = 3$. If the included angle of their corresponding vectors is 60° then $\left|\frac{z_1+z_2}{z_1-z_2}\right|$ can be expressed as $\frac{\sqrt{N}}{7}$ where N is natural number then N equals**

(A) 126 **(B) 119** **(C) 133** **(D) 19**

Sol. Using cosine rule.

$$|z_1+z_2| = \sqrt{|z_1|^2 + |z_2|^2 - 2|z_1||z_2|\cos 120^\circ}$$

$$= \sqrt{4+9+2\cdot 3} = \sqrt{19}$$



$$\text{and } |z_1-z_2| = \sqrt{|z_1|^2 + |z_2|^2 - 2|z_1||z_2|\cos 60^\circ}$$

$$= \sqrt{4+9-6} = \sqrt{7}$$

$$\therefore \left|\frac{z_1+z_2}{z_1-z_2}\right| = \sqrt{\frac{19}{7}} = \frac{\sqrt{133}}{7} \Rightarrow N = 133$$

Q. **If $z = \frac{1+i\sqrt{3}}{2i\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)}$ then find |z| & amp(z)**

Sol. $$z = \frac{1+i\sqrt{3}}{(-\sqrt{3}+i)} \Rightarrow |z| = \frac{|1+i\sqrt{3}|}{|-\sqrt{3}+i|} = 1$$

$$\text{ampz} = \text{amp}(1+i\sqrt{3}) - \text{amp}(-\sqrt{3}+i) \Rightarrow \frac{\pi}{3} - \frac{5\pi}{6} \Rightarrow \frac{-\pi}{2}$$

Q. **Find the value of $\text{Cis}\pi.\text{Cis}\frac{\pi}{2}.\text{Cis}\frac{\pi}{2^2}...\infty$**

Sol. $$\text{Cis}\pi.\text{Cis}\frac{\pi}{2}.\text{Cis}\frac{\pi}{2^2}...\infty = \text{Cis}\left(\pi + \frac{\pi}{2} + \frac{\pi}{2^2} + ...\infty\right)$$

$$. = \text{Cis}\left(\frac{\pi}{1-\frac{1}{2}}\right)$$

$$. = \text{Cis}(2\pi)$$

$$. = \cos(2\pi) + i\sin(2\pi)$$

$$= 1$$

Q. **Find $\text{amp}(z)$ if $z = \dfrac{(1+i)(1+2i)(1+3i)}{(1-i)(2-i)(3-i)}$**

Sol. z is of the form $\dfrac{z_1z_2z_3}{z_4z_5z_6}$

Here amp(z)

$$= \text{amp}(z_1) + \text{amp}(z_2) + \text{amp}(z_3) - \text{amp}(z_4) - \text{amp}(z_5) - \text{amp}(z_5) + 2k\pi$$

$$= \tan^{-1}(1) + \tan^{-1}2 + \tan^{-1}3 - \tan^{-1}(-1) - \tan^{-1}\left(-\frac{1}{2}\right) - \tan^{-1}\left(-\frac{1}{3}\right) + 2k\pi$$

$$= \pi + \left(\tan^{-1}(1) + \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)\right) + 2k\pi$$

$$= \pi + \frac{\pi}{2} + 2k\pi$$

$$\Rightarrow k = -1$$

$$\Rightarrow \text{amp}(z) = \frac{3\pi}{2} - 2\pi = -\frac{\pi}{2}.$$

Q. **$z = \dfrac{\left(2\sqrt{3}+2i\right)^6}{(1-i)^6} + \dfrac{(1+i)^6}{\left(2\sqrt{3}-2i\right)^6}$ has argument**

(A) $-\dfrac{\pi}{3}$ **(B)** $\dfrac{5\pi}{6}$ **(C)** $-\dfrac{2\pi}{3}$ **(D)** $\dfrac{5\pi}{12}$

Sol. $$z = \frac{\left(2\sqrt{3}+2i\right)^8}{(1-i)^6} + \frac{(1+i)^6}{\left(2\sqrt{3}-2i\right)^8}$$

$$= \frac{\left(2\sqrt{3}+2i\right)^8}{(1-i)^6} + \frac{\left(2\sqrt{3}+2i\right)^8}{16^8} \times \frac{(2)^6}{(1-i)^6}$$

$$= \left(1+\frac{1}{2^{26}}\right)2^8 \frac{\left(\sqrt{3}+i\right)^8}{(1-i)^6}$$

$$\Rightarrow \text{Arg}(z) = \text{Arg}\left(\frac{\left(\sqrt{3}+i\right)^8}{(1-i)^6}\right)$$

$$= 8\text{Arg}\left(\sqrt{3}+i\right) - 6\text{Arg}(1-i) + 2k\pi$$

$$= 8\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) - 6\tan^{-1}(-1) + 2k\pi$$

$$= \frac{4\pi}{3} + \frac{3\pi}{2} + 2k\pi$$

$$= \frac{17\pi}{6} + 2k\pi$$

$$= \frac{5\pi}{6} \qquad (k = -1)$$

Previous Year's Question

If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}\left(i = \sqrt{-1}\right)$. Then $\left(1 + iz + z^5 + iz^8\right)^9$ is equal to: **(JEE Main 2019)**

(A) -1 (B) 1 (C) 0 (D) $(-1+2i)^9$

Previous Year's Question

If $\frac{z-\alpha}{z+\alpha}(\alpha \in \mathbf{R})$ is a purely imaginary number and $|z| = 2$, then a value of α is:

(JEE Main 2019)

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\sqrt{2}$

Previous Year's Question

Let $z=\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5$. If R (z) and I(z) respectively denote the real and imaginary parts of z, then:

(JEE Main 2019)

(A) $I(z)=0$

(B) $R(z)>0$ and $I(z)>0$

(C) $R(z)<0$ an $I(z)>0$

(D) $R(z)=-3$

Previous Year's Question

Let z_1 and z_2 be any two non-zero complex numbers such that $3|z_1|=2|z_2|$. If $z=\frac{3z_1}{2z_2}+\frac{2z_2}{3z_1}$, then

(JEE Main 2019)

(A) $Re(z)=0$

(B) $|z|=\sqrt{\frac{5}{2}}$

(C) $|z|=\frac{1}{2}\sqrt{\frac{17}{2}}$

(D) $Im(z)=0$

Previous Year's Question

If z and w are two complex numbers such that $|zw|=1$ and $arg(z)-arg(w)=\frac{\pi}{2}$, then:

(JEE Main 2019)

(A) $\bar{z}w=i$

(B) $\bar{z}w=-i$

(C) $z\bar{w}=\frac{1-i}{\sqrt{2}}$

(D) $z\bar{w}=\frac{-1+i}{\sqrt{2}}$

Previous Year's Question

If $\frac{3+i\sin\theta}{4-i\cos\theta}, \theta\in[0,2\pi]$, is a real number, then an argument of $\sin\theta+i\cos\theta$ is:

(JEE Main 2020)

(A) $\pi-\tan^{-1}\left(\frac{4}{3}\right)$

(B) $-\tan^{-1}\left(\frac{3}{4}\right)$

(C) $\pi-\tan^{-1}\left(\frac{3}{4}\right)$

(D) $\tan^{-1}\left(\frac{4}{3}\right)$

Previous Year's Question

The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is : **(JEE Main 2020)**

(A) $-\frac{1}{2}\left(1-i\sqrt{3}\right)$ (B) $\frac{1}{2}\left(1-i\sqrt{3}\right)$ (C) $-\frac{1}{2}\left(\sqrt{3}-i\right)$ (D) $\frac{1}{2}\left(\sqrt{3}-i\right)$

Previous Year's Question

The imaginary part of $\left(3+2\sqrt{-54}\right)^{1/2}-\left(3-2\sqrt{-54}\right)^{1/2}$ can be:

(JEE Main 2020)

(A) $\sqrt{6}$ (B) $-2\sqrt{6}$ (C) 6 (D) $-\sqrt{6}$

Previous Year's Question

If a and b are real numbers such that $(2+\alpha)^4=a+b\alpha$, where $\alpha=\frac{-1+i\sqrt{3}}{2}$ then $a+b$ is equal to: **(JEE Main 2020)**

(A) 33 (B) 57 (C) 9 (D) 24

Previous Year's Question

The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is: **(JEE Main 2020)**

(A) $2^{15}i$ (B) -2^{15} (C) $-2^{15}i$ (D) 6^5

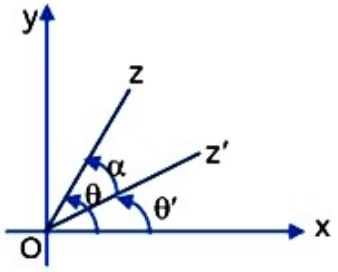
Previous Year's Question

Let $z \in C$ be such that $|z| < 1$. If $\omega = \dfrac{5+3z}{5(1-z)}$, then: **(JEE Main 2019)**

(A) $5\,\mathrm{Im}(\omega) < 1$ (B) $4\,\mathrm{Im}(\omega) > 5$ (C) $5\,\mathrm{Re}(\omega) > 1$ (D) $5\,\mathrm{Re}(\omega) > 4$

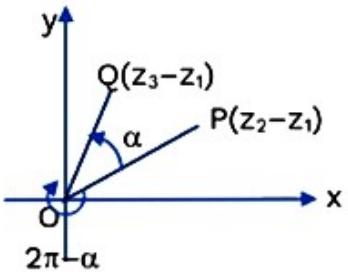
CONCEPT OF ROTATION

If z and z' are two complex numbers, then argument of $\dfrac{z}{z'}$ is the angle through which Oz' must be turned in order that it may lie along Oz.



$$\frac{z}{z'} = \frac{|z|e^{i\theta}}{|z'|e^{i\theta'}} = \frac{|z|}{|z'|}e^{i(\theta-\theta')} = \frac{|z|}{|z'|}e^{i\alpha}$$

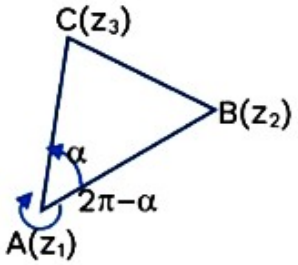
In general, let z_1, z_2, z_3 be the three vertices of a triangle ABC described in the counter-clock wise sense. Draw OP and OQ parallel and equal to AB and AC respectively. Then the point P is $z_2 - z_1$ and Q is $z_3 - z_1$ and



$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{OQ}{OP}(\cos\alpha + i\sin\alpha)$$

$$= \frac{CA}{BA}\cdot e^{i\alpha} = \frac{|z_3 - z_1|}{|z_2 - z_1|}\cdot e^{i\alpha}$$

Note that $\arg.(z_3 - z_1) - \arg(z_2 - z_1) = \alpha$ is the angle through which OP must be rotated in the anti-clockwise direction so that it concides with OQ.



Here we can write $\dfrac{z_3 - z_1}{z_2 - z_1} = \dfrac{|z_3 - z_1|}{|z_2 - z_1|}\cdot e^{-i(2\pi-\alpha)}$

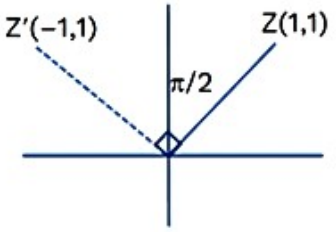
also. In case we are rotating OP in clockwise direction by an angle $(2\pi - \alpha)$. Since the rotation is in clockwise direction, we are taking negative sign with angle $(2\pi - \alpha)$.

Note:

If a complex number (z) is multiplied by i, it means z has been rotated through an angle $\frac{\pi}{2}$ in anticlockwise sense.

e.g. $z = 1 + i$

$z' = (1+i)e^{i\pi/2} = (1+i)i = -1+i$



Q. **Consider a square ABCD such that z_1, z_2, z_3 and z_4 represent its vertices A,B,C and D respectively. Express 'z_3' and 'z_4' in terms of z_1 and z_2.**

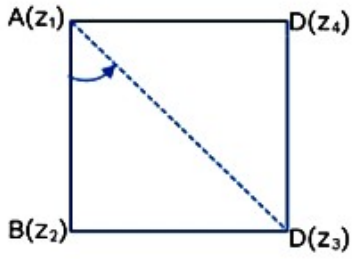
Sol. Consider the rotation of AB about A through an angle $\frac{\pi}{4}$. We get

$$\frac{z_3 - z_1}{z_2 - z_1} = \left|\frac{z_3 - z_1}{z_2 - z_1}\right| e^{i\pi/4}$$

$$= \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) \Rightarrow z_3 = z_1 + (z_2 - z_1)(1+i)$$

Similarly, $\frac{z_4 - z_1}{z_2 - z_1} = \left|\frac{z_4 - z_1}{z_2 - z_1}\right| e^{i\pi/2} = i$

$\Rightarrow \quad z_4 = z_1 + i(z_2 - z_1).$



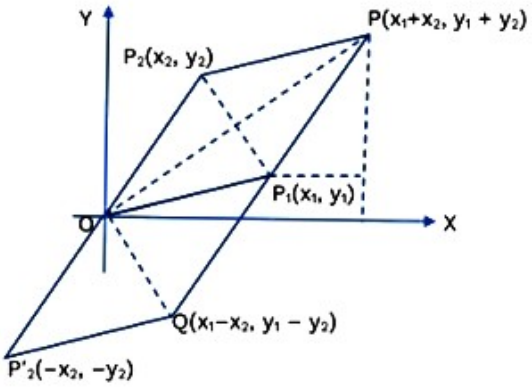
GEOMETRY OF COMPLEX NUMBER:

Geometrical meaning of addition and subtraction:

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers represented by the point $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ respectively. By definition $z_1 + z_2$ should be represented by the point $(x_1 + x_2, y_1 + y_2)$ This point is the vertex which completes the parallelogram with the line segments joining the origin with OP_1 and OP_2 as the adjacent sides.

$\Rightarrow \quad |z_1 + z_2| = OP$

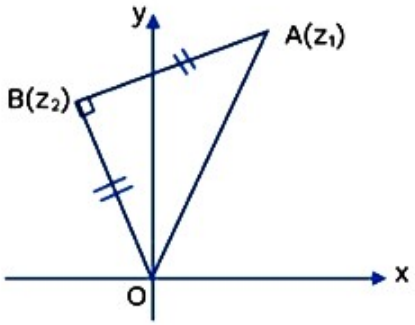
Also by definition $z_1 - z_2$ should be



represented by the point $(x_1 - x_2, y_1 - y_2)$.

This point is the vertex which completes the parallelogram with the line segments joining the origin with OP_1 and OP_2' (where the point P_2' represents $-z_2$; the point $-z_2$ can be obtained by producing the directed line P_2O by length ($|z_2|$) as the adjacent sides.

$\Rightarrow \quad |z_1 - z_2| = OQ = P_2P_1$



Geometrical Meaning of product and Division:

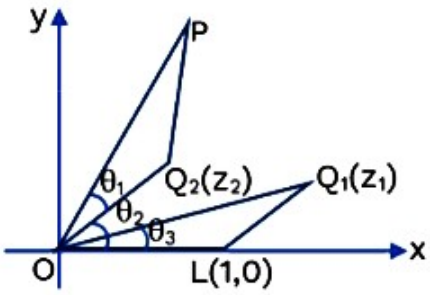
Let

$z_1 = r_1(\cos\theta_1 + i\sin\theta_1), z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

be complex numbers represented by Q_1 and Q_2.

(i) Construction for the point representing the z_1z_2 :

Let L be the point on OX which represents unity, so that OL=1. Draw the triangle OQ_2P directly similar to the triangle OLQ_1. Then point P represents the product z_1z_2.



Explanation:

Due to similar triangles

$\frac{OP}{OQ_1} = \frac{OQ_2}{OL}$, that is $\frac{OP}{r_1} = \frac{r_2}{1} \Rightarrow OP = r_1r_2$

Also $\angle Q_2OP = \angle LOQ_1 = \theta_1 \Rightarrow \angle LOP = \theta_1 + \theta_2$

Since

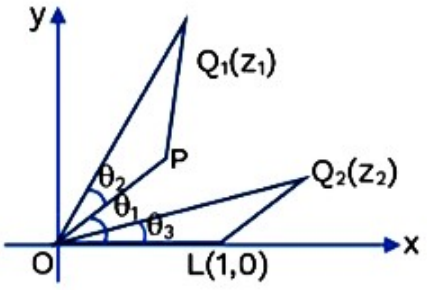
$z_1z_2 = r_1r_2\{\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)\}$, P

represents z_1z_2.

(ii) Construction for the point representing the quotient z_1/z_2:

Draw the triangle OQ_1P directly similar to the triangle OQ_2L

Then P represents the quotient $\frac{z_1}{z_2}$



Explanation:

From the last construction,

$\frac{OQ_1}{OQ_2} = \frac{OP}{OL} \Rightarrow \frac{r_1}{r_2} = \frac{OP}{1}$

Number represented by $P \cdot z_2 = z_1 \Rightarrow$ number represented by $P = \frac{z_1}{z_2}$

Remark:

(i) If

$z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

then $z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$ and $\frac{z_1}{z_2} = \frac{r_1}{r_2} \cdot e^{i(\theta_1-\theta_2)}$

(ii) $\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2)$

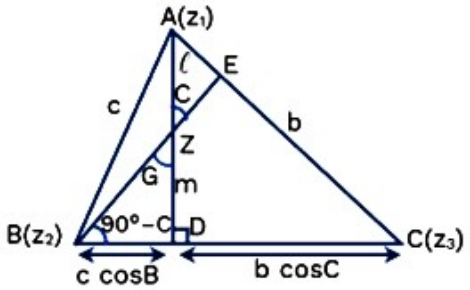
(iii) $\arg\left(\frac{z_1}{z_2}\right) = \theta_1 - \theta_2 = \arg(z_1) - \arg(z_2)$

Centroid, Incenter, Orthocenter & Circumcenter of a triangle on a complex plane:

(i) Centroid 'G' $= \frac{z_1 + z_2 + z_3}{3}$

(ii) Incenter 'I' $= \frac{az_1 + bz_2 + cz_3}{a + b + c}$

(iii) Orthocentre:



$Z_D = \frac{b\cos C z_2 + c\cos B z_3}{a}$

Now AE = c cos A;

ℓ = AE cosec C = c cos A cosec C

$\ell = 2R\cos A \left(\frac{c}{\sin C} = 2R\right)$

And m = c cos B cot C or m = 2R cos B cos C

Hence $Z_H = \frac{mz_1 + \ell Z_D}{\ell + m}$

$$= \frac{2R\cos B\cos C z_1 + 2R\cos A\left(\frac{b\cos C z_2 + c\cos B z_3}{a}\right)}{2R(\cos A + \cos B\cos C)}$$

$$= \frac{a\cos B\cos C z_1 + b\cos A\cos C z_2 + c\cos A\cos B z_3}{a(-\cos(B+C) + \cos B\cos C)}$$

$$= \frac{z_1(\sin A\cos B\cos C) + (\sin B\cos C\cos A)z_2 + (\sin C\cos A\cos B)z_3}{\sin A(\sin B\sin C)}$$

$$Z_H = \frac{z_1\tan A + z_2\tan B + z_3\tan C}{\Sigma\tan A} \text{ or } \frac{z_1\tan A + z_2\tan B + z_3\tan C}{\Pi\tan A}$$

(iv) Circumcenter:

We have z_0 being equidistant from the vertices gives,

$|z_1 - z_0| = |z_2 - z_0| = |z_3 - z_0|$

Consider $|z_1 - z_0|^2 = |z_2 - z_0|^2$

$(z_1 - z_0)(\bar{z}_1 - \bar{z}_0) = (z_2 - z_0)(\bar{z}_2 - \bar{z}_0)$

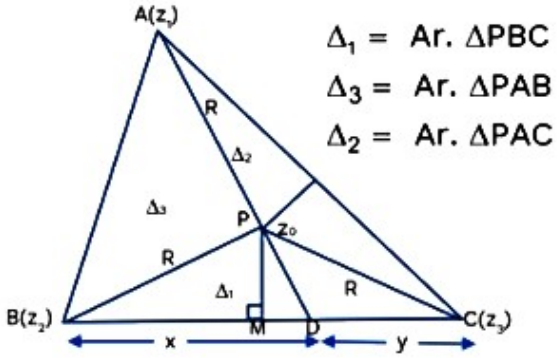
$\bar{z}_1(z_1 - z_0) - \bar{z}_2(z_2 - z_0) = \bar{z}_0[(z_1 - z_0) - (z_2 - z_0)]$

$\bar{z}_1(z_1 - z_0) - \bar{z}_2(z_2 - z_0) = \bar{z}_0(z_1 - z_2)$ ....(1)

Similarly, 1st & 3rd gives

$\bar{z}_1(z_1 - z_0) - \bar{z}_3(z_3 - z_0) = \bar{z}_0(z_1 - z_3)$ ...(2)

Dividing (1) by (2) eliminate $\bar{z}_0$ and get z_0.



Δ_1 = Ar. ΔPBC

Δ_3 = Ar. ΔPAB

Δ_2 = Ar. ΔPAC

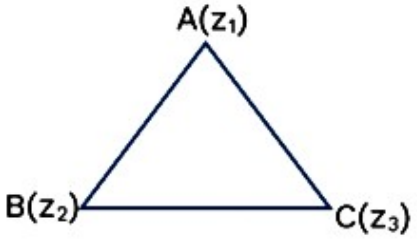
Q. **If z_1, z_2, z_3 are the vertices of an equilateral triangle then prove that $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$ and if z_0 is the circumcentre of the triangle then also prove that $3z_0^2 = z_1^2 + z_2^2 + z_3^2$.**

Sol. Taking vertices in anti-clockwise manner.

ΔABC is equilateral

$\therefore$ AB = BC = CA

$|z_1 - z_2| = |z_2 - z_3| = |z_3 - z_1|$ ... (1)



Also $\angle$CBA = $\angle$ACB = α

Rotate BC about $B(z_2)$ in anticlockwise sense then

$$\Rightarrow \frac{z_1 - z_2}{z_3 - z_2} = \frac{|z_1 - z_2|}{|z_3 - z_2|} \cdot e^{i\alpha} \quad \text{... (2)}$$

again rotate side AB about $A(z_1)$ in anticlockwise sense then

$$\Rightarrow \frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_2 - z_1}{|z_2 - z_1|} \cdot e^{i\alpha} \quad \text{... (3)}$$

By equation (2) and (3)

$$\frac{z_1 - z_2}{z_3 - z_2} = \frac{z_3 - z_1}{z_2 - z_1}$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1 \quad \text{... (4)}$$

and if z_0 is the circumcentre for equilateral triangle

$$z_0 = \frac{z_1 + z_2 + z_3}{3} \quad \text{...(5)}$$

or by equation (5)

$9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1z_2 + z_2z_3 + z_3z_1)$

or by equation (4)

$z_1^2 + z_2^2 + z_3^2 = 3z_0^2$

Q. **If z_1 & z_2 be non-zero complex numbers satisfying the equation, $z_1^2 - 2z_1z_2 + 2z_2^2 = 0$ then the geometrical nature of the triangle whose vertices are the origin and the points representing z_1 & z_2 is**
(A) an isosceles right-angled triangle
(B) a right-angled triangle which is not isosceles
(C) an equilateral triangle
(D) an isosceles triangle which is not right angled.

Sol. $\frac{z_1}{z_2} = z \Rightarrow z^2 - 2z + 2 = 0 \Rightarrow z = 1 \pm i$

$\Rightarrow \frac{z_1}{z_2} = 1 \pm i$

$\Rightarrow z_1 = z_2 \pm z_2 i$

$\Rightarrow z_1 - z_2 = \pm z_2 i$

$\Rightarrow$ $z_1 - z_2$ is perpendicular to z_2

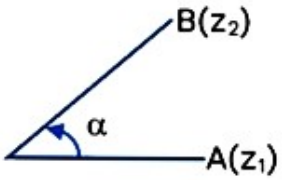
and $|z_1 - z_2| = |z_2|$

Q. **Let z_1 & z_2 be the roots of the equation $z^2 + pz + q = 0$, where the coefficients p and q may be complex number. Let A and B represent z_1 & z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and OA = OB, where O is the origin, prove that $p^2 = 4q\cos^2\left(\frac{\alpha}{2}\right)$.**

Sol. $z_1 + z_2 = -p$ ...(1)

and $z_1z_2 = q$ ...(2)

Also $z_2 = z_1e^{i\alpha}$ ...(3)



Now $p^2 = (z_1 + z_2)^2 = \left(z_1 + z_1e^{i\alpha}\right)^2$

$= z_1^2\left(1 + 2e^{i\alpha} + e^{i2\alpha}\right)$

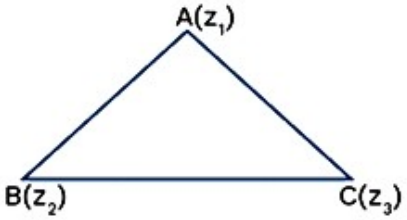
$= qe^{-i\alpha}\left(1 + 2e^{i\alpha} + e^{i2\alpha}\right) = q\left(2 + e^{-i\alpha} + e^{i\alpha}\right)$

$= q(2 + 2\cos\alpha) = 4q\cos^2\frac{\alpha}{2}$

Q. If z_1, z_2, z_3 and z_1', z_2', z_3' represent the vertices of two similar triangles ABC and PQR, respectively then prove that $\left|\frac{\bar{z}_1'}{\bar{z}_2 - \bar{z}_1}\right|\left|\frac{z_2 - z_3}{z_3'}\right| + \left|\frac{z_2'}{z_2 - z_1}\right|\left|\frac{\bar{z}_3 - \bar{z}_1}{\bar{z}_3'}\right| \geq 1$

Sol. Since ΔABC and ΔPQR are similar

$\frac{AC}{AB} = \frac{PR}{PQ}$ and $\angle BAC = \angle QPR = \theta$.



In ΔABC, $\frac{z_3 - z_1}{z_2 - z_1} = \frac{AC}{AB} e^{i\theta}$ ...(1)

In ΔPQR, $\frac{z_3' - z_1'}{z_2' - z_2'} = \frac{PR}{PQ} e^{i\theta}$ ...(2)

From (1) and (2), $\frac{z_3 - z_1}{z_2 - z_1} = \frac{z_3' - z_1'}{z_2' - z_1'}$

$\Rightarrow z_1'(z_2 - z_3) + z_2'(z_3 - z_1) = z_3'(z_2 - z_1)$

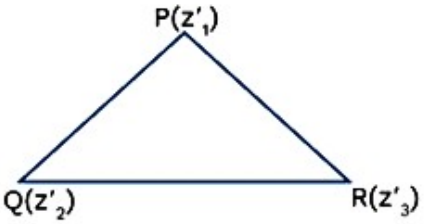
$\Rightarrow \left|z_1'(z_2 - z_3) + z_2'(z_3 - z_1)\right| = \left|z_3'(z_2 - z_1)\right|$

$\Rightarrow \left|z_1'(z_2 - z_3)\right| + \left|z_2'(z_3 - z_1)\right| \geq \left|z_3'(z_2 - z_1)\right|$

$\Rightarrow \left|\frac{z_1'}{z_2 - z_1} \cdot \frac{z_2 - z_3}{z_3'}\right| + \left|\frac{z_2'}{z_2 - z_1}\right|\left|\frac{z_3 - z_1}{z_3'}\right| \geq 1$

$\Rightarrow \left|\frac{z_1'}{z_2 - z_1}\right|\left|\frac{z_2 - z_3}{z_3'}\right| + \left|\frac{z_2'}{z_2 - z_1}\right|\left|\frac{z_3 - z_1}{z_3'}\right| \geq 1$

$\Rightarrow \left|\frac{\bar{z}_1'}{\bar{z}_2 - \bar{z}_1}\right|\left|\frac{z_2 - z_3}{z_3'}\right| + \left|\frac{z_2'}{z_2 - z_1}\right|\left|\frac{\bar{z}_3 - \bar{z}_1}{\bar{z}_3'}\right| \geq 1$



Q. If $z_r\ (r = 1, 2, \ldots 6)$ are the vertices of a regular hexagon then prove that $\sum_{r=1}^{6} z_r^2 = 6z_0^2$ where z_0 is the circumcentre.

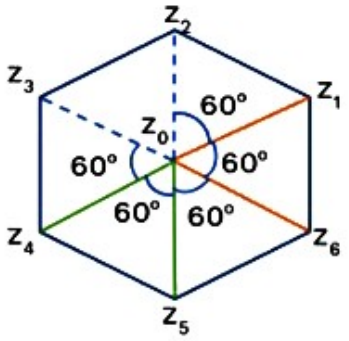
Sol. Let $z_1 - z_0 = z$

$\Rightarrow z_2 - z_0 = ze^{i\frac{\pi}{3}}$

$z_3 - z_0 = ze^{i\frac{2\pi}{3}}$

$z_4 - z_0 = ze^{i\pi}$

$z_5 - z_0 = ze^{i\frac{4\pi}{3}}$



$$z_6 - z_0 = ze^{i\frac{5\pi}{3}}$$

$$\Rightarrow \sum_{r=1}^{6} z_r^2 = (z_0 + z)^2 + \left(z_0 + ze^{i\frac{\pi}{3}}\right)^2 + \ldots + \left(z_0 + ze^{i\frac{5\pi}{3}}\right)^2$$

$$= 6z_0^2 + z^2\left(1 + e^{i\frac{2\pi}{3}} + e^{i\frac{4\pi}{3}} + e^{i2\pi} + e^{i\frac{8\pi}{3}} + e^{i\frac{10\pi}{3}}\right) + 2z_0 z\left(1 + e^{i\frac{\pi}{3}} + e^{i\frac{2\pi}{3}} + e^{i\pi} + e^{i\frac{4\pi}{3}} + e^{i\frac{5\pi}{3}}\right)$$

$$\Rightarrow \sum_{r=1}^{6} z_r^2 = 6z_0^2 + 2z^2\left(1 + e^{i\frac{2\pi}{3}} + e^{i\frac{4\pi}{3}}\right) + 2z_0 z(0)$$

$$\Rightarrow \sum_{r=1}^{6} z_r^2 = 6z_0^2 + 0$$

$$= 6z_0^2$$

Hence Proved.

Q. **Prove that the triangle whose vertices are the points z_1, z_2, z_3 on the Argand plane is an equilateral triangle if and only if $\frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2} = 0$.**

Sol. If ΔABC is equilateral, then

$$|z_1 - z_2| = |z_2 - z_3| = |z_3 - z_1| \qquad \ldots(i)$$

$$\frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2}$$

$$= \frac{\bar{z}_2 - \bar{z}_3}{(z_2 - z_3)(\bar{z}_2 - \bar{z}_3)} + \frac{\bar{z}_3 - \bar{z}_1}{(z_3 - z_1)(\bar{z}_3 - \bar{z}_1)} + \frac{\bar{z}_1 - \bar{z}_2}{(z_1 - z_2)(\bar{z}_1 - \bar{z}_2)}$$

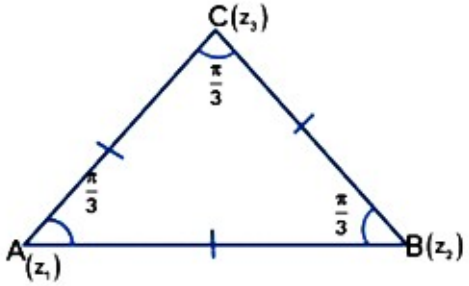
$$= \frac{\bar{z}_2 - \bar{z}_3}{|z_2 - z_3|^2} + \frac{\bar{z}_3 - \bar{z}_1}{|z_3 - z_1|^2} + \frac{\bar{z}_1 - \bar{z}_2}{|z_1 - z_2|^2}$$

$$= \frac{\bar{z}_2 - \bar{z}_3 + \bar{z}_3 - \bar{z}_1 + \bar{z}_1 - \bar{z}_2}{|z_1 - z_2|^2} \quad \text{(from (i))}$$

$$\Rightarrow \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2} = 0$$

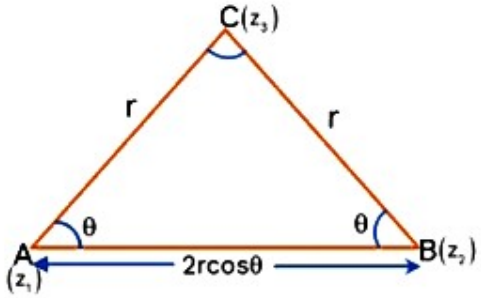
$\therefore$ if $\frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2} \neq 0$,

triangle can not be equilateral. **Hence proved.**



Q. **On the Argand plane z_1, z_2 and z_3 are respectively the vertices of an isosceles triangle ABC with AC=BC and equal angles are θ. If z_4 is the incentre of the triangle then prove that $(z_2 - z_1)(z_3 - z_1) = (1 + \sec\theta)(z_4 - z_1)^2$**

Sol. Let $AC = BC = r$



$\Rightarrow AB = 2r\cos\theta$

$$\Rightarrow z_4 = \frac{rz_1 + rz_2 + 2r\cos\theta z_3}{r + r + 2r\cos\theta}$$

$$\Rightarrow z_4 = \frac{z_1 + z_2 + 2\cos\theta z_3}{2 + 2\cos\theta}$$

$$\Rightarrow z_4 - z_1 = \frac{z_1 + z_2 + 2\cos\theta z_3 - 2z_1 - 2\cos\theta z_1}{2 + 2\cos\theta}$$

$$\Rightarrow z_4 - z_1 = \frac{(z_2 - z_1) + 2\cos\theta(z_3 - z_1)}{2 + 2\cos\theta}$$

Let $z_3 - z_1$ be z

$$\Rightarrow z_2 - z_1 = (2\cos\theta)ze^{-i\theta}$$

$$\Rightarrow z_4 - z_1 = \frac{2\cos\theta ze^{-i\theta} + 2\cos\theta z}{2(1 + \cos\theta)}$$

$$= \frac{z\cos\theta\left(1 + e^{-i\theta}\right)}{1 + \cos\theta}$$

$$(z_4 - z_1)^2(1 + \sec\theta) = \frac{z^2\cos^2\theta\left(1 + e^{-i\theta}\right)^2}{(1 + \cos\theta)^2}\left(\frac{1 + \cos\theta}{\cos\theta}\right)$$

$$= \frac{z^2\cos\theta(1 + \cos\theta - i\sin\theta)^2}{1 + \cos\theta}$$

$$= \frac{z^2\cos\theta\left(2\cos^2\frac{\theta}{2} - i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)^2}{2\cos^2\frac{\theta}{2}}$$

$$= 2z^2\cos\theta\left(\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\right)^2 \qquad = 2z^2\cos\theta\left(e^{-i\frac{\theta}{2}}\right)^2$$

$$= 2z^2\cos\theta e^{-i\theta} \quad = z\left(2z\cos\theta e^{-i\theta}\right)$$

$= (z_2 - z_1)(z_3 - z_1) \Rightarrow LHS = RHS$ **Hence Proved.**

Q. **Check whether True of False**

If z_1, z_2, z_3, z_4 in order are the vertices of the square taken in order then.

(i) $\frac{z_1 - z_3}{z_2 - z_4}$ **is purely imaginary**

(ii) $\frac{z_1 - z_2}{z_3 - z_4}$ **is purely real**

(iii) $\frac{z_4 - z_3}{z_2 - z_3}$ **is purely imaginary**

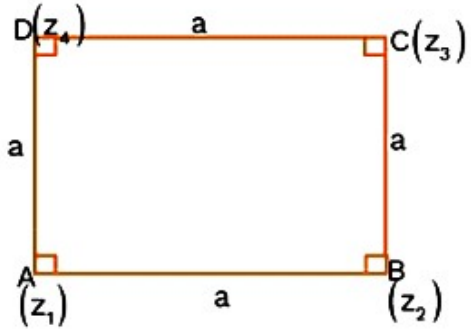
(iv) $\frac{z_2 + z_4}{z_1 + z_3} = 1$ **as mid point is same**

Sol. **(i)** Diagonals of square are equal and perpendicular

$\Rightarrow z_1 - z_3 = (z_4 - z_2)e^{i\frac{\pi}{2}}$

$\Rightarrow \frac{z_1 - z_3}{z_2 - z_4} = -i$ = purely imaginary

$\Rightarrow$ Statement is True.



(ii) AB || CD and AB = CD

$\therefore |z_1 - z_2| = |z_3 - z_4|$

$\Rightarrow z_1 - z_2 = (z_3 - z_4)e^{i\pi}$

$\Rightarrow \frac{z_1 - z_2}{z_3 - z_4} = -1$ = purely real.

$\Rightarrow$ Statement is true.

(iii) Just like (i), CD $\perp$ BC

$\Rightarrow \frac{z_4 - z_3}{z_2 - z_3} = e^{\frac{-i\pi}{2}} = -i$

$\Rightarrow$ Statement is true.

(iv) Midpoint of AC : $\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2}$

$\Rightarrow \frac{z_1 + z_3}{z_2 + z_4} = 1$

$\Rightarrow$ Statement is true.

Q. **Let $A \equiv z_1; B \equiv z_2; C \equiv z_3$ are three complex numbers denoting the vertices of an acute angled triangle. If the origin 'O' is the orthocentre of the triangle, then prove that**

$z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1$

hence show that the ΔABC is a right angled triangle.

$\Leftrightarrow z_1\bar{z}_2 + \bar{z}_1 z_2 = z_2\bar{z}_3 + \bar{z}_2 z_3 = z_3\bar{z}_1 + \bar{z}_3 z_1 = 0$

Sol. $OA = 2R\cos A$

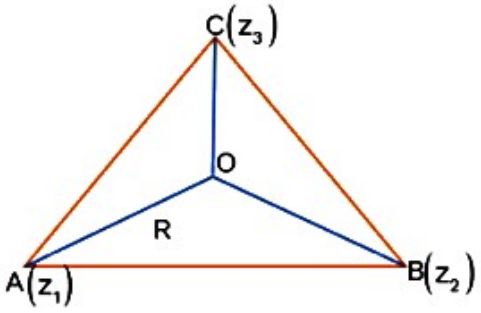
$OB = 2R\cos B$

$OC = 2R\cos C$

$\angle AOB = \pi - C$

$\angle AOC = \pi - B$

$\angle BOC = \pi - A$



Now, let $z_1 = (2R\cos A)e^{i\theta}$

$z_2 = (2R\cos B)e^{i\theta}e^{i(\pi - C)} = -2R\cos Be^{i(\theta - C)}$

$z_3 = (2R\cos C)e^{i\theta}e^{i(B-\pi)} = -2R\cos Ce^{i(\theta + B)}$

$z_1\bar{z}_2 + \bar{z}_1z_2 = \left(2R\cos Ae^{i\theta}\right)\left(-2R\cos Be^{-i(\theta - C)}\right) + \left(2R\cos Ae^{-i\theta}\right)\left(-2R\cos Be^{i(\theta - C)}\right)$

$\Rightarrow z_1\bar{z}_2 + z_2\bar{z}_1 = -4R^2\cos A\cos B\left[e^{-iC} + e^{iC}\right]$

$= -8R^2\cos A\cos B\cos C$

Similarly,

$z_2\bar{z}_3 + z_3\bar{z}_2 = (-2R\cos B)e^{i(\theta - C)}(-2R\cos C)e^{-i(B+\theta)} + (-2R\cos C)e^{i(\theta + B)}(-2R\cos B)e^{-i(\theta - C)}$

$= 4R^2\cos B\cos C\left(e^{i(B+C)} + e^{-i(B+C)}\right)$

$= 8R^2\cos B\cos C\cos(B + C)$

$= -8R^2\cos A\cos B\cos C$

Similarly, $z_3\bar{z}_1 + z_1\bar{z}_3 = -8R^2\cos A\cos B\cos C$

$\Rightarrow z_1\bar{z}_2 + z_2\bar{z}_1 = z_2\bar{z}_3 + \bar{z}_2z_3 = z_3\bar{z}_1 + \bar{z}_3z_1$

Now , $8R^2\cos A\cos B\cos C = 0 \Rightarrow \cos A$ or $\cos B$ or $\cos C = 0$

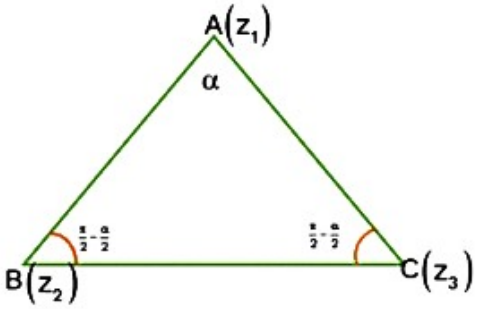
$\Rightarrow$ A or B or C must be 90°.

$\therefore \Delta ABC$ is right angle triangle $\Leftrightarrow z_1\bar{z}_2 + z_2\bar{z}_1 = 0 = z_1\bar{z}_3 + z_3\bar{z}_1 = z_2\bar{z}_3 + \bar{z}_2z_3$

Hence proved.

Q. **The points A, B, C depict the complex numbers z_1, z_2, z_3 respectively on a complex plane & the angle B & C of the triangle ABC are each equal to $\frac{1}{2}(\pi - \alpha)$. Show that $(z_2 - z_3)^2 = 4(z_3 - z_1)(z_1 - z_2)\sin^2\frac{\alpha}{2}$.**

Sol. $\frac{BC}{\sin\alpha} = \frac{AB}{\cos\frac{\alpha}{2}}$ (Using sine law)



$$\Rightarrow AB = \frac{\cos\frac{\alpha}{2}BC}{\sin\alpha} = \frac{BC}{2\sin\frac{\alpha}{2}}$$

And AC=AB (Given)

$$z_2 - z_3 = \frac{BC}{AC}(z_1 - z_3)e^{i\left(\frac{\pi}{2}-\frac{\alpha}{2}\right)}$$

$$\text{Also, } z_1 - z_2 = \frac{AB}{BC}(z_3 - z_2)e^{i\left(\frac{\pi}{2}-\frac{\alpha}{2}\right)}$$

$$\Rightarrow z_2 - z_3 = -(z_1 - z_2)\frac{BC}{AB}e^{-i\left(\frac{\pi}{2}-\frac{\alpha}{2}\right)}$$

$$\Rightarrow (z_2 - z_3)^2 = \left(\frac{BC}{AC}(z_1 - z_3)e^{i\left(\frac{\pi}{2}-\frac{\alpha}{2}\right)}\right)\left(-(z_1 - z_2)\frac{BC}{AB}e^{-i\left(\frac{\pi}{2}-\frac{\alpha}{2}\right)}\right)$$

$$= (z_3 - z_1)(z_1 - z_2)\frac{BC^2}{AB^2}$$

$$\Rightarrow (z_2 - z_3)^2 = (z_3 - z_1)(z_1 - z_2)4\sin^2\frac{\alpha}{2}$$

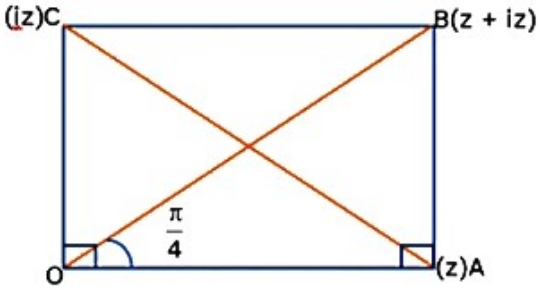
Hence proved.

Q. **If the area of the triangle formed by z, iz and z + iz is 8 sq. unit then find $|z|$.**

Sol. $iz = z.e^{i\frac{\pi}{2}}$

$$z + iz = z(1+i) = \sqrt{2}z\ e^{\left(i\frac{\pi}{4}\right)}$$

From above figure, OABC is a square with side length $= |z|$



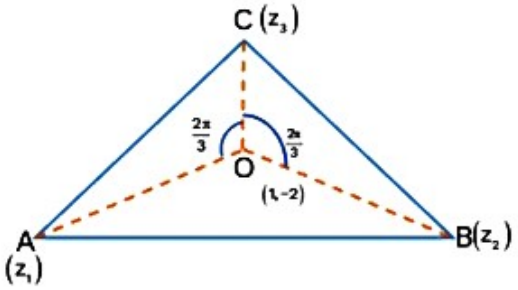
$$\text{Area of } \Delta ABC = \frac{1}{2}(\text{area of square } OABC)$$

$$\Rightarrow 8 = \frac{1}{2}|z|^2$$

$$\Rightarrow |z| = 4$$

Q. **z_1, z_2, z_3 are the vertices of an equilateral triangle taken anti-clockwise with circumcentre at $(1-2i)$. Find z_2 and z_3 if $z_1 = 2+i$.**

Sol. $z_1 = 2 + i$



Let $z_0 = 1 - 2i$

$$z_2 - z_0 = (z_1 - z_0)e^{i\frac{2\pi}{3}}$$

$$\Rightarrow z_2 = (1-2i) + (1+3i)\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$= (1-2i) + \left(-\frac{1}{2} - \frac{3\sqrt{3}}{2} + \frac{\sqrt{3}}{2}i - \frac{3i}{2}\right)$$

$$= \frac{1-3\sqrt{3}}{2} + \frac{\left(\sqrt{3}-7\right)}{2}i$$

$$z_3 - z_0 = (z_1 - z_0)e^{-i\frac{2\pi}{3}}$$

$$\Rightarrow z_3 = (1-2i) + (1+3i)\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)$$

$$= \left(\frac{1+3\sqrt{3}}{2}\right) - \left(\frac{\sqrt{3}+7}{2}\right)i$$

DEMOIVRE'S THEOREM(DMT)

Case-I:

Statement: If n is any integer then

(i) $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

(ii) $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)(\cos\theta_3 + i\sin\theta_3).....(\cos\theta_n + i\sin\theta_n)$

$= \cos(\theta_1 + \theta_2 + \theta_3 + ...\theta_n) + i\sin(\theta_1 + \theta_2 + \theta_3 + ...\theta_n)$

Case-II:

Statement: If $p, q \in Z$ and $q \neq 0$ then

$$(\cos\theta + i\sin\theta)^{p/q} = \cos\left(\frac{2k\pi + p\theta}{q}\right) + i\sin\left(\frac{2k\pi + p\theta}{q}\right)$$

Where k = 0, 1, 2, 3, ..., q – 1

Note:

When index 'n' is integer then $(\cos\theta + i\sin\theta)^n$ has exactly one value which is $\cos n\theta + i\sin n\theta$ but when n is rational number (say p/q, q≠0) other than integer then $(\cos\theta + i\sin\theta)^{p/q}$ has different values.

Q. **Prove that** $\left(\frac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\right)^n = \cos n\left(\frac{\pi}{2}-\theta\right)+i\sin n\left(\frac{\pi}{2}-\theta\right)$

Sol. $\left(\frac{(1+\sin\theta)+i\cos\theta}{(1+\sin\theta)-i\cos\theta}\right)^n$

$$= \left[\frac{1+\cos\left(\frac{\pi}{2}-\theta\right)+i\sin\left(\frac{\pi}{2}-\theta\right)}{1+\cos\left(\frac{\pi}{2}-\theta\right)-i\sin\left(\frac{\pi}{2}-\theta\right)}\right]^n = \left(\frac{1+\cos\alpha+i\sin\alpha}{1+\cos\alpha-i\sin\alpha}\right)^n \text{ where } \alpha = \frac{\pi}{2}-\theta$$

$$= \left(\frac{2\cos^2\frac{\alpha}{2}+i2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{2\cos^2\frac{\alpha}{2}-i2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}\right)^n = \left(\frac{\cos\frac{\alpha}{2}+i\sin\frac{\alpha}{2}}{\cos\frac{\alpha}{2}-i\sin\frac{\alpha}{2}}\right)^n = \left(\frac{e^{i\alpha/2}}{e^{-i\alpha/2}}\right)^n = \left(e^{i\alpha}\right)^n = e^{in\alpha}$$

$$= \cos n\left(\frac{\pi}{2}-\theta\right)+i\sin n\left(\frac{\pi}{2}-\theta\right)$$

Q. **If** $z = (1+i\sqrt{3})^6 + (1-i\sqrt{3})^6$ **then find** $|z|$.

Sol. $1+i\sqrt{3} = 2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)$

$$1-i\sqrt{3} = 2\left(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right)$$

$$\therefore z = (1+i\sqrt{3})^6 + (1-i\sqrt{3})^6 = 2^6\left[\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)^6 + \left(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right)^6\right]$$

$$= 2^6\left[\cos 2\pi + i\sin 2\pi + \cos 2\pi - i\sin 2\pi\right] = 2^7$$

$$\therefore |z| = 2^7$$

Application of Demoivre's Theorem:

To find roots of a complex quantity is the main application of DMT.

Working rule for finding roots of a complex quantity :

Q. **Find all the roots of the equation** $z^4 - (1+i) = 0$.

Sol. $z^4 = 1+i \Rightarrow z = (1+i)^{1/4}$

Step-1: Express 1+i in the polar form

$$z = \left(\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)\right)^{1/4} = 2^{1/8}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)^{1/4}$$

Step-2: Add $2m\pi$ in the principal argument.

$$z = 2^{1/8}\left(\cos\left(2m\pi + \frac{\pi}{4}\right) + i\sin\left(2m\pi + \frac{\pi}{4}\right)\right)^{1/4}$$

Step-3: Apply DMT

$$z = 2^{1/8}\left(\cos\left(2m\pi + \frac{\pi}{4}\right)\cdot\frac{1}{4} + i\sin\left(2m\pi + \frac{\pi}{4}\right)\cdot\frac{1}{4}\right)$$

Step-4: Put m = 0,1,2,3 to get all four roots of the equation

$$m = 0, z_1 = 2^{1/8}\left(\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right), \quad m = 1, z_2 = 2^{1/8}\left(\cos\frac{9\pi}{16} + i\sin\frac{9\pi}{16}\right)$$

$$m = 2, z_3 = 2^{1/8}\left(\cos\frac{17\pi}{16} + i\sin\frac{17\pi}{16}\right), \quad m = 3, z_4 = 2^{1/8}\left(\cos\frac{25\pi}{16} + i\sin\frac{25\pi}{16}\right)$$

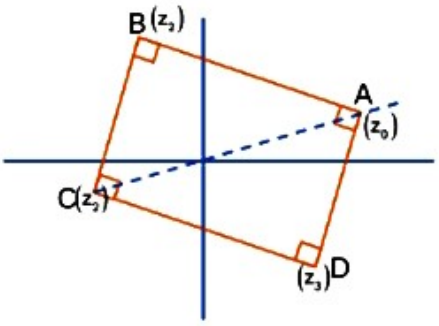
Q. **Find z if $z = \left(1 + i\sqrt{3}\right)^{1/4}$ .Also find the area bounded by figure joining all such z.**

Sol. $z = \left(1 + i\sqrt{3}\right)^{\frac{1}{4}}$

$$= 2^{\frac{1}{4}}\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{\frac{1}{4}}$$

$$= 2^{\frac{1}{4}}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{\frac{1}{4}}$$

$$\Rightarrow z = 2^{\frac{1}{4}}\left(e^{i\left(\frac{\pi}{3}+2k\pi\right)}\right)^{\frac{1}{4}} \qquad (k=0, 1, 2, 3)$$



$$k = 0 \Rightarrow z_0 = 2^{\frac{1}{4}}e^{i\frac{\pi}{12}} = 2^{\frac{1}{4}}\cos\left(\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$$

$$k = 1 \Rightarrow z_1 = 2^{\frac{1}{4}}e^{i\frac{7\pi}{12}} = 2^{\frac{1}{4}}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$$

$$k = 2 \Rightarrow z_2 = 2^{\frac{1}{4}}e^{i\frac{13\pi}{12}} = 2^{\frac{1}{4}}\left(\cos\frac{13\pi}{12} + i\sin\frac{13\pi}{12}\right)$$

$$k = 3 \Rightarrow z_3 = 2^{\frac{1}{4}}e^{i\frac{19\pi}{12}} = 2^{\frac{1}{4}}\left(\cos\frac{19\pi}{12} + i\sin\frac{19\pi}{12}\right)$$

$\Rightarrow$ABCD is a square with diagonal length $= 2\times 2^{\frac{1}{4}}$

$$\Rightarrow \text{Length of side} = \frac{2\times 2^{\frac{1}{4}}}{\sqrt{2}} = 2^{\frac{3}{4}} \Rightarrow \text{Area of square} = \left(2^{\frac{3}{4}}\right)^2 = 2^{\frac{3}{2}} = 2\sqrt{2}.$$

Q. $2\sqrt{2}z^4 = (\sqrt{3}-1)+i(\sqrt{3}+1)$

Sol. $z^4 = \dfrac{\sqrt{3}-1}{2\sqrt{2}} + i\dfrac{\sqrt{3}+i}{2\sqrt{2}}$

$\Rightarrow z^4 = \cos\dfrac{5\pi}{12} + i\sin\dfrac{5\pi}{12}$

$\Rightarrow z = \left(e^{i\left(\frac{5\pi}{12}+2k\pi\right)}\right)^{\frac{1}{4}}, k = 0,1,2,3$

$\Rightarrow k = 0 \Rightarrow z_0 = e^{\left(i\frac{5\pi}{48}\right)}$

$\Rightarrow k = 1 \Rightarrow z_1 = e^{\left(i\frac{29\pi}{48}\right)}$

$\Rightarrow k = 2 \Rightarrow z_2 = e^{\left(i\frac{53\pi}{48}\right)}$

$\Rightarrow k = 3 \Rightarrow z_3 = e^{\left(i\frac{77\pi}{48}\right)}$

Q. **Find the roots of the equation $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$**

Sol. $z^5 + z^4 + z^3 + z^2 + z^1 + 1 = 0$

$\Rightarrow \dfrac{z^6 - 1}{z - 1} = 0, z \neq 1$ (G. P)

$\Rightarrow z = 1^{\frac{1}{6}}, z \neq 1.$

$\Rightarrow z = \left(e^{i(2\pi+2k\pi)}\right)^{1/6}, z \neq 1, k = 0,1,2,3,4$

$k = 0 \Rightarrow z_0 = e^{i\frac{\pi}{3}}$

$k = 1 \Rightarrow z_1 = e^{i\frac{2\pi}{3}}$

$k = 2 \Rightarrow z_2 = e^{i\pi}$

$k = 3 \Rightarrow z_3 = e^{\left(i\frac{4\pi}{3}\right)}$

$k = 4 \Rightarrow z_4 = e^{\left(i\frac{5\pi}{3}\right)}$

Where $z_0, z_1, \ldots, z_4$ are roots of given equation.

Q. $z^4 - z^3 + z^2 - z + 1 = 0$

Sol. $z^4 - z^3 + z^2 - z + 1 = 0$

$$\Rightarrow \frac{z^5 + 1}{z + 1} = 0, z \neq -1$$

$$\Rightarrow z = (-1)^{\frac{1}{5}} = \left(e^{i(\pi + 2k\pi)}\right)^{\frac{1}{5}}, z \neq -1$$

$k = 0 \Rightarrow z_0 = e^{i\frac{\pi}{5}}$

$k = 1 \Rightarrow z_1 = e^{i\frac{3\pi}{5}}$

$k = 2 \Rightarrow z_2 = e^{i\pi} = -1 \Rightarrow$ Rejected.

$k = 3 \Rightarrow z_3 = e^{\left(i\frac{7\pi}{5}\right)}$

$k = 4 \Rightarrow z_4 = e^{\left(\frac{9\pi}{5}\right)}$

Q. **Let $Z = 18 + 26i$ where $z_0 = x_0 + iy_0 \left(x_0, y_0 \in R\right)$ is the cube root of Z having least positive argument . Find the value of $x_0 y_0 \left(x_0 + y_0\right)$.**

Sol. $z_0 = (18 + 26i)^{\frac{1}{3}}$ (Least positive argument)

$$z_0 = \left[10^{\frac{3}{2}}\left(\frac{18}{10^{\frac{3}{2}}} + \frac{26}{10^{\frac{3}{2}}}i\right)\right]^{\frac{1}{3}}$$

$$\Rightarrow z_0 = \sqrt{10}\left(\frac{18}{10\sqrt{10}} + \frac{26i}{10\sqrt{10}}\right)^{\frac{1}{3}}$$

$$\Rightarrow z_0 = \sqrt{10}\,(\cos\alpha + \sin\alpha)^{\frac{1}{3}}$$

Where $\cos\alpha = \frac{18}{10\sqrt{10}}$

$$\Rightarrow z_0 = \sqrt{10}\left(\cos\frac{\alpha}{3} + i\sin\frac{\alpha}{3}\right)$$

If $z_0 = x_0 + iy_0$,

$$x_0 = \sqrt{10}\cos\frac{\alpha}{3}, y_0 = \sqrt{10}\sin\frac{\alpha}{3}, \cos\alpha = \frac{18}{10\sqrt{10}}$$

$$\Rightarrow x_0y_0(x_0+y_0) = 10\sqrt{10}\sin\frac{\alpha}{3}\cos\frac{\alpha}{3}\left(\sin\frac{\alpha}{3}+\cos\frac{\alpha}{3}\right)$$

Now, from hit and trial, $\sin\frac{\alpha}{3} = \frac{1}{\sqrt{10}}$ and $\cos\frac{\alpha}{3} = \frac{3}{\sqrt{10}}$

$$\Rightarrow x_0y_0(x_0+y_0) = 10\sqrt{10}\left(\frac{1}{\sqrt{10}}\right)\left(\frac{3}{\sqrt{10}}\right)\left(\frac{1}{\sqrt{10}}+\frac{3}{\sqrt{10}}\right) = 12$$

Q. **Find the number of roots of the equation $z^{10} - z^5 - 992 = 0$ with real part -ve.**

Sol. $z^{10} - z^5 - 992 = 0$

$\Rightarrow (z^5 - 32)(z^5 + 31) = 0$

$\Rightarrow z^5 = 32$ or $z^5 = -31$

$z = 2e^{i\frac{2k\pi}{5}}$ or $z = (31)^{\frac{1}{5}}e^{i(2k+1)\frac{\pi}{5}}$

$\Rightarrow z = 2,$	$\Rightarrow z = (31)^{\frac{1}{5}}e^{i\frac{\pi}{5}},$
$= 2e^{i\frac{2\pi}{5}},$	$= (31)^{\frac{1}{5}}e^{i\frac{3\pi}{5}}$
$= 2e^{i\frac{4\pi}{5}},$	$= (31)^{\frac{1}{5}}e^{i\pi},$
$= 2e^{\left(i\frac{6\pi}{5}\right)},$	$= (31)^{\frac{1}{5}}e^{i\frac{7\pi}{5}},$
$= 2e^{\left(i\frac{8\pi}{5}\right)},$	$= (31)^{\frac{1}{5}}e^{i\frac{9\pi}{5}}$

$\Rightarrow$ z with negative real values are:

$2e^{i\frac{4\pi}{5}}, 2e^{i\frac{6\pi}{5}}, (31)^{\frac{1}{5}}e^{i\frac{3\pi}{5}}, (31)^{\frac{1}{5}}e^{i\frac{7\pi}{5}}, (31)^{\frac{1}{5}}e^{i\pi}.$

$\Rightarrow$ Total 5 roots with real part negative.

Cube Roots of unity:

$$z^3 - 1 = 0 \Rightarrow z = (1)^{1/3} = (\cos 0 + i\sin 0)^{1/3} = (\cos 2m\pi + i\sin 2m\pi)^{1/3}$$

$$= \cos\frac{2m\pi}{3} + i\sin\frac{2m\pi}{3}, m = 0,1,2$$

$m = 0, \quad z_1 = \cos 0 + i\sin 0 = 1$

$$m = 1, z_2 = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} = \frac{-1}{2} + i\frac{\sqrt{3}}{2} = \omega$$

$$m = 2, z_3 = \cos\frac{4\pi}{3}, +i\sin\frac{4\pi}{3} = \frac{-1}{2} - i\frac{\sqrt{3}}{2} = \omega^2$$

Note:

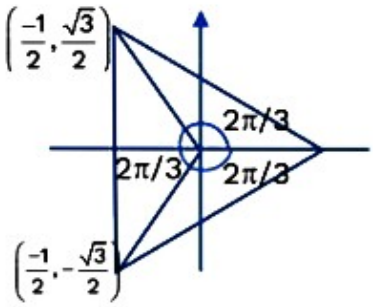
(i) $\omega^3 = 1$

(ii) $\omega^{3n} = 1, n \in I$

(iii) $1 + \omega + \omega^2 = 0$

(iv) $1 + \omega^r + \omega^{2r} =$

- 0 if r is not a multiple of 3
- 3 if r is a multiple of 3



(v) Representation of cube roots of unity on argand plane. Cube roots of unity form an equilateral Δ whose side is $\sqrt{3}$ units.

(vi) Some important facts

$$a^3 + b^3 = (a+b)\left(a^2 - ab + b^2\right)$$

$$= (a+b)\left(a^2 + \left(\omega + \omega^2\right)ab + b^2\right)$$

$$= (a+b)\left[a^2 + ab\omega + ab\omega^2 + b^2\omega^3\right]$$

$$= (a+b)\left[a(a+b\omega) + bw^2(a+b\omega)\right]$$

$$\Rightarrow (a+b)\left(a+b\omega^2\right)(a+b\omega)$$

$$a^3 - b^3 = (a-b)\left(a-b\omega^2\right)(a-b\omega)$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)\left(a+b\omega+c\omega^2\right)\left(a+b\omega^2+c\omega\right)$$

Q. **Find the sum of series**

$1\cdot(2-\omega)\left(2-\omega^2\right) + 2\cdot(3-\omega)\left(3-\omega^2\right) + \ldots\ldots + (n-1)(n-\omega)\left(n-\omega^2\right)$ **where ω is one of the imaginary cube root of unity.**

Sol. $T_n = (n-1)(n-\omega)\left(n-\omega^2\right)$

$$= (n-1)\left(n^2 - n\left(\omega+\omega^2\right) + \omega^3\right)$$

$$= (n-1)\left(n^2 + n + 1\right)$$

$$T_n = n^3 - 1$$

$$\therefore S_n = \sum_1^n T_n = \sum n^3 - \sum 1 = \left(\frac{n(n+1)}{2}\right)^2 - n$$

Q. **If $x = 1 - i\sqrt{3}, y = 1 + i\sqrt{3}$ and z = 2 then prove that $x^p + y^p = z^p$ where p is a prime number > 3.**

Sol. $x = (1 - i\sqrt{3}) = -2\left(\frac{-1 + i\sqrt{3}}{2}\right) = -2\omega$

$\therefore x^p = (-2\omega)^p = -(2)^p\omega^p$

$y = 1 + i\sqrt{3} = -2\left(\frac{-1 - i\sqrt{3}}{2}\right) = -2\omega^2$

$\therefore y^p = \left(-2\omega^2\right)^p = -(2)^p\omega^{2p}$

$x^p + y^p = -2^p\left(\omega^p + \omega^{2p}\right) = -(2)^p(-1) = 2^p$

Q. **If $(a+\omega)^{-1} + (b+\omega)^{-1} + (c+\omega)^{-1} = 2\omega^{-1}$ and $\left(a+\omega^2\right)^{-1} + \left(b+\omega^2\right)^{-1} + \left(c+\omega^2\right)^{-1} = 2\omega^{-2}$ where ω is the complex cube root of unity then show that $(a+1)^{-1} + (b+1)^{-1} + (c+1)^{-1} = 2$. (a, b, c $\in$ R)**

Sol. Let $\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} = \frac{2}{x}$

then $\frac{1}{a+x} + \frac{1}{b+x} = \frac{2}{x} - \frac{1}{c+x}$

$\Rightarrow \quad x^3 - x(ac + bc + ab) - 2abc = 0$

(Let the roots of the equation are ω_1, ω_2 and α)

If equation has three roots, then sum of roots = 0

$\therefore \quad \omega + \omega^2 + \alpha = 0$ or $\alpha = 1$

$\therefore \quad \frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = \frac{2}{1}$

Q. **Let x_1 and x_2 be the roots of the equation $x^2 - x + 1 = 0$. Compute**
(i) $x_1^{2000} + x_2^{2000}$ (ii) $x_1^{1999} + x_2^{1999}$ (iii) $x_1^n + x_2^n$ for all n$\in$N

Sol. For equation $x^2 - x + 1 = 0$

Solutions are $-\omega$ and $-\omega^2$

$x_1 = -\omega$ and $x_2 = -\omega^2$

(i) $x_1^{2000} + x_2^{2000} = (-\omega)^{2000} + \left(-\omega^2\right)^{2000} = \omega^2 + \omega = -1$

(ii) $x_1^{1999} + x_2^{1999} = (-\omega)^{1999} + \left(-\omega^2\right)^{1999}$

$\Rightarrow -\left(\omega + \omega^2\right) = +1$

(iii) $x_1^n + x_2^n = (-\omega)^n + \left(-\omega^2\right)^n, n \in N$

$n = 3\lambda \Rightarrow \pm(1+1) = \pm 2$ depending on if n is even or odd

$n = 3\lambda + 1 \Rightarrow \pm\left(\omega + \omega^2\right) = \mp 1$ depending on if n is even or odd

$n = 3\lambda + 2 \Rightarrow \pm\left(\omega^2 + \omega\right) = \mp 1$ depending on if n is even or odd

Q. **$1, \omega, \omega^2$ are cube roots of unity. Find least possible degree of polynomial with real coefficient having $2\omega^2, 3+4\omega, 3+4\omega^2, 5-\omega-\omega^2$ as roots.**

Sol. $3 + 4\omega = 3 + 3\omega + \omega = \omega - 3\omega^2$

$3 + 4\omega^2 = \omega^2 - 3\omega$

$5 - \omega - \omega^2 = 6$

Now $\overline{\omega - 3\omega^2} = \omega^2 - 3\omega$

$\therefore \omega - 3\omega^2$ and $\omega^2 - 3\omega$ will have their product and sum real.

So, only '$2\omega^2$' is left. So to neutralize it, there must be another root '$\overline{2\omega^2}$' or 2ω

$\therefore$ roots are $2\omega, 2\omega^2, \omega - 3\omega^2, \omega^2 - 3\omega$ and 6.

Since there are 5 roots, least possible degree of polynomial will also be 5.

Previous Year's Question

Let $\mathbf{z_0}$ be a root of the quadratic equation $\mathbf{x^2 + x + 1 = 0}$. If $\mathbf{z = 3 + 6iz_0^{81} - 3iz_0^{93}}$, then arg z is equal to: **(JEE Main 2019)**

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) 0 (D) $\frac{\pi}{6}$

Previous Year's Question

Let $\omega \neq 1$ be a cube root of unify. Then the minimum of the set $\left\{\left|a + b\omega + c\omega^2\right|^2 : a, b, c \text{ distinct non-zero integers}\right\}$ equals____. **(JEE Advance 2019)**

Previous Year's Question

If the four complex numbers $z, \bar{z}, \bar{z} - 2\text{Re}(\bar{z})$ and $z - 2\text{Re}(z)$ represents the vertices of a square of side 4 units in the Argand plane, then $|z|$ is equal to: **(JEE Advanced 2019)**

(A) 2 (B) 4 (C) $4\sqrt{2}$ (D) $2\sqrt{2}$

Q. **If the area of the triangle in the Argand diagram, formed by $z, \omega z$ and $z + \omega z$ where ω is the complex cube root of unity is $16\sqrt{3}$ square units, then $|z|$ is**

(A) 16 (B) 4 (C) 8 (D) 3

Sol. $z + \omega z = (1 + \omega)z = -\omega^2 z$

In figure $OA = OB = BC = AC = |z|$

$\Rightarrow$ OABC is a rhombus

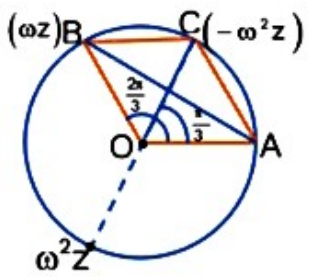
$\Rightarrow \angle ACB = \dfrac{2\pi}{3}$

Area of $\Delta ABC = \dfrac{1}{2} \times |z| \times |z| \times \sin\dfrac{2\pi}{3}$

$\Rightarrow 16\sqrt{3} = \dfrac{|z|^2\sqrt{3}}{4}$

$\Rightarrow |z|^2 = 64$

$\Rightarrow |z| = 8$



n^{th} root of unity:

$z^n - 1 = 0 \Rightarrow z = (1)^{1/n} = (\cos 0 + i\sin 0)^{1/n} = (\cos 2m\pi + i\sin 2m\pi)^{1/n}$

$\Rightarrow z = \cos\left(\dfrac{2m\pi}{n}\right) + i\sin\left(\dfrac{2m\pi}{n}\right), \quad m = 0, 1, 2, 3, \ldots\ldots, (n-1)$

$m = 0, z_1 = 1$

$m = 1, z_2 = \cos\left(\dfrac{2\pi}{n}\right) + i\sin\left(\dfrac{2\pi}{n}\right) = e^{i\frac{2\pi}{n}} = \alpha$ (Let)

$m = 2, z_3 = \cos\left(\dfrac{4\pi}{n}\right) + i\sin\left(\dfrac{4\pi}{n}\right) = e^{i\frac{4\pi}{n}} = \alpha^2$

$\vdots$

$\vdots$

$$m = n-1, z_n = \cos\left(\frac{2(n-1)\pi}{n}\right) + i\sin\left(\frac{2(n-1)\pi}{n}\right) = e^{\frac{i2(n-1)\pi}{n}} = \alpha^{n-1}$$

(1) $1, \alpha, \alpha^2, \alpha^3 + \ldots\ldots, \alpha^{n-1}$ are n, n^{th} roots of unity which are in G.P. with common ratio α where $\alpha = e^{\frac{i2\pi}{n}}$

(2) Sum of n, n^{th} roots of unity is always zero

$1 + \alpha + \alpha^2 + \alpha^3 + \ldots\ldots\alpha^{n-1} = 0$

Proof : $S = \dfrac{1-\alpha^n}{1-\alpha}$

$$\alpha = e^{\frac{i2\pi}{n}} \Rightarrow \alpha^n = \left(e^{\frac{i2\pi}{n}}\right)^n = e^{i2\pi} = 1$$

$$\therefore S = \frac{1-1}{1-\alpha} = 0$$

(3) Sum of pth powers of n, n^{th} roots of unity $= \begin{cases} n \text{ , if p is an integral multiple of n} \\ 0 \text{ , if p is not an integral multiple of n} \end{cases}$

Proof :

$$1^p + (\alpha)^p + \left(\alpha^2\right)^p + \left(\alpha^3\right)^p + \ldots\ldots + \left(\alpha^{n-1}\right)^p$$

Case-I:

When 'p' is an integral multiple of $p = n\lambda, \lambda \in I$

$$\alpha^p = \left(e^{\frac{i2\pi}{n}}\right)^p = e^{\frac{i2p\pi}{n}} = e^{i2\lambda\pi} = 1$$

$$\alpha^{2p} = 1, \alpha^{3p} = 1, \ldots\ldots\ldots, \alpha^{(n-1)p} = 1$$

$$\therefore \quad 1 + \alpha^p + \alpha^{2p} + \alpha^{3p} + \ldots\ldots + \alpha^{(n-1)p}$$

$$\Rightarrow 1 + 1 + 1 + 1 + \ldots + 1 = n$$

Case-II:

When p is not an integral multiple of n:

$$\Rightarrow \quad 1 + \alpha^p + \alpha^{2p} + \alpha^{3p} + \ldots\ldots\ldots + \alpha^{(n-1)p}$$

$$= \frac{1-\left(\alpha^p\right)^n}{1-\alpha^p} \Rightarrow \frac{1-1}{1-\alpha^p} = 0 \quad \left\{\because \alpha^{np} = 1\right\}$$

Q. **If $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α_5 are 5, 5th roots of unity then find the value of**

(i) $\alpha_1^4 + \alpha_2^4 + \alpha_3^4 + \alpha_4^4 + \alpha_5^4$ **(ii)** $\alpha_1^{20} + \alpha_2^{20} + \alpha_3^{20} + \alpha_4^{20} + \alpha_5^{20}$

Sol. **(i)** $\because$ 4 is not a multiple of 5

$$\therefore \sum_{i=1}^{5} i^4 = 0$$

(ii) 20 is multiple of 5

$$\therefore \sum_{i=1}^{5} i^{20} = 5$$

Q. **Evaluate:** $\sum_{\lambda=1}^{12}\left(\sin\frac{2\pi\lambda}{13} - i\cos\frac{2\pi\lambda}{13}\right)$

Sol. $\sum_{\lambda=1}^{12}\left(\sin\frac{2\pi\lambda}{13} - i\cos\frac{2\pi\lambda}{13}\right) \Rightarrow (-i)\sum_{\lambda=1}^{12}\left(\cos\frac{2\pi\lambda}{13} + i\sin\frac{2\pi\lambda}{13}\right)$

$$\Rightarrow (-i)\sum_{\lambda=1}^{12} e^{\frac{i2\pi\lambda}{13}} \Rightarrow (-i)\left[1 + e^{\frac{i2\pi}{3}} + e^{\frac{i4\pi}{13}} + e^{\frac{i6\pi}{13}} + \ldots\ldots + e^{\frac{i24\pi}{13}} - 1\right]$$

$\Rightarrow (-i)\,(-1) = i$

Q. **If $\alpha_1, \alpha_2, \alpha_3, \cdots \alpha_{n-1}$ are non-real n^{th} roots of unity, then prove that**

(i) $(1-\alpha_1)(1-\alpha_2)(1-\alpha_3)\ldots\ldots\ldots(1-\alpha_{n-1}) = n$

(ii) $(1+\alpha_1)(1+\alpha_2)(1+\alpha_3) + \ldots\ldots + (1+\alpha_{n-1}) = \begin{cases} 0, & \text{if n is even} \\ 1, & \text{if n is odd} \end{cases}$

Sol. **(i)** $z^n - 1 = (z-1)(z-\alpha_1)(z-\alpha_2)(z-\alpha_3)\ldots\ldots(z-\alpha_{n-1})$

$$\frac{z^n - 1}{z-1} = (z-\alpha_1)(z-\alpha_2)(z-\alpha_3)\ldots\ldots\ldots(z-\alpha_{n-1})$$

$$1 + z + z^2 + \ldots\ldots\ldots z^{n-1} = (z-\alpha_1)(z-\alpha_2)\ldots\ldots(z-\alpha_{n-1})$$

Put $z = 1$

$$\Rightarrow \underbrace{1+1+1+\ldots\ldots\ldots 1}_{\text{n times}} = (1-\alpha_1)(1-\alpha_2)\ldots\ldots\ldots(1-\alpha_{n-1})$$

Alternative method:

$$\Rightarrow (1-\alpha_1)(1-\alpha_2)\ldots(1-\alpha_{n-1}) = n$$

$$\lim_{z\to 1}\frac{z^n - 1}{z-1} = (1-\alpha_1)(1-\alpha_2)\ldots\ldots(1-\alpha_{n-1})$$

$$\Rightarrow \lim_{z\to 1}\frac{nz^n - 0}{1} = (1-\alpha_1)(1-\alpha_2)\ldots\ldots\ldots(1-\alpha_{n-1})$$

$\Rightarrow (1-\alpha_1)(1-\alpha_2).....(1-\alpha_{n-1}) = n$

(ii) $\dfrac{z^n-1}{z-1} = (z-\alpha_1)(z-\alpha_2)(z-\alpha_3).....(z-\alpha_{n-1})$

Put $z = -1$

$(-1-\alpha_1)(-1-\alpha_2)(-1-\alpha_3).....(-1-\alpha_{n-1}) = \dfrac{(-1)^n-1}{-1-1}$

$\Rightarrow (-1)^{n-1}\left[(1+\alpha_n)(1+\alpha_2)+.....(1+\alpha_{n-1})\right] = \dfrac{(-1)^n-1}{-2}$

$\Rightarrow (1+\alpha_1)(1+\alpha_2).....(1+\alpha_{n-1}) = \dfrac{(-1)^n-1}{-2}\dfrac{1}{(-1)^{n-1}} \Rightarrow \begin{cases} 0, \text{ if n is even} \\ 1, \text{ if n is odd} \end{cases}$

Q. **If z_1, z_2, z_3, z_4 & z_5 are roots of the equation $z^5+z^4+z^3+z^2+z+1=0$ then find them and also evaluate**

(i) $\sum_{i=1}^{5} z_i$ **(ii)** $\sum_{i=1}^{5} z_i^4$ **(iii)** $\sum_{i=1}^{5} z_i^{12}$ **(iv)** $\prod_{i=1}^{5}(2-z_i)$

Sol. $\left(z^5+z^4+z^3+z^3+z^2+z+1\right)(z-1) = 0$

$\Rightarrow z^6-1=0 \Rightarrow z = (1)^{1/6} \Rightarrow z = e^{\frac{i2m\pi}{6}}, m = 0,1,2,.....,5$

Let roots of the equation are $1, z_1, z_2, z_3, ... z_5$

$z_1 = e^{\frac{12\pi}{6}} = e^{\frac{i\pi}{3}}$, $z_2 = e^{\frac{i4\pi}{6}} = e^{\frac{i2\pi}{3}}$, $z_3 = e^{\frac{i6\pi}{6}} = e^{i\pi}$, $z_4 = e^{\frac{i8\pi}{6}} = e^{\frac{i4\pi}{3}}, z_5 = e^{\frac{i10\pi}{6}} = e^{\frac{i5\pi}{3}}$

(i) $1+z_1+z_2+z_3+z_4+z_5 = 0 \Rightarrow \sum_{i=1}^{5} z_i = -1$

(ii) $1^4+z_1^4+.....+z_5^4 = 0 \Rightarrow \sum_{i=1}^{5} z_i^4 = -1$ {$\because$ 4 is not a mutliple of 6}

(iii) $\sum_{i=1}^{5} z_i^{12}$

$\Rightarrow 1+z_1^{12}+z_2^{12}+.....+z_5^{12} = 6$ {$\because$ 12 is a mutliple of 6}

$\Rightarrow z_1^{12}+z_2^{12}+.....+z_5^{12} = 5$

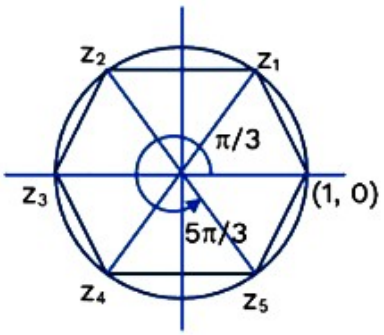
(iv) $\prod_{i=1}^{5}(2-z_i)$

$z^6-1 = (z-1)(z-z_1)(z-z_2)(z-z_3)(z-z_4)(z-z_5)$

Put $z = 2$

$\dfrac{2^6-1}{2-1} = (2-z_1)(2-z_2).....(2-z_5) \Rightarrow \prod_{i=1}^{5}(2-z_i) = 63$

Representation of all the roots of the equation on argand plane:



Q. **Factorize z^7+1 into linear and quadratic factors with real coefficients.**

Sol. $z^7+1=0$

$\Rightarrow z=(-1)^{1/7}=(\cos\pi+i\sin\pi)^{1/7}=(\cos(2m\pi+\pi)+i\sin(2m\pi+\pi))^{1/7}$

$=\cos\left(\frac{2m\pi+\pi}{7}\right)+i\sin\left(\frac{2m\pi+\pi}{7}\right)=e^{\left(\frac{2m\pi+\pi}{7}\right)},m=0,1,2,.....,6$

$z_1=e^{\frac{i\pi}{7}},z_2=e^{\frac{i3\pi}{7}},z_3=e^{\frac{i5\pi}{7}},z_4=e^{\frac{i7\pi}{7}},z_5=e^{\frac{i9\pi}{7}},z_6=e^{\frac{i11\pi}{7}},z_7=e^{\frac{i13\pi}{7}}$

$z_7=e^{\frac{i13\pi}{7}}=e^{\frac{-i\pi}{7}}=\bar{z}_1$

$z_6=e^{\frac{i11\pi}{7}}=e^{\frac{-3i\pi}{7}}=\bar{z}_2$

$z_5=e^{\frac{i9\pi}{7}}=e^{\frac{-i5\pi}{7}}=\bar{z}_3$

Now $z^7+1=(z-z_1)(z-z_2)\ldots(z-z_7)$

$=(z-z_1)(z-z_2)(z-z_3)(z-z_4)(z-\bar{z}_3)(z-\bar{z}_2)(z-\bar{z}_1)$

$=\left(z^2-z(z_1+\bar{z}_1)+|z_1|^2\right)\left(z^2-z(z_2+\bar{z}_2)+|z_2|^2\right)\left(z^2-z(z_3+\bar{z}_3)+|z_3|^2\right)(z-z_4)$

$=\left(z^2-z\cdot 2\cos\frac{\pi}{7}+1\right)\left(z^2-z\cdot 2\cos\frac{3\pi}{7}+1\right)\left(z^2-z\cdot 2\cos\frac{5\pi}{7}+1\right)(z+1)$

$=(z+1)\left(z^2-2z\cos\frac{\pi}{7}+1\right)\left(z^2-2z\cos\frac{3\pi}{7}+1\right)\left(z^2-2z\cos\frac{5\pi}{7}+1\right)$

Q. **Prove that all roots of the equation $\left(\frac{z+1}{z}\right)^n=1$ are collinear on the complex plane.**

Sol. $\left(1+\frac{1}{z}\right)^n = e^{i2k\pi}, k \in \{0,1,2,...,n-1\}$

$$\Rightarrow 1+\frac{1}{z} = e^{\frac{2k\pi}{n}i}$$

$$\Rightarrow z = \frac{1}{e^{\frac{2k\pi}{n}i}-1}$$

$$= \frac{1}{\cos\left(\frac{2k\pi}{n}\right)-1+i\sin\left(\frac{2k\pi}{n}\right)}$$

$$= \frac{-\left[1-\cos\left(\frac{2k\pi}{n}\right)+i\sin\left(\frac{2k\pi}{n}\right)\right]}{\left(\cos\left(\frac{2k\pi}{n}\right)-1\right)^2+\sin^2\left(\frac{2k\pi}{n}\right)}$$

$$= \frac{-\left[1-\cos\left(\frac{2k\pi}{n}\right)+i\sin\left(\frac{2k\pi}{n}\right)\right]}{2-2\cos\left(\frac{2k\pi}{n}\right)}$$

$$\Rightarrow z = -\frac{1}{2} - \frac{i\cot\left(\frac{k\pi}{n}\right)}{2}$$

$$\Rightarrow \text{Re}(z) = -\frac{1}{2}$$

$\therefore$, z always line on line $x = -\frac{1}{2}$.

$\therefore$All possible values of z lie on $x = -\frac{1}{2}$ and so are collinear. **Hence proved.**

Q. **If z_r, $r = 1,2,3,...2m$, $m \in N$ are the roots of the equation $z^{2m} + z^{2m-1} + z^{2m-2} + ... + z + 1 = 0$ then prove that $\sum_{r=1}^{2m} \frac{1}{z_r - 1} = -m$.**

Sol. $z^{2m} + z^{2m-1} + z^{2m-2} + ... + z + 1 = (z-z_1)(z-z_2)...(z-z_{2m})$

Taking log both side,

$$\sum_{r=1}^{2m} \log(z-z_r) = \log\left(1+z+z^2+...+z^{2m}\right)$$

Differentiating both side.

$$\sum_{r=1}^{2m}\left(\frac{1}{z-z_r}\right)=\frac{1+2z+3z^2+\ldots+2mz^{2m-1}}{1+z+z^2+\ldots+z^{2m}}$$

Putting $z = 1$ on both side.

$$\sum_{r=1}^{2m}\frac{1}{1-z_r}=\frac{1+2+3+4+\ldots+2m}{1+1+1+\ldots+(2m+1)\text{ times}}$$

$$\Rightarrow -\sum_{r=1}^{2m}\left(\frac{1}{z_r-1}\right)=\frac{(2m)(2m+1)}{2\times(2m+1)}=m$$

$$\Rightarrow \sum_{r=1}^{2m}\left(\frac{1}{z_r-1}\right)=-m. \qquad \textbf{Hence proved.}$$

Different forms of equation of a straight line:

(i) Equation of straight line with the help of coordinate geometry:

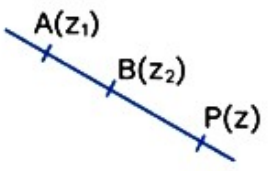
Writing $x=\frac{z+\bar{z}}{2}, y=\frac{z-\bar{z}}{2i}$ etc. in $\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}$

and re-arranging terms, we find that the equation of the line through z_1 and z_2 is given by

$$\frac{z-z_1}{z_2-z_1}=\frac{\bar{z}-\bar{z}_1}{\bar{z}_2-\bar{z}_1} \quad \text{or} \quad \begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix}=0$$

(ii) Equation of a straight line with the help of rotation formula:

Let $A(z_1)$ and $B(z_2)$ be any two points lying on any line and we have to obtain the equation of this line. For this purpose, let us take any point $P(z)$ lying on this line. Since



$$\arg\left(\frac{z-z_1}{z_2-z_1}\right)=0 \text{ or } \pi,$$

$\frac{z-z_1}{z_2-z_1}$ is purely real.

$$\Rightarrow \quad \frac{z-z_1}{z_2-z_1}=\frac{\bar{z}-\bar{z}_1}{\bar{z}_2-\bar{z}_1}$$

Condition for which two lines which are parallel or perpendicular:

(a) For parallel:

Since AB and CD lines are parallel

$$\therefore \arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = 0 \text{ or } \pi \Rightarrow \frac{z_1 - z_2}{z_3 - z_4}$$

is purely real $\Rightarrow \frac{z_1 - z_2}{z_3 - z_4} = \frac{\bar{z}_1 - \bar{z}_2}{\bar{z}_3 - \bar{z}_4}$



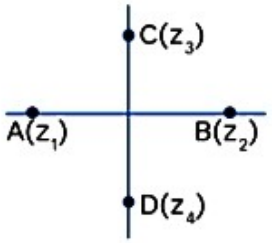
(b) For perpendicular:

Since AB and CD lines are perpendicular

$$\arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = \pm\frac{\pi}{2}$$

$\Rightarrow \frac{z_1 - z_2}{z_3 - z_4}$ is purely imaginary

$$\Rightarrow \frac{z_1 - z_2}{z_3 - z_4} = -\left(\frac{\bar{z}_1 - \bar{z}_2}{\bar{z}_3 - \bar{z}_4}\right)$$



(c) General equation of the line:

From equation (i) we get,

$$z(\bar{z}_2 - \bar{z}_1) - z_1\bar{z}_2 + z_1\bar{z}_1 = \bar{z}(z_2 - z_1) - \bar{z}_1 z_2 + z_1\bar{z}_1$$

$$\Rightarrow z(\bar{z}_2 - \bar{z}_1) + \bar{z}(z_1 - z_2) + \bar{z}_1 z_2 - z_1\bar{z}_2 = 0$$

Here $\bar{z}_1 z_2 - z_1\bar{z}_2$ is a purely imaginary number

as $\bar{z}_1 z_2 - z_1\bar{z}_2 = 2i\,\text{Im}(\bar{z}_1 z_2)$.

Let $\bar{z}_1 z_2 - z_1\bar{z}_2 = ib,\ b \in R$

$$\Rightarrow z(\bar{z}_2 - \bar{z}_1) + \bar{z}(z_1 - z_2) + ib = 0$$

$$\Rightarrow zi(\bar{z}_1 - \bar{z}_2) + \bar{z}i(z_2 - z_1) + b = 0$$

Let $a = i(z_2 - z_1) \Rightarrow \bar{a} = i(\bar{z}_1 - \bar{z}_2)$

$$\Rightarrow \bar{a}z + a\bar{z} + b = 0$$

This is the general equation of a line in the complex plane.

Complex slope of a line:

(a) Complex slope of a line $\bar{a}z + a\bar{z} + b = 0$ is defined as

$$\omega = -\frac{a}{\bar{a}}$$

(b) Complex slope of a line passing through $A(z_1)$ & $B(z_2)$ is given by $\omega = \dfrac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$

Note:

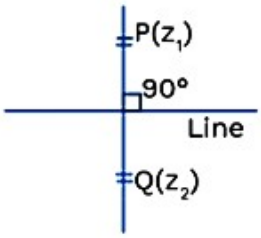
Let ω_1 and ω_2 be the complex slopes of two lines.

(i) If $\omega_1 = \omega_2$ then lines are parallel

(ii) If $\omega_1 + \omega_2 = 0$ then lines are perpendicular

Reflection Points for A Line (Image of a point in a line):

Two given points P & Q denoted by complex numbers z_1 and z_2 respectively are the image of each other in given straight line $\bar{\alpha}z + \alpha\bar{z} + r = 0$ the given line is the right bisector of the line segment PQ. Two points $P(z_1)$ & $Q(z_2)$ will be the reflection points of each other in the given straight line, iff $\bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$ where 'r' is real and α is non-zero complex constant.



Proof:

M is the mid-point of PQ which lies on the line mirror

$$\therefore \quad \bar{\alpha}\left(\frac{z_1 + z_2}{2}\right) + \alpha\left(\frac{\bar{z}_1 + \bar{z}_2}{2}\right) + r = 0$$

$$\bar{\alpha}z_1 + \bar{\alpha}z_2 + \alpha\bar{z}_1 + \alpha\bar{z}_2 + 2r = 0 \qquad \ldots(i)$$

Line PQ and the given line both are perpendicular.

$$\therefore \quad \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} + \left(-\frac{\alpha}{\bar{\alpha}}\right) = 0$$

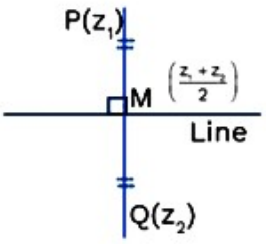
$$z_1\bar{\alpha} - z_2\bar{\alpha} = \alpha\bar{z}_1 - \alpha\bar{z}_2$$

$$\alpha\bar{z}_1 + \bar{\alpha}z_2 = \alpha\bar{z}_2 + \bar{\alpha}z_1 \qquad \ldots(ii)$$

From (i) & (ii)

$$2(\bar{\alpha}z_2 + \alpha\bar{z}_1) + 2r = 0$$

$$\Rightarrow \quad \alpha\bar{z}_1 + \bar{\alpha}z_2 + r = 0$$



Q. **Find the image of the point P(1–i) in the line mirror** $(1+i)z-(1-i)\bar{z}+2i=0$

Sol. The given line mirror is

$(1+i)z-(1-i)\bar{z}+2i=0$

$(-1+i)z-(1+i)\bar{z}-2=0$

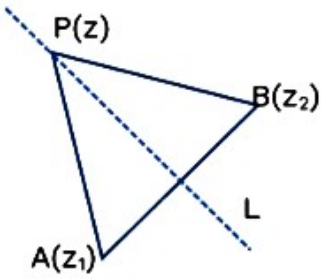
Let z_2 be the image of the given point P(1–i)

$\Rightarrow \quad (1-i)z_2+(1+i)(1+i)+2=0 \quad \Rightarrow \quad (1-i)z_2+1-1+2i+2=0$

$\Rightarrow \quad z_2=\dfrac{-2(1+i)}{1-i}=\dfrac{-2(1-1+2i)}{2}=-2i$

Equation of perpendicular bisector:

Consider a line segment joining $A(z_1)$ and $B(z_2)$.
Let the line 'L' be its perpendicular bisector.
If P(z) be any point on the 'L'. we have



$PA=PB \Rightarrow |z-z_1|=|z-z_2|$

$\Rightarrow \quad |z-z_1|^2=|z-z_2|^2$

$\Rightarrow \quad (z-z_1)(\bar{z}-\bar{z}_1)=(z-z_2)(\bar{z}-\bar{z}_2)$

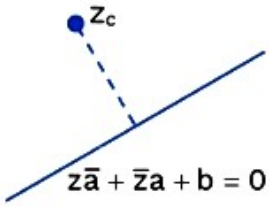
$\Rightarrow \quad z\bar{z}-z\bar{z}_1-z_1\bar{z}+z_1\bar{z}_1=z\bar{z}-z\bar{z}_2-z_2\bar{z}+z_2\bar{z}_2$

$\Rightarrow \quad z(\bar{z}_2-\bar{z}_1)+\bar{z}(z_2-z_1)+z_1\bar{z}_1-z_2\bar{z}_2=0$

Distance of a given point from a given line:

Let the given line be $z\bar{a}+\bar{z}a+b=0$ and the given point be z_c

Say $z_c=x_c+iy_c$



Replacing z by x + iy, in the given equation, we get, $x(a+\bar{a})+iy(\bar{a}-a)+b=0$

Distance of (x_c, y_c) from this line is,

$$\frac{|x_c(a+\bar{a})+iy_c(\bar{a}-a)+b|}{\sqrt{(a+\bar{a})^2-(a-\bar{a})^2}}=\frac{|z_c\bar{a}+\bar{z}_ca+b|}{\sqrt{4(Re(a))^2+4(im(a))^2}}=\frac{|z_c\bar{a}+\bar{z}_ca+b|}{2|a|}$$

Q. **Consider a line** $(1+i)z-(1-i)\bar{z}+2i=0$**. Find the distance of a point 2i from the above line.**

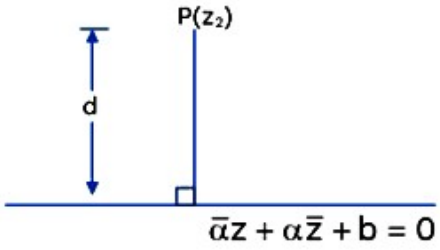
Sol. The given line can be written as

$(1-i)z + (1+i)\bar{z} + 2 = 0$

Applying distance formula

$$d = \frac{|z_c\bar{\alpha} + \bar{z}_c\alpha + b|}{2|\alpha|}$$

$$= \left|\frac{(1-i)2i + (1+i)(-2i) + 2}{2\sqrt{2}}\right| = \left|\frac{2i + 2 - 2i + 2 + 2}{2\sqrt{2}}\right| = \frac{3}{\sqrt{2}}$$



Q. **Find the equation of a line on complex plane which passes through a point A denoted by complex number α and is perpendicular to the vector $\overline{OA}$.**

Sol. Vector perpendicular to $\overline{OA} = \alpha e^{i\frac{\pi}{2}} = \alpha i$

Since it passes through α and equation of line is : $z = \alpha + \lambda(\alpha i - \alpha)$ (parametric form)

Q. **Find the area bounded by the curves**

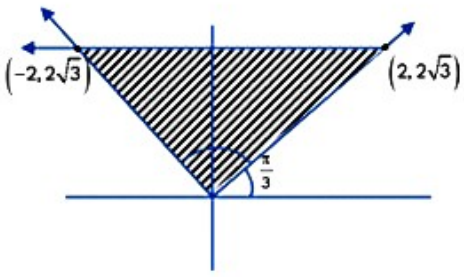
$\text{Arg}\,z = \frac{\pi}{3}$, $\text{Arg}\,z = \frac{2\pi}{3}$ & $\text{Arg}\left(z - 2 - 2\sqrt{3}i\right) = \pi$ on the complex plane.

Sol. $\text{Arg}(z) = \frac{\pi}{3} \Rightarrow$ Line $y = \sqrt{3}x$ in Ist quadrant

$\text{Arg}(z) = \frac{2\pi}{3} \Rightarrow$ Line $y = -\sqrt{3}x$ in IInd quadrant

Also $2 + 2\sqrt{3}i$ lies on $y = \sqrt{3x}$

$\text{Area} = \frac{1}{2} \times \left(2\sqrt{3}\right) \times 4 = 4\sqrt{3}$



Q. **Find all the points in the complex plane which satisfy the equations**

$\log_5(|z| + 3) - \log_{\sqrt{5}}|z| - 1| = 1$ and $\arg(z - 1 - i) = \frac{\pi}{4}$

Sol. $\arg(z - 1 - i) = \frac{\pi}{4}$ ...(i)

Let $z = x + yi$

$\Rightarrow z - 1 - i = (x - 1) + (y - 1)i$

$\Rightarrow \frac{y-1}{x-1} = 1$ (from (i))

$\Rightarrow x = y$

So $z = x(1+i), x > 1$

$\Rightarrow |z| = \sqrt{2}x$

$\log_5(|z|+3) - \log_{\sqrt{5}}||z|-1| = 1$

$\Rightarrow \log_5(\sqrt{2}x+3) - 2\log_5|\sqrt{2}x-1| = 1$

$\Rightarrow \frac{\sqrt{2}x+3}{(\sqrt{2}x-1)^2} = 5$

$\Rightarrow \sqrt{2}x+3 = 5(2x^2 - 2\sqrt{2}x+1)$

$\Rightarrow 10x^2 - 11\sqrt{2}x + 2 = 0$

$\Rightarrow 10x^2 - 10\sqrt{2}x - \sqrt{2}x + 2 = 0$

$\Rightarrow (10x-\sqrt{2})(x-\sqrt{2}) = 0$

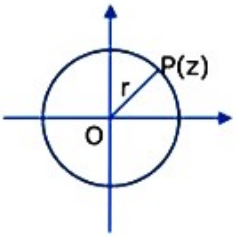
$\Rightarrow x = \frac{\sqrt{2}}{10}$ (rejected) or $x = \sqrt{2}$

$\Rightarrow x = \sqrt{2}$

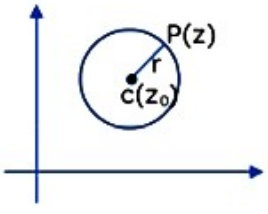
$\Rightarrow z = \sqrt{2}(1+i)$

Different forms of equation of a circle:

(i) $|z| = r$, $r \in R^+$ then locus of z represents a circle whose center is the origin and radius is equal to r.



(ii) $|z - z_0| = r$, $r \in R^+$ then locus of z represents a circle whose center is z_0 and radius is equal to r .



(iii) Equation $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$ represent a circle whose center is $-a$ and radius is equal $\sqrt{|a|^2 - b}$.

Equation of the circle with center z_0 and

radius 'r'.

$|z - z_0| = r$

$\Rightarrow |z - z_0|^2 = r^2$

$\Rightarrow (z - z_0)(\bar{z} - \bar{z}_0) = r^2$

$\Rightarrow z\bar{z} - z\bar{z}_0 - \bar{z}z_0 + |z_0|^2 - r^2 = 0$

Putting $-z_0 = a$ and $|z_0|^2 - r^2 = b$ equation

becomes

$z\bar{z} + \bar{a}z + a\bar{z} + b = 0$

$\therefore$ Center $z_0 = -a$ and radius $r = \sqrt{|z_0|^2 - b} = \sqrt{|a|^2 - b}$

Q. **Find the center and radius of the circle.**

$z\bar{z} - (3 - 4i)z - (3 + 4i)\bar{z} + 9 = 0$

Sol. $z\bar{z} + (-3 + 4i)z + (-3 - 4i)\bar{z} + 9 = 0$

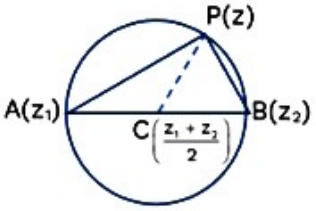
Here, $a = -3 - 4i$, $b = 9$

$\therefore$ center $= -a = 3 + 4i$ or $(3, 4)$

radius $r = \sqrt{|a|^2 - b} = \sqrt{|3 + 4i|^2 - 9} = \sqrt{25 - 9} = 4$

(iv) Diametric form of equation of circle:

Let $A(z_1)$ and $B(z_2)$ are the extremities of diameter of a circle and $P(z)$ be a variable point then



(a) CP = r (r = radius of the circle)

$$\left|z - \frac{z_1 + z_2}{2}\right| = \left|\frac{z_1 - z_2}{2}\right|$$

(b) AP & BP are perpendicular

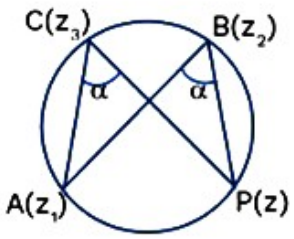
$$\therefore \omega_1 + \omega_2 = 0 \Rightarrow \frac{z - z_1}{\bar{z} - \bar{z}_1} + \frac{z - z_2}{\bar{z} - \bar{z}_2} = 0$$

(c) $AP^2 + BP^2 = AB^2$

$$|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$$

(v) Equation of circle passing through three non-collinear points:

Let $A(z_1), B(z_2)$ and $C(z_3)$ be three given non-collinear points and P(z) be a variable point. Angle subtended by the arc AP at B and C are equal (say 'α')



Using rotation theorem:

$$\frac{z - z_2}{z_1 - z_2} = \lambda_1 e^{i\alpha} \quad \ldots(i)$$

$$\frac{z - z_3}{z_1 - z_3} = \lambda_2 e^{i\alpha} \quad \ldots(ii)$$

(i) ÷ (ii)

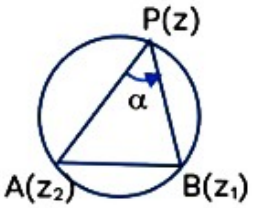
$$\frac{z - z_2}{z_1 - z_2} \cdot \frac{z_1 - z_3}{z - z_3} = \frac{\lambda_1}{\lambda_2} = \text{Purely real}$$

$$\therefore \left(\frac{z - z_2}{z - z_3}\right)\left(\frac{z_1 - z_3}{z_1 - z_2}\right) = \left(\frac{\bar{z} - \bar{z}_2}{\bar{z} - \bar{z}_3}\right)\left(\frac{\bar{z}_1 - \bar{z}_3}{\bar{z}_1 - \bar{z}_2}\right)$$

The above equation represents a circle passing through $A(z_1), B(z_2)$ and $C(z_3)$.

Important Note:

Let z_1 and z_2 be two given complex numbers and z be any complex number such that $\arg\left(\frac{z - z_1}{z - z_2}\right) = \alpha$, where $\alpha \in (0, \pi)$



Then 'z' would lie on an arc of segment of a circle on z_1z_2, containing angle α. Clearly if $\alpha \in \left(0, \frac{\pi}{2}\right)$, z would lie on the major arc (excluding the points z_1 and z_2) and if $\alpha \in \left(\frac{\pi}{2}, \pi\right)$. 'z' would lie on the minor arc (excluding the points z_1 and z_2).

Q. If $z_1 = 2 + 3i, z_2 = 3 - 2i$ and $z_3 = -1 - 2\sqrt{3}i$ **then which of the following is true?**

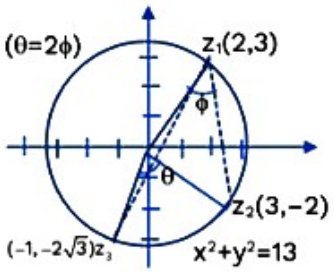
(A) $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$ **(B)** $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_2}{z_1}\right)$

(C) $\arg\left(\frac{z_3}{z_2}\right) = 2\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$ **(D)** $\arg\left(\frac{z_3}{z_2}\right) = \frac{1}{2}\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$

Sol. **(C)**

Note that $|z_1| = |z_2| = |z_3| = \sqrt{13}$

Hence z_1, z_2, z_3 lies on a circle with centre (0,0) and $r = \sqrt{13}$ as shown



now $\text{Arg}\frac{z_2}{z_3} = 2\text{Arg}\frac{z_2 - z_1}{z_3 - z_1}$

$\therefore \quad \text{Arg}\frac{z_3}{z_2} = 2\,\text{Arg}\frac{z_3 - z_1}{z_2 - z_1}$

Condition of concyclic of four points:

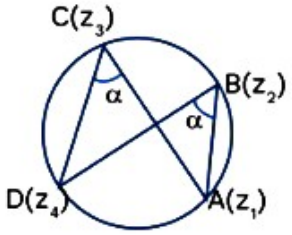
Four points $A(z_1), B(z_3), C(z_3)$ and $D(z_4)$ taken in order are concyclic then

$$\frac{z_1 - z_2}{z_4 - z_2} = \lambda_1 e^{i\alpha} \quad \ldots(i)$$

$$\frac{z_1 - z_3}{z_4 - z_3} = \lambda_2 e^{i\alpha x} \quad \ldots(ii)$$

From (i) & (ii)

$\frac{z_1 - z_2}{z_4 - z_2} \cdot \frac{z_4 - z_3}{z_1 - z_3}$ = Purely real (which is the required condition)

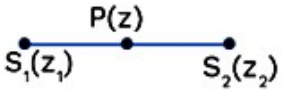


Inverse Point w.r.t. a circle:

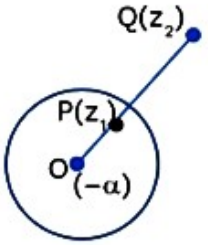
Two points $P(z_1)$ & $Q(z_2)$ are said to be the inverse point of each other w.r.t. the circle if

(i) O,P,Q are collinear and lie on the same side of the center 'O' and

(ii) $(OP) \cdot (OQ) = \rho^2$ where ρ is the radius of the circle.

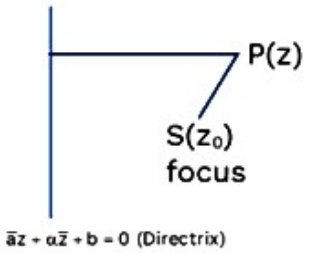


The points z_1 and z_2, will be the inverse point of each other
w.r.t the circle $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$
if $z_1\bar{z}_2 + \bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$

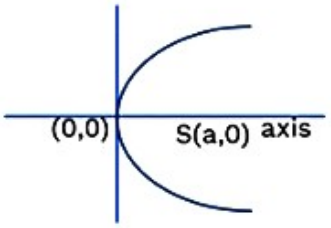


General locii on complex plane:

(a) **(i)** If $|z - z_0| = \left|\frac{\bar{\alpha}z + \alpha\bar{z} + b}{2|\alpha|}\right|$ then locus of z is a parabola whose focus is z_0 and directrix is the line $\bar{\alpha}z + \alpha\bar{z} + b = 0$ provided $\bar{\alpha}z_0 + \alpha\bar{z}_0 + b \neq 0$



(ii) If $(z - \bar{z})^2 + 8a(z + \bar{z}) = 0$, the locus of z represents a parabola whose focus is (a, 0), vertex is (0,0) and real axis is the axis of parabola.



Proof: Put $z = x + iy$

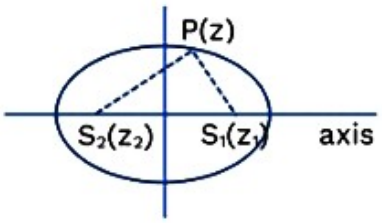
$\Rightarrow (i2y)^2 + 8a.2x = 0$

$\Rightarrow -4y^2 + 16ax = 0$

$\Rightarrow y^2 = 4ax$

(b) $|z - z_1| + |z - z_2| = k$

(i) $k > |z_1 - z_2|$ then locus of z represents an ellipse whose foci are z_1 and z_2.



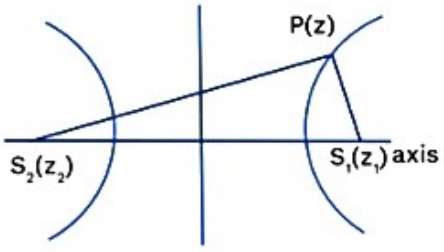
(ii) $k = |z_1 - z_2|$ then locus of z is line segment joining z_1 and z_2.

$PS_1 + PS_2 = S_1S_2 = |z_1 - z_2| \Rightarrow$ P lies on the line segement joining $S_1(z_1)$ and $S_2(z_2)$

(c) $||z - z_1| - |z - z_2|| = k$

(i) $k < |z_1 - z_2|$ then locus of z represents a hyperbola whose foci are z_1 and z_2.

$||z - z_1| - |z - z_2| < |z_1 - z_2|$



(ii) $k = |z_1 - z_2|$ then locus of z is union of two rays emanating from z_1 and z_2



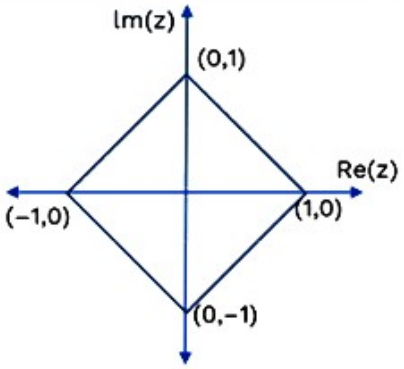
Q. **Number of complex numbers satisfying the relation** $|z+\bar{z}|+|z-\bar{z}|=2$ **and** $|z+i|+|z-i|=2$, **is**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol. We have $\left|\frac{z+\bar{z}}{2}\right|+\left|\frac{z-\bar{z}}{2}\right|=1 \Rightarrow |x|+|y|=1$

Also $|z-i|+|z+i|=2$

$\Rightarrow$ A line segment between (0,1) and (0,-1)
So, number of solution is 2
i.e., z = i and –i



LOGARITHM OF COMPLEX QUANTITY

Let u and z be two complex number such that $u = e^z$ where $u = \alpha + i\beta$ and $z = x + iy$ or $\alpha + i\beta = e^{x+iy}$ then z is called the logarithm of u to the base e

$z = \log_e u$
or $x + iy = \log_e(\alpha + i\beta)$
Now $\alpha + i\beta = e^{x+iy} = e^x \cdot e^{iy} = e^x(\cos y + i \sin y)$

Hence corresponding to some values of x and y there is one and only one value of $\alpha+i\beta$. Now we will prove that for a given value of u, there can be infinite values of z.

Here $z = \log u$
Or $x + iy = \log(\alpha+i\beta)$
$= \log(r(\cos\theta + i\sin\theta))$

[where $\alpha = r\cos\theta, \beta = r\sin\theta$ and $r = \sqrt{(\alpha^2+\beta^2)}, \theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$]

$= \log r\{\cos(\theta + 2n\pi) + i\sin(\theta + 2n\pi)\}$

$= \log re^{i(\theta+2n\pi)}$, where $n \in I$

$= \log(r + i(\theta + 2n\pi))$

$= \log\sqrt{(\alpha^2 + \beta^2)} + i(2n\pi + \tan^{-1}\beta/\alpha)$ ...(1)

From (1) it is clear that for different values of n, we have different values of x+iy Hence logarithim of a complex quantity is a multi-valued function and it is expressed as log u i.e.,

$\log u = \log(\alpha + i\beta) = \frac{1}{2}\log(\alpha^2 + \beta^2) + i\tan^{-1}\beta/\alpha + 2n\pi i$

If we put n=0, we obtain the principal value of u.

Hence the principal value of $\log u = \frac{1}{2}\log(\alpha^2 + \beta^2) + i\theta$ and $\theta(= \tan^{-1}\beta/\alpha)$ should be such that $-\pi < \theta \le \pi$

Q. **Separate into real and imaginary parts the following:**
(i) log(1 + i) **(ii) log(−5)**

Sol. **(i)** Let $1 = r\cos\theta$ and $1 = r\sin\theta$

then $r = \sqrt{1^2 + 1^2} = \sqrt{2}$

and $\tan\theta = 1$ or $\theta = \pi/4$

$\therefore \log(1 + i) = \log(r\cos\theta + i\sin\theta)$

$= \log r(\cos\theta + i\sin\theta)$

$= \log\sqrt{2}(\cos\pi/4 + i\sin\pi/4)$

$= \log(\sqrt{2}e^{i\pi/4})$

$= \log\sqrt{2} + \log e^{i\frac{\pi}{4}}$

$= \frac{1}{2}\log 2 + \frac{1}{4}i\pi$

Here $\log(1 + i) = \frac{1}{2}\log 2 + \frac{1}{4}i\pi + 2n\pi i$

$= \frac{1}{2}\log 2 + i(2n\pi + \pi/4)$, $n \in I$

(ii) $\log(-5) = \log(-5) + 2n\pi i$

Now $\log(-5) = \log(-5 + i.0)$

$= \log(r\cos\theta + ir\sin\theta)$

Where $r\cos\theta = -5, r\sin\theta = 0$

$\therefore \quad r = \sqrt{\left[(-5)^2 + 0^2\right]} = 5, r\sin\theta = 0$

$\therefore \quad \log(-5) = \log 5(\cos\pi + i\sin\pi)$

$= \log\left(5e^{i\pi}\right)$

$= \log 5 + i\pi$

$\log(-5) = \log 5 + i\pi + 2n\pi i$

$= \log 5 + i(2n+1)\pi$

Q. **If** $\tan\log(x+iy) = a+ib$ and $a^2+b^2 \neq 1$, **then prove that** $\tan\log\left(x^2+y^2\right) = \dfrac{2a}{1-a^2-b^2}$.

Sol. Here $\tan\log(x+iy) = a+ib$

$\therefore \log(x+iy) = \tan^{-1}(a+ib)$ ...(1)

and $\log(x-iy) = \tan^{-1}(a-ib)$ ...(2)

$\therefore \log(x+iy) + \log(x-iy) = \tan^{-1}(a+ib) + \tan^{-1}(a-ib)$

Or $\log\{(x+iy)(x-iy)\} = \tan^{-1}\left\{\dfrac{(a+ib)+(a-ib)}{1-(a+ib)(a-ib)}\right\}$

Or $\log\left(x^2+y^2\right) = \tan^{-1}\left\{\dfrac{2a}{1-a^2-b^2}\right\}$

Or $\tan\log\left(x^2+y^2\right) = \dfrac{2a}{1-a^2-b^2}$

Q. **Prove that :** $i^i = \exp\{-(4n+1)\pi/2\}$

Sol. $i^i = \exp(i\log i)$

$= \exp\{i(\log i + 2n\pi i)\}$

$= \exp[i\{2n\pi i + \log(0+1.i)\}]$

$= \exp[i\{2n\pi i + \log(\cos\pi/2 + i\sin\pi/2)\}]$

$= \exp\left[i\left(2n\pi i + \log e^{(i\pi)/2}\right]\right.$

$= \exp\{i(2n\pi i + i\pi/2)\} = \exp(-2n\pi - \pi/2)$

$= \exp\{-(4n+1)\pi/2\}$ Where $n \in I$

Q. **If $i^{i^{i^{\infty}}} = A + iB$ (Principal values only being considered) then prove that $\tan\left(\frac{\pi A}{2}\right) = \frac{B}{A}$ and $A^2 + B^2 = e^{-B\pi}$**

Sol. Here $i^{i^{i^{\infty}}} = A + iB$

$\therefore \; i^{A+iB} = A + iB$

$\Rightarrow \; (A + iB)\log\{\cos(\pi / 2) + i\sin(\pi / 2)\} = \log(A + iB)$

$\Rightarrow \; (A + iB)\log e^{(i\pi 2)} = \frac{1}{2}\log\left(A^2 + B^2\right) + i\tan^{-1}(B / A)$

$\Rightarrow \; (A + iB)(i\pi / 2) = \frac{1}{2}\log\left(A^2 + B^2\right) + i\tan^{-1}(B / A)$

Equating real and imaginary parts of both sides

$-\left(\frac{B\pi}{2}\right) = \frac{1}{2}\log\left(A^2 + B^2\right)$

$\Rightarrow \; A^2 + B^2 = e^{-B\pi}$

And $A\pi / 2 = \tan^{-1}(B / A) \Rightarrow B / A = \tan(A\pi / 2)$

$\Rightarrow \; \tan\frac{A\pi}{2} = \frac{B}{A}.$ **Hence proved.**

Q. **If $\sin(A + iB) = x + iy$, where $A, B, x, y \in R$ prove that $\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = 1$**

Sol. $\sin(A + iB) = x + iy$ ...(i)

$\Rightarrow \sin(A - iB) = x - iy$ ...(ii)

(i) + (ii)

$\Rightarrow 2\sin A\cos(iB) = 2x \quad \Rightarrow \frac{x}{\sin A} = \cos(iB)$

(i)– (ii).

$2\sin(iB)\cos A = 2iy$

$\Rightarrow \frac{y}{\cos A} = \frac{\sin(iB)}{i} = -i\sin(iB)$

$\Rightarrow \frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = \left(\cos(iB)\right)^2 - i^2\left(\sin(iB)\right)^2$

$= \cos^2(iB) + \sin^2(iB) = 1$ **Hence proved.**

Q. **Find the ampz if $z = (1+i)^i$.**

Sol. $z = (1+i)^i$

$$\Rightarrow z = \left(\sqrt{2}e^{i\frac{\pi}{4}}\right)^i$$

$$\Rightarrow z = \left(\sqrt{2}\right)^i e^{-\frac{\pi}{4}}$$

$$\Rightarrow z = e^{-\frac{\pi}{4}} e^{i\ell n\left(\sqrt{2}\right)}$$

$$\Rightarrow z = e^{-\frac{\pi}{4}}\left(\cos\left(\ell n\sqrt{2}\right) + i\sin\left(\ell n\sqrt{2}\right)\right)$$

$$\Rightarrow \text{Amp}(z) = \ell n\sqrt{2} = \frac{1}{2}\ell n2$$

Miscellaneous Questions

Q. **The complex number sinx + icos 2x and cosx –isin2x are conjugate to each other, for**

(A) $x = n\pi$ **(B) $x = 0$**

(C) $x = \left(n + \frac{1}{2}\right)\pi$ **(D) no value of x (where $n \in I$)**

Sol. Since, both complex numbers are same,
sinx = cosx and cos2x = –(–sin2x) = sin2x
Solving both equation, tanx =1 and tan2x = 1 at the same time which is not possible.
$\Rightarrow x \in \phi$

Q. **The equation $z^2 + \alpha z + \beta = 0$, where α, β are complex numbers has a**

(A) real root, if $(\bar{\alpha} - \alpha)(\alpha\bar{\beta} - \bar{\alpha}\beta) = (\beta - \bar{\beta})^2$

(B) real root if $(\bar{\alpha} - \alpha)(\alpha\bar{\beta} + \bar{\alpha}\beta) = (\beta - \bar{\beta})^2$

(C) a purely imaginary root if $(\alpha - \bar{\alpha})(\alpha\bar{\beta} + \bar{\alpha}\beta) + (\beta - \bar{\beta})^2 = 0$

(D) a purely imaginary root if $(\alpha + \bar{\alpha})(\alpha\bar{\beta} - \bar{\alpha}\beta) + (\beta - \bar{\beta})^2 = 0$

Sol. **(A)**

If z is real,
$z^2 + \alpha z + \beta = 0$ ...(1)

$\bar{z}^2 + \overline{\alpha z} + \bar{\beta} = 0$

$z^2 + \bar{\alpha} z + \bar{\beta} = 0$ ...(2)

(1)–(2) $\Rightarrow z = \dfrac{\bar{\beta} - \beta}{\alpha - \bar{\alpha}}$

Put in (1) we get $(\beta - \bar{\beta})^2 = (\bar{\alpha} - \alpha)(\alpha\bar{\beta} - \bar{\alpha}\beta)$

(C) If z is purely imaginary,

$z^2 + \alpha z + \beta = 0$ ...(1)

$z^2 - \bar{\alpha} z + \bar{\beta} = 0$ ...(2)

(1)–(2) $\Rightarrow z = \dfrac{\bar{\beta} - \beta}{\alpha + \bar{\alpha}}$

Put in (1) $(\bar{\beta} - \beta)^2 + \alpha(\bar{\beta} - \beta)(\alpha + \bar{\alpha}) + \beta(\alpha + \bar{\alpha})^2 = 0$

$\Rightarrow (\beta - \bar{\beta})^2 + (\alpha + \bar{\alpha})(\alpha\bar{\beta} + \overline{\alpha\beta}) = 0$

Q. **If the imaginary part of the expression $\dfrac{z - e}{e^{i\theta}} + \dfrac{e^{i\theta}}{z - 1}$ be zero, $z \in C$, $\theta \in R$, then**

(A) $|Z|_{min} = 1$ **(B)** $|Z|_{max} = 3$ **(C)** $|Z|_{min} = 0$ **(D)** $|Z|_{max} = 2$

Sol. Let $Z = \dfrac{z - 1}{e^{i\theta}}$,

$Z + \dfrac{1}{Z} = \bar{Z} + \dfrac{1}{\bar{Z}}$, (Given)

$\Rightarrow (Z - \bar{Z})\left(1 - \dfrac{1}{|Z|^2}\right) = 0$

$\Rightarrow |Z| = 1$

$\Rightarrow \left|\dfrac{z - 1}{e^{i\theta}}\right| = 1$

$\Rightarrow |z - 1| = 1$

It represents a circle with center at (1,0) and radius 1.

Q. **If $z_1 = a + ib$ and $z_2 = c + id$ are complex number such that $|z_1| = |z_2| = 1$ and $\mathrm{Re}\left(z_1\bar{z}_2\right) = 0$, then the pair of complex numbers $\omega_1 = a + ic$ and $\omega_2 = b + id$ satisfies:**

(A) $|\omega_1| = 1$ **(B)** $|\omega_2| = 1$ **(C)** $\mathrm{Re}(\omega_1\bar{\omega}_2) = 0$ **(D)** $\mathrm{Im}(\omega_1\bar{\omega}_2) = 0$

Sol. $z_1 = e^{i\theta_1}, z_2 = e^{i\theta_2}$

$\mathrm{Re}\left(z_1\bar{z}_2\right) = 0 \Rightarrow \cos(\theta_1 - \theta_2) = 0$

$$\theta_1 - \theta_2 = (2n+1)\frac{\pi}{2}, n \in I$$

$$|\omega_1|^2 = a^2 + c^2 = \cos^2\theta_1 + \cos^2\theta_2$$

$$= \cos^2\theta_1 + \cos^2\left(\theta_1 - (2n+1)\frac{\pi}{2}\right)$$

$$= \cos^2\theta_1 + \sin^2\theta_1 = 1$$

$$|\omega_2|^2 = b^2 + d^2 = \sin^2\theta_1 + \sin^2\theta_2 = \sin^2\theta_1 + \sin^2\left(\theta_1 - (2n+1)\frac{\pi}{2}\right)$$

$$= \sin^2\theta_1 + \cos^2\theta_1 = 1$$

$$\mathrm{Re}\left(\omega_1\bar{\omega}_2\right) = ab + cd = \frac{1}{2}\left(\sin 2\theta_1 + \sin 2\theta_2\right)$$

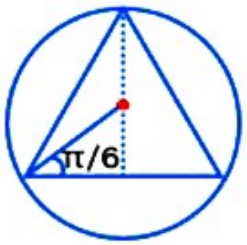
$$= \frac{1}{2}\left(\sin 2\theta_1 + \sin\left(2\theta_1 - (2n+1)\pi\right)\right) = 0$$

$$\mathrm{Im}\left(\omega_1\bar{\omega}_2\right) = bc - ad = \sin\left(\theta_1 - \theta_2\right) = \sin\left((2n+1)\frac{\pi}{2}\right) = 1$$

Q. **Find the perimeter of the triangle whose vertices are the roots of the equation $(z+\alpha\beta)^3 = \alpha^3$, where α, β are given complex numbers**

(A) $3\sqrt{3}\,|\beta|$ **(B)** $\sqrt{3}\,|\alpha|$ **(C)** $3|\alpha|$ **(D)** $3\sqrt{3}\,|\alpha|$

Sol. If we shift the origin to $(-\alpha\beta, 0)$, we will observe that equation becomes of the form $Z^3 = \alpha^3$.
Roots are vertices of equilateral triangle lying on circle with radius $|\alpha|$



Length of side $= \left(2|\alpha|\cos\frac{\pi}{6}\right) = \sqrt{3}\,|\alpha|$

Q. **Let A = {z : (1 + 2i) $\bar{z}$ + (1 − 2i) z + 2 = 0} and B = {z : (3 + 2i) $\bar{z}$ + (3−2i)z + 3=0}, then**

(A) $n(A \cap B) = 0$ **(B)** $A \subseteq B$ **(C)** $n(A \cap B) = 1$ **(D)** $B \subseteq A$

Sol. We can observe that A and B sets represents two non-parallel lines.
Hence, number of intersection points = 1.

Q. **The number of solutions of the equation $z^{2017} = p\bar{z}$, where p is a non-zero give complex number is equal to**

(A) 2017 **(B) 2018** **(C) 2019** **(D) 2020**

Sol. $|Z|^{2017} = |p|\ |\bar{z}|$

$\Rightarrow$ $|z| = 0$ or $|z| = |p|^{\frac{1}{2016}}$

$\therefore$ $z^{2018} = p|z|^2$

$\Rightarrow$ $z^{2018} = 0$ or $|p|^{\frac{1}{2008}}p$

$\Rightarrow$ Number of solutions = 1 + 2018 = 2019.

Q. **If the lines $\alpha_1\bar{z} + \bar{\alpha}_1 z + \beta_1 = 0$ and are mutually perpendicular, then**

(A) $\alpha_1\bar{\alpha}_2 + \bar{\alpha}_1\alpha_2 = 0$ **(B)** $\alpha_1\bar{\alpha}_2 = \bar{\alpha}_1\alpha_2$

(C) $\left|\arg\left(\frac{\alpha_1}{\alpha_2}\right)\right| = \frac{\pi}{2}$ **(D)** $\arg\left(\frac{\alpha_1}{\alpha_2}\right) = 0$ **or** π

Sol. Since both lines are mutually perpendicular, $\frac{\alpha_1}{\alpha_2}$ is purely imaginary.

$\Rightarrow \left|\arg\left(\frac{\alpha_1}{\alpha_2}\right)\right| = \frac{\pi}{2}$

$\Rightarrow \frac{\alpha_1}{\alpha_2} + \frac{\bar{\alpha}_1}{\bar{\alpha}_2} = 0 \Rightarrow \alpha_1\bar{\alpha}_2 + \bar{\alpha}_1\alpha_2 = 0$

Q. **If $\alpha^3 = 5 + i\sqrt{2}$ and $\beta^3 = 5 - i\sqrt{2}$, then possible real values of $\alpha + \beta$ is/are**

(A) 2 **(B) –2** **(C)** $1 - \sqrt{6}$ **(D)** $1 + \sqrt{6}$

Sol. $\alpha^3 + \beta^3 = 10$, $(\alpha\beta)^3 = 27$

$\Rightarrow$ $\alpha\beta = 3, 3\omega, 3\omega^2$

$(\alpha + \beta)^3 - 3(\alpha+\beta)\alpha\beta = 10$

Let $\alpha+\beta = t$

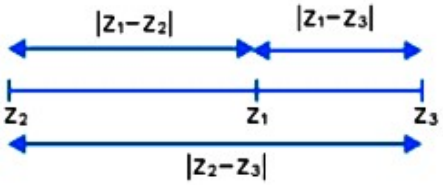
$\Rightarrow$ $t^3 - 9t - 10 = 0 = (t + 2)(t^2 - 2t - 5)$

$\Rightarrow$ $\alpha+\beta = -2, 1 \pm \sqrt{6}$

Q. **Let $A(z_1)$, $B(z_2)$, $C(z_3)$ are points in complex plane such that $z_1|z_2-z_3|-z_2|z_3-z_1|-z_3|z_1-z_2| = 0$, then which of the following may be correct?**
(A) A, B, C are collinear such that A lies between B And C
(B) A, B, C are collinear such that B lies between A and C
(C) A, B, C are collinear such that C lies between A and B
(D) O(0) is the centre of circle which touches the sides of triangle ABC.

Sol. If A lies between B and C

$$z_1 = \frac{|z_1 - z_2|z_3 + |z_1 - z_3|z_2}{|z_2 - z_3|}$$



$\Rightarrow z_1|z_2-z_3|-z_2|z_3-z_1|-z_3|z_1-z_2| = 0$
Hence, option (A) is correct.
If A, B, C are non collinear, then excentre o

$$I_1 = \frac{-|z_2 - z_3|z_1 + |z_1 - z_2|z_3 + |z_1 - z_3|z_2}{-|z_2 - z_3| + |z_1 - z_2| + |z_1 - z_3|} = 0$$

Hence, option (D) is also correct.

Comprehension (Q. 1 to Q.3)

Suppose A, B, C are three collinear points corresponding complex numbers $z_1 = ai$, $z_2 = \frac{1}{2} + bi, z_3 = 1 + ci$ (a, b, c being real numbers), respectively, consider a curve 'C' whose equation is given by $z = z_1 \cos^4 t + 2z_2 \cos^2\sin^2 t + z_3 \sin^4 t$, $t \in R$.

Q.1 **'C' in argand plane represents**
(A) Straight line **(B) Circle** **(C) Parabola** **(D) Ellipse**

Ans. C

Q.2 **A line bisecting AB and parallel to AC meets curve 'C' in point P, then point P is given by**
(A) $\left(-\frac{1}{2}, \frac{a+c-2b}{2}\right)$ **(B)** $\left(\frac{1}{2}, \frac{a+c-2b}{4}\right)$
(C) $\left(\frac{1}{2}, \frac{a+c+2b}{2}\right)$ **(D)** $\left(\frac{1}{2}, \frac{a+c+2b}{4}\right)$

Ans. D

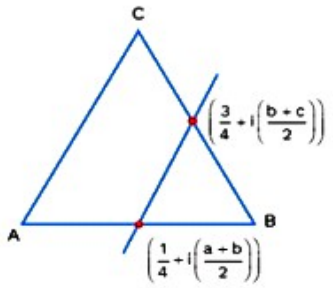
Q.3 **Point P lies**

(A) Inside ΔABC **(B) Outside ΔABC**

(C) On the side of ΔABC **(D) Nothing can be said**

Ans. **A**

Sol. Curve 'C' is $x + iy = (a\cos^4 t)i + (1 + 2bi)\cos^2 t \sin^2 t + (1 + ci)\sin^4 t$



$x = \sin^4 t + \sin^2 t \cos^2 t = \sin^2 t$

$y = a\cos^4 t + 2b\sin^2 t\cos^2 t + c\sin^4 t$

$\Rightarrow y = a(1-x)^2 + 2bx(1-x) + cx^2$

which represents a parabola.

Equation of line through mid point of AB and parallel to AC is

$$z = \lambda\left(\frac{1}{4} + i\frac{a+b}{2}\right) + (1-\lambda)\left(\frac{3}{4} + i\left(\frac{b+c}{2}\right)\right)$$

For point of intersection 'P'

$$\frac{3-2\lambda}{4} + i\left(\frac{b+c}{2} + \lambda\left(\frac{a-c}{2}\right)\right) = i\left(a\cos^4 t + 2b\sin^2 t\cos^2 t + c\sin^4 t\right) + \sin^2 t$$

$$\Rightarrow \quad \frac{3-2\lambda}{4} = \sin^2 t$$

$$\Rightarrow \quad \frac{3}{2} - 2\sin^2 t = \lambda$$

$$\text{and } \frac{(b+c)}{2} + \lambda\left(\frac{a-c}{2}\right) = a\cos^4 t + 2b\sin^2 t\cos^2 t + c\sin^4 t$$

$$\frac{b+c}{2} + \left(\frac{3}{2} - 2\sin^2 t\right)\left(\frac{a-c}{2}\right) = a\cos^4 t + 2b\sin^2 t\cos^2 t + c\sin^4 t$$

$$\frac{3a+2b-c}{4} + (c-a)\sin^2 t = (2b-a-c)\sin^2\cos^2 t + a\cos^2 t + c\sin^2 t$$

$$\Rightarrow \quad 0 = (2b-a-c)(4\sin^2 t\cos^2 t - 1)$$

$$\Rightarrow \quad 4\sin^4 t - 4\sin^2 t + 1 = 0$$

$$\Rightarrow \quad \sin^2 t = \frac{1}{2} \quad \Rightarrow \lambda = \frac{1}{2}$$

$$P \equiv \left(\frac{1}{2}, \frac{2b+a+c}{4}\right)$$

Comprehension (Q. 4 to Q. 6)

If $\frac{1}{a+\omega}+\frac{1}{b+\omega}+\frac{1}{c+\omega}+\frac{1}{d+\omega}=\frac{2}{\omega}$ and $\frac{1}{a+\omega^2}+\frac{1}{b+\omega^2}+\frac{1}{c+\omega^2}+\frac{1}{d+\omega^2}=\frac{2}{\omega^2}$, where ω and ω^2 are imaginary cube roots of unity, then

Q.4 **$a+b+c+d=$**

(A) 0 **(B) abcd** **(C) −2abcd** **(D) 2abcd**

Ans. D

Q.5 **$abc+abd+acd+bcd=$**

(A) 0 **(B) 1** **(C) 2** **(D) 4**

Ans. C

Q.6 $\frac{1}{a+1}+\frac{1}{b+1}+\frac{1}{c+1}+\frac{1}{d+1}=$

(A) 0 **(B) 1** **(C) 2** **(D) 4**

Ans. C

$$\frac{1}{a+x}+\frac{1}{b+x}+\frac{1}{c+x}+\frac{1}{d+x}=\frac{2}{x}$$ ω, ω^2

$\Rightarrow 2x^4+(a+b+c+d)x^3-(abc+abd+acd+bcd)x-2abcd=0$

Put $x=\omega$, $(2-\Sigma abc)\,\omega+\Sigma a-2abcd=0$

Put $x=\omega^2$ $(2-\Sigma abc)\,\omega^2+\Sigma a-2abcd=0$

Subtracting, we get

$(2-\Sigma abc)(\omega-\omega^2)=0$

$\Rightarrow \Sigma abc=2$

$\Rightarrow \Sigma a=2abcd$

Now, using these values, we can see that $x=1$, satisfy the equation

$\therefore \frac{1}{a+1}+\frac{1}{b+1}+\frac{1}{c+1}+\frac{1}{d+1}=2$

Q. **If α, β, γ are cube roots of unity, then the value of $\begin{vmatrix} e^{\alpha} & e^{2\alpha} & e^{3\alpha}-1 \\ e^{\beta} & e^{2\beta} & e^{3\beta}-1 \\ e^{\gamma} & e^{2\gamma} & e^{3\gamma}-1 \end{vmatrix}$ is equal to:**

Sol. $$\Delta = e^{\alpha+\beta+\gamma}\begin{vmatrix} 1 & e^{\alpha} & e^{2\alpha}-e^{-\alpha} \\ 1 & e^{\beta} & e^{2\beta}-e^{-\beta} \\ 1 & e^{\gamma} & e^{2\gamma}-e^{-\gamma} \end{vmatrix} = \left(e^{\beta}-e^{\alpha}\right)\left(e^{\gamma}-e^{\alpha}\right)\begin{vmatrix} 1 & e^{\alpha} & e^{2\alpha}-e^{-\alpha} \\ 0 & 1 & e^{\beta}+e^{\alpha}+e^{-\alpha}e^{-\beta} \\ 0 & 1 & e^{\gamma}+e^{\alpha}+e^{-\alpha}e^{-\gamma} \end{vmatrix}$$

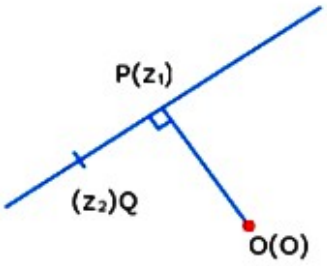
$$\Delta = \left(e^{\beta}-e^{\alpha}\right)\left(e^{\gamma}-e^{\alpha}\right)\left(e^{\gamma}-e^{\beta}\right)\left(1-e^{-(\alpha+\beta+\gamma)}\right) = 0$$

Q. **Let L be any line not passing through origin and $P(z_1)$ be the foot of perpendicular from origin to the line. Let $Q(z_2)$ be any point different from P on L. Then $\frac{z_2}{z_1}+\frac{\bar{z}_2}{\bar{z}_1}$ is equal to :**

Sol. $\frac{z_2-z_1}{z_1}$ is purely imaginary

$$\Rightarrow \quad \frac{z_2-z_1}{z_1}+\frac{\bar{z}_2-\bar{z}_1}{\bar{z}_1}=0$$

$$\Rightarrow \quad \frac{z_2}{z_1}+\frac{\bar{z}_2}{\bar{z}_1}=2$$



Q. **$\omega_1, \omega_2, \omega_3, \ldots, \omega_n$ be complex numbers. A line L in the argand plane is called a mean line for the points $\omega_1, \omega_2, \ldots, \omega_n$ if L contains points (complex numbers) $z_1, z_2, \ldots, z_n$ such that $\sum_{i=1}^{n}(z_i-\omega_i)=0$. For the numbers $\omega_1 = 32 + 170i$, $\omega_2 = -7 + 64i$, $\omega_3 = -9 + 200i$, $\omega_4 = 1 + 27i$ and $\omega_5 = -14 + 43i$, there is a unique mean line with y-intercept 3 . Then the slope of this mean line is equal to**

Sol. Let the line be $y = mx + 3$

Put $z_k = x_k + iy_k$

$$\Rightarrow \quad y_k = mx_k + 3$$

$$\Rightarrow \quad \sum_{k=1}^{5} y_k = m\sum_{k=1}^{5} x_k + 15$$

$$\sum_{k=1}^{5} z_k = \sum_{k=1}^{5} \omega_k = 3 + 504i$$

$$\Rightarrow \quad \sum y_k = 504, \quad \sum x_k = 3$$

$$\Rightarrow \quad 504 = m(3) = 15 \qquad \Rightarrow m = 163$$

Previous Year's Question

If z_1, z_2 are complex numbers such that $Re(z_1) = |z_1 - 1|$, $Re(z_2) = |z_2 - 1|$ and $\arg(z_1 - z_2) = \frac{\pi}{6}$, then $Im(z_1 + z_2)$ is equal to: **(JEE Main 2020)**

(A) $2\sqrt{3}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{2}$

Previous Year's Question

Let $u = \frac{2z + i}{z - ki}$, $z = x + iy$ and $k > 0$. If the curve represented by $Re(u) + Im(u) = 1$ intersects the y-axis at the points P and Q where PQ = 5, then the value of k is: **(JEE Main 2020)**

(A) 4 (B) 1/2 (C) 2 (D) 3/2